



FEATURE ENGINEERING AND REGULARIZATION

Adding and Selecting Features in Linear Regression

Reminder



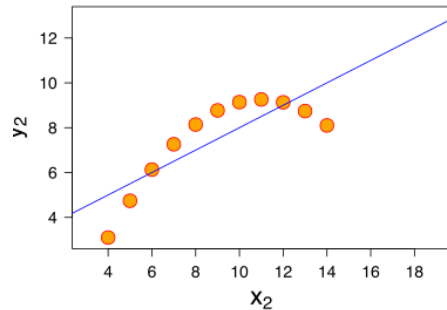
- Wednesday office hour: 2-3pm



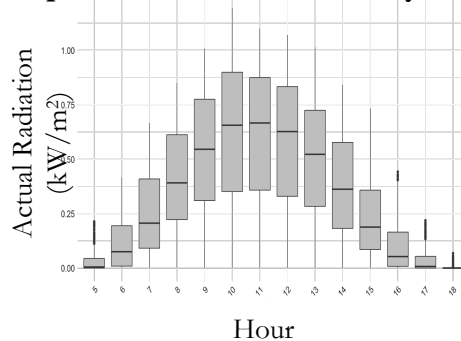
Part I: Feature Engineering

NYC Taxi Ride Prediction

Back to Linear Regression



Boxplot of Actual Radiation By Hour



Non-Linearities

Real-world data often includes “bends” and ”interactions”.

```
=====
Regression Results
=====
R-squared: 0.94
Adjusted R-squared: 0.94
No. Observations: 1536
=====
Coefficients:
=====
```

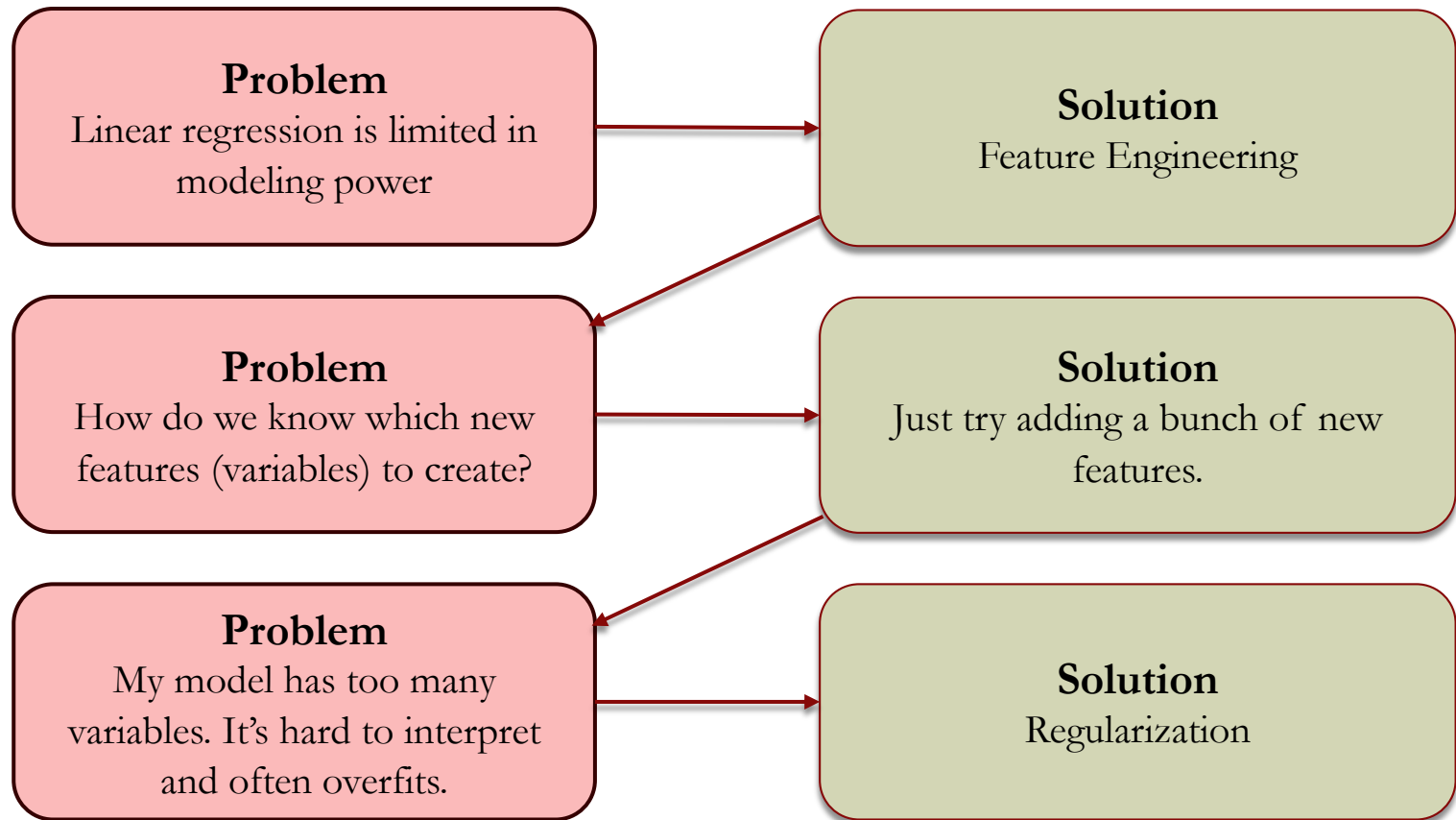
	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
Intercept	1828.91	76.75	23.83	0.00	1678.35	1979.47	***
hour	-878.49	113.90	-7.71	0.00	-1101.90	-655.08	***
hour_after_1	536.01	193.58	2.77	0.01	156.30	915.72	**
hour_after_2	110.64	183.26	0.60	0.55	-248.83	470.12	
hour_after_3	217.38	182.65	1.19	0.23	-140.89	575.65	
hour_after_4	75.62	182.61	0.41	0.68	-282.58	433.82	
hour_after_5	882.78	182.61	4.83	0.00	524.58	1240.97	***
hour_after_6	734.45	182.61	4.02	0.00	376.26	1092.65	***
hour_after_7	-335.57	182.61	-1.84	0.07	-693.76	22.63	.
hour_after_8	-1497.03	182.61	-8.20	0.00	-1855.23	-1138.84	***
hour_after_9	-439.79	182.61	-2.41	0.02	-797.99	-81.60	*
hour_after_10	344.96	182.61	1.89	0.06	-13.23	703.16	.
hour_after_11	636.18	182.61	3.48	0.00	277.98	994.37	***
hour_after_12	-476.22	182.61	-2.61	0.01	-834.41	-118.02	**
hour_after_13	380.17	182.61	2.08	0.04	21.98	738.37	*
hour_after_14	120.60	182.61	0.66	0.51	-237.60	478.79	
hour_after_15	-907.25	182.61	-4.97	0.00	-1265.44	-549.05	***
hour_after_16	619.22	182.61	3.39	0.00	261.03	977.42	***
hour_after_17	1016.96	182.61	5.57	0.00	658.77	1375.16	***
hour_after_18	-1199.01	182.61	-6.57	0.00	-1557.21	-840.82	***
hour_after_19	-513.61	182.61	-2.81	0.00	-871.81	-155.41	**
hour_after_20	634.94	182.62	3.48	0.00	276.72	993.16	***
hour_after_21	-164.00	182.82	-0.90	0.37	-522.62	194.61	
hour_after_22	-999.81	186.27	-5.37	0.00	-1365.19	-634.43	***
hour_after_23	-179.79	238.15	-0.75	0.45	-646.93	287.36	

```
=====
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====
```

Variable Selection

Model includes many insignificant variables, making it hard to interpret.

The Big Picture



Feature Engineering

Goal: Demonstrate how feature engineering can allows us to capture *non-linear* relationships using *linear* regression.

- NYC Taxi Rides Dataset and Baseline Models
- Feature Engineering: “Non-Linear” Regression
 - Interaction Terms
 - Piece-wise
 - Polynomial

NYC Taxi Rides Dataset

NYC Taxi Rides Dataset and Baseline Models

Feature Engineering for Non-Linear Regression

Interaction Terms

Piece-wise

Polynomial

Variable Selection Methods

New York City Taxi Data

- The New York City Taxi & Limousine Commission (NYCTLC) maintains a complete database of yellow cab rides
 - Over 170 million rides per year
- Granular information on mobility patterns: temporal patterns, spatial patterns, fares, etc.
- This information is central to many business & policy decisions



Aggregate Ride Data

k : # of independent variables (**small k**); n : # of observations (**large n**)



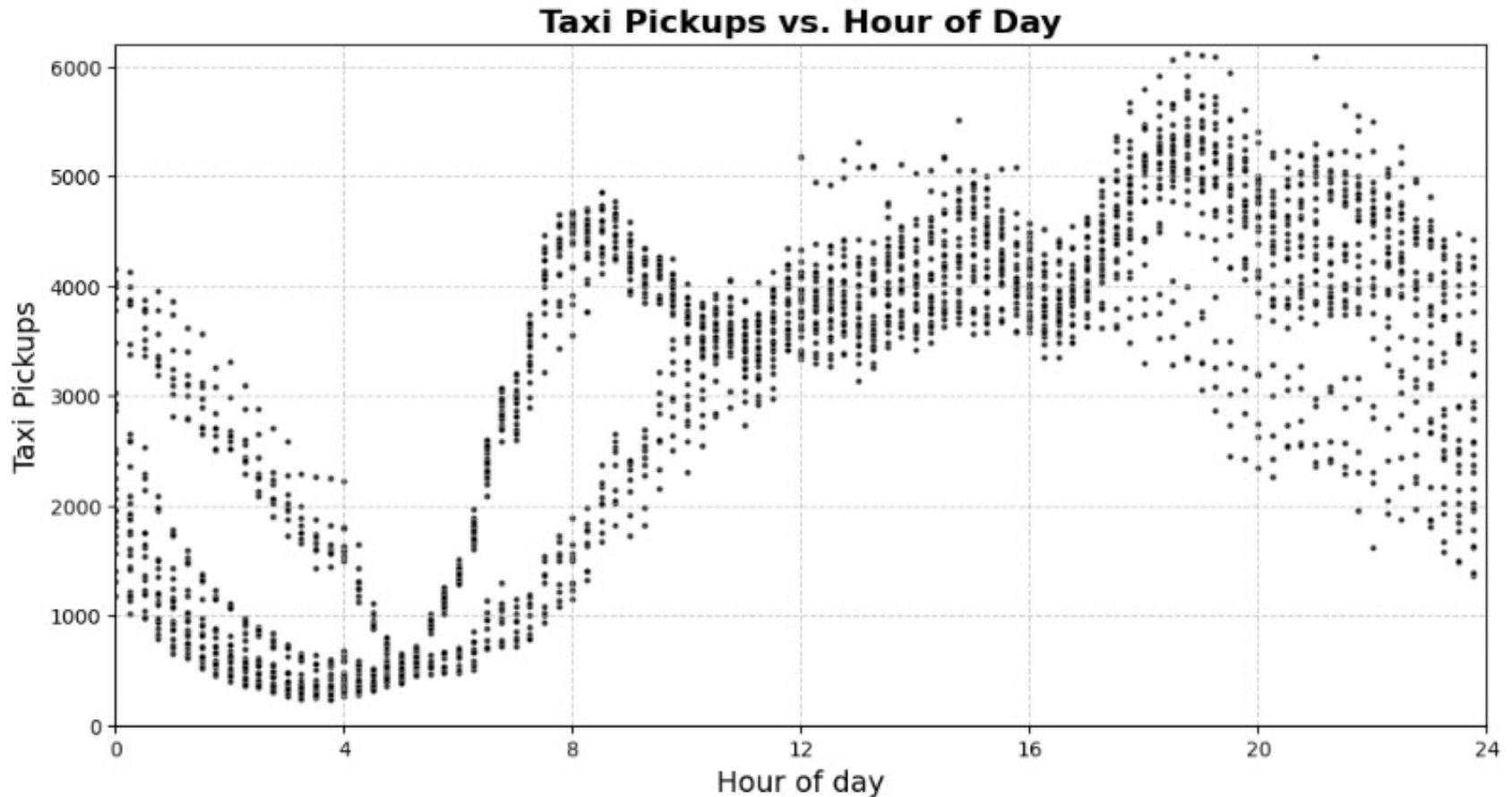
	pickup_month	pickup_day	pickup_wday	pickup_hour	pickup_hour_qtr	count	passengers	fare
0	1	1.0	2.0	0.0	0.00	2597.0	4516.0	34085.15
1	1	1.0	2.0	0.0	0.25	4455.0	7723.0	55377.60
2	1	1.0	2.0	0.0	0.50	4667.0	8064.0	58139.85
3	1	1.0	2.0	0.0	0.75	4738.0	8326.0	60099.30
4	1	1.0	2.0	1.0	1.00	4923.0	8491.0	62071.80
5	1	1.0	2.0	1.0	1.25	5023.0	8723.0	64117.61
6	1	1.0	2.0	1.0	1.50	4623.0	8029.0	59550.30
7	1	1.0	2.0	1.0	1.75	4416.0	7535.0	58492.20
...	...	...	...	...	...	...	...	...
34938	12	30.0	1.0	23.0	23.50	1770.0	2902.0	25385.35
34939	12	30.0	1.0	23.0	23.75	1823.0	3023.0	24292.10

Cleaned Dataset Structure

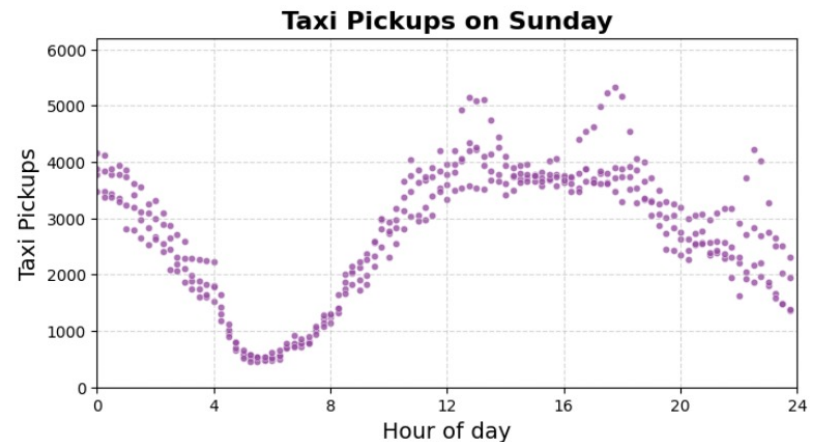
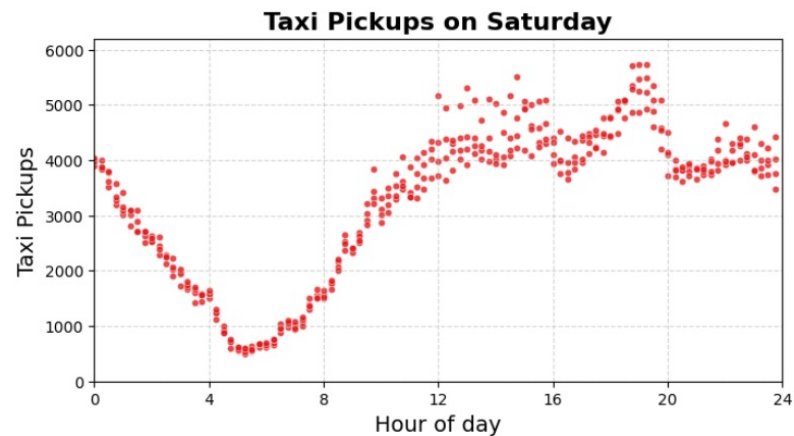
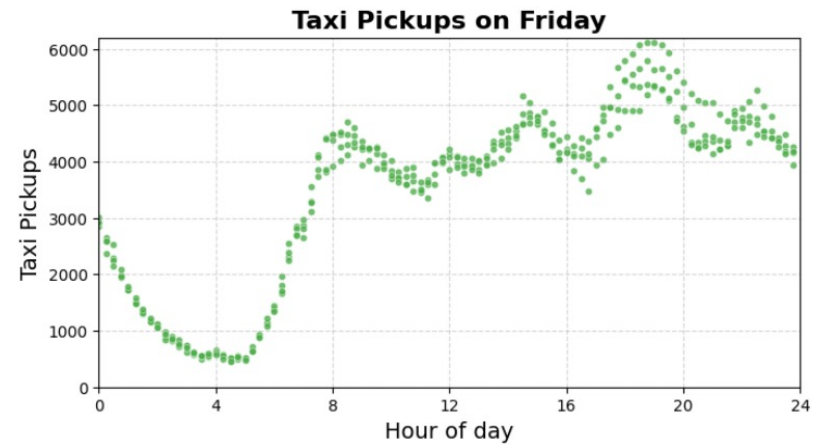
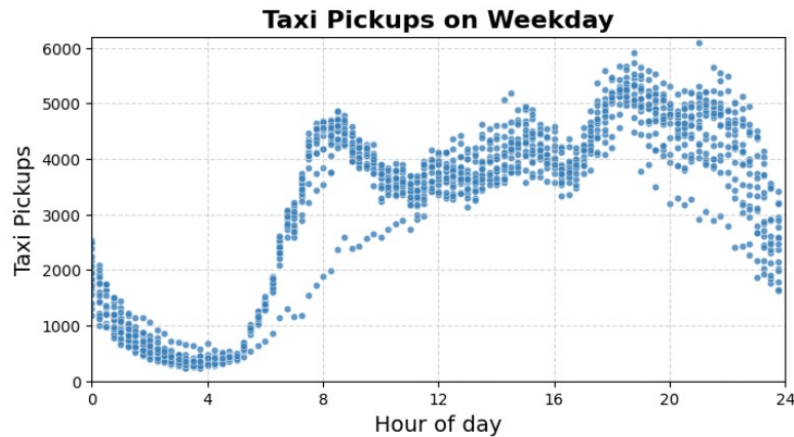
- **Hour**
 - 15 minute intervals from 0.00 (midnight) to 23.75 (11:45pm)
- **Day of Week**
 - Weekday (Monday – Thursday)
 - Friday
 - Saturday
 - Sunday
- **Pickups:** Total number of passenger pickups
- **Test-Train Split**
 - Train: February 2018
 - Test: March 2018

	hour	pickups	day
0	0.00	2161	Weekday
1	0.25	1747	Weekday
2	0.50	1620	Weekday
3	0.75	1428	Weekday
4	1.00	1161	Weekday
5	1.25	1010	Weekday
6	1.50	925	Weekday
7	1.75	850	Weekday
8	2.00	739	Weekday
9	2.25	607	Weekday

Pickups Overall (All Days of the Week)



Pickups by Day of the Week



Baseline Models

- **Baseline 1.** Linear Regression with Hour Only

$$y = \beta_0 + \beta_{\text{hour}} \cdot x_{\text{hour}}$$

- **Baseline 2.** Linear Regression with Hour + Day of Week (Categorical)

$$\begin{aligned} y = & \beta_0 + \beta_{\text{hour}} \cdot x_{\text{hour}} \\ & + \beta_{\text{Fri}} \cdot x_{\text{Fri}} \\ & + \beta_{\text{Sat}} \cdot x_{\text{Sat}} \\ & + \beta_{\text{Sun}} \cdot x_{\text{Sun}} \end{aligned}$$

The reference level is weekday

Baseline 1: Hour Only

```
model_hour = smf.ols("pickups ~ hour", data = nyc_train).fit()
```

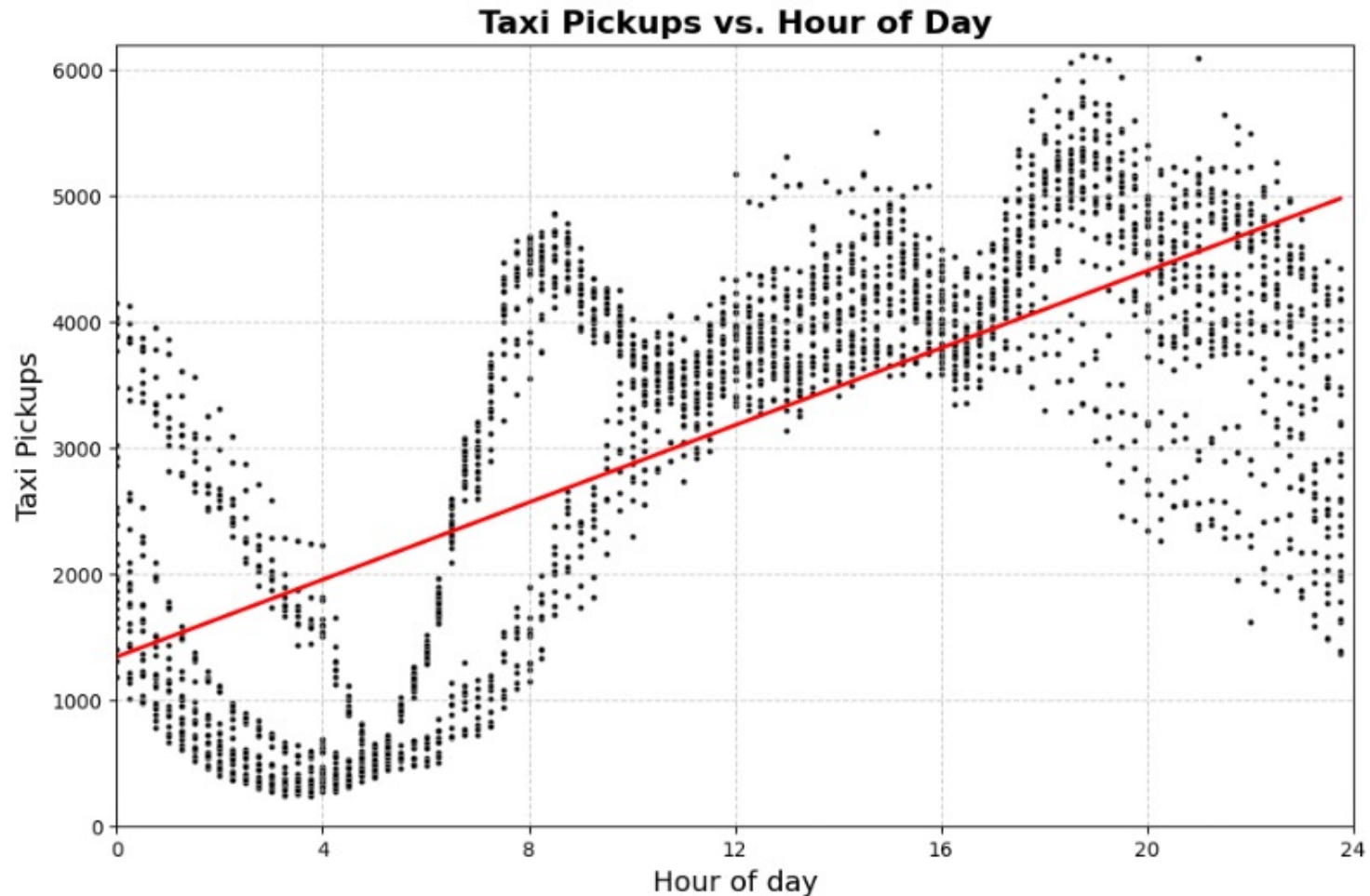
```
=====
Regression Results
=====
R-squared: 0.48
No. Observations: 2688
=====
Coefficients:
-----

```

	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
Intercept	1341.51	41.92	32.0	0.0	1259.30	1423.71	***
hour	153.09	3.05	50.2	0.0	147.11	159.06	***

```
=====
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====
```

Baseline 1: Hour Only



Baseline 2: LinReg with Hour and Day

- Adding Day of Week to Our Model:
 - Weekday will be our reference level.

Pickups	Hour	Day	Week day	Fri	Sat	Sun
234	5.00	Weekday	1	0	0	0
134	5.25	Weekday	1	0	0	0
...	...	...	...	...	...	...
622	12.50	Friday	0	1	0	0
575	12.75	Friday	0	1	0	0
...	...	...	...	...	...	...
117	23.75	Sunday	0	0	0	1

Baseline 2: LinReg with Hour and Day

```
model_day = smf.ols("pickups ~ hour + day", data = nyc_train).fit()
```

```
=====
Regression Results
=====
```

```
R-squared: 0.50
```

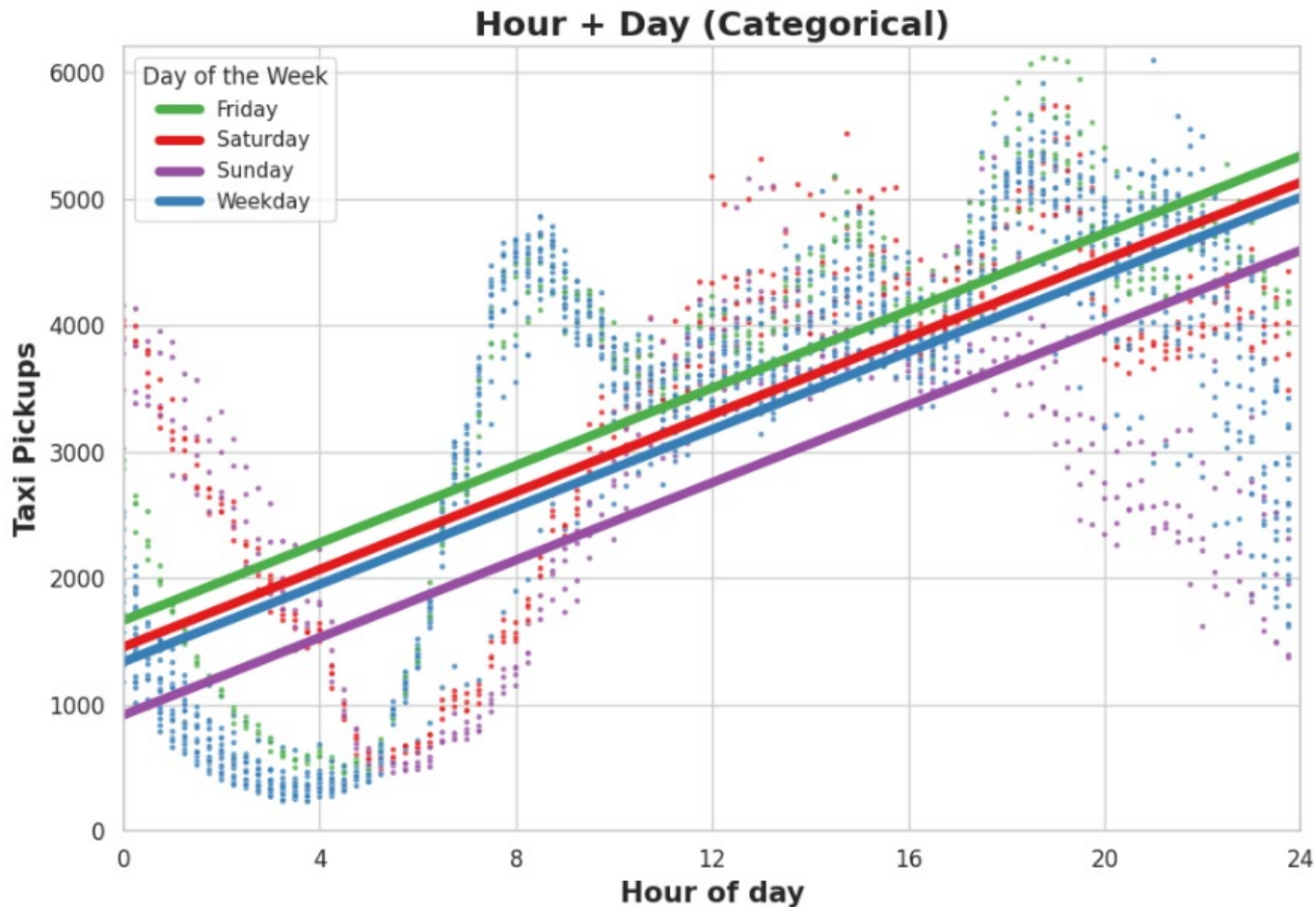
```
No. Observations: 2688
=====
```

```
Coefficients:
```

```
-----
              Coef.  Std.Err.      t  P>|t|  [0.025   0.975] Signif
Intercept    1337.98    44.95   29.77   0.00   1249.84   1426.11   ***
day[T.Friday]  327.46    61.41    5.33   0.00    207.05    447.86   ***
day[T.Saturday] 117.56    61.41    1.91   0.06    -2.85    237.96    .
day[T.Sunday] -420.31    61.41   -6.84   0.00   -540.72   -299.90   ***
hour          153.09     3.00   51.09   0.00    147.21    158.96   ***
=====
```

```
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====
```

Baseline 2: LinReg with Hour and Day



Model Performance

Model	Number of Parameters	In-sample R^2	Out-of-sample R^2
LR with hour	2	0.48	0.47
LR with hour + day	5	0.50	0.48

Improvement Opportunities

- **Observation 1.** Each day of the week should be treated separately, i.e. the best-fit line for each day of the week should have different slopes, in addition to having different intercepts.
 - **Solution 1:** Adding interaction terms between hour and day of the week.
- **Observation 2.** A single line is just not a good fit. Allow the curve to “bend” by:
 - **Solution 2a:** Breaking up the x-axis into segments (AM rush hour, PM rush hour...etc.)
 - **Solution 2b:** Adding higher-order polynomial terms (x^2 , x^3 ...etc.)
 - **Solution 2:** Put them together!

Feature Engineering

NYC Taxi Rides Dataset and Baseline Models

Feature Engineering for Non-Linear Regression

Interaction Terms

Piece-wise

Polynomial

What is Feature Engineering?

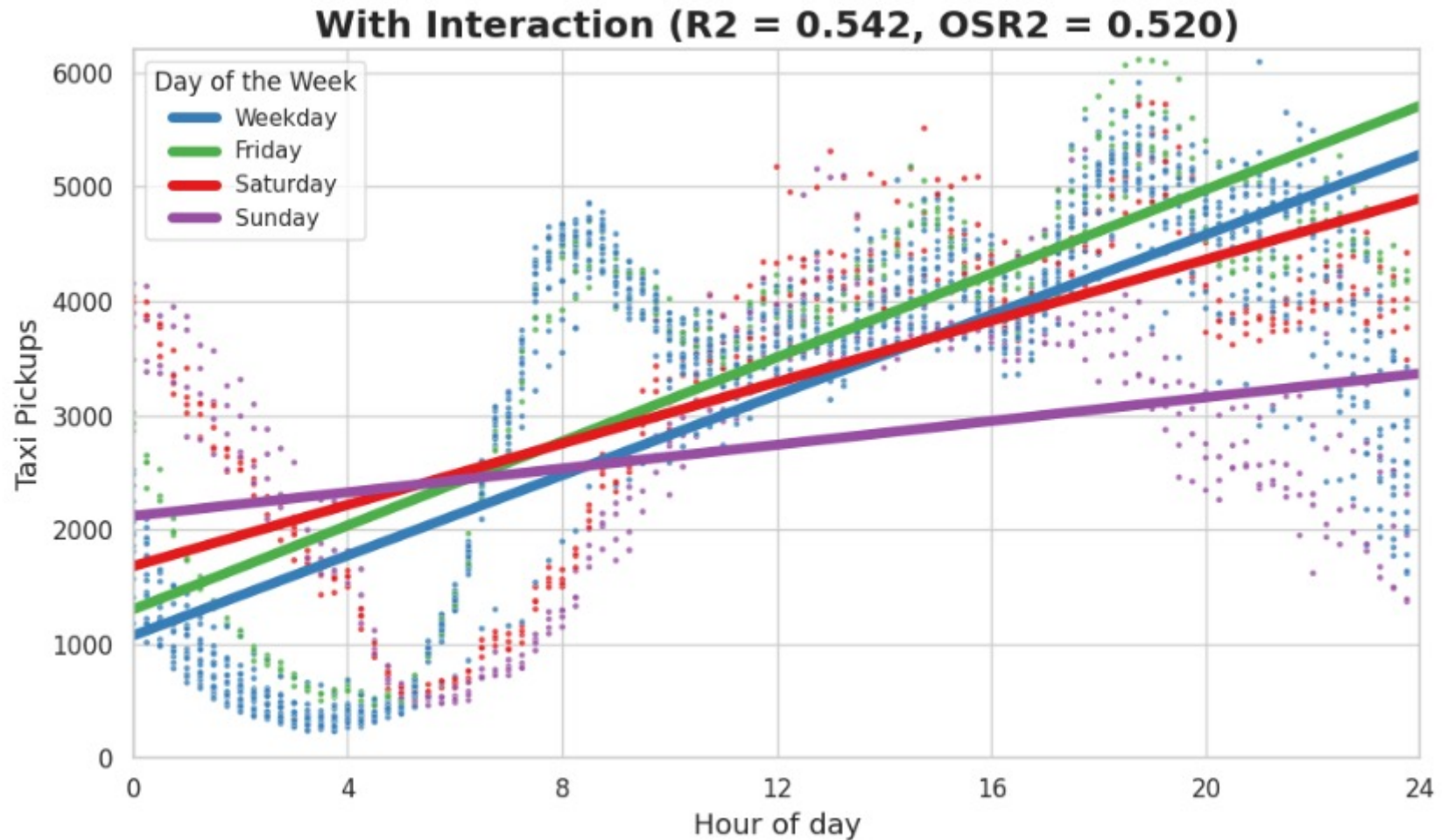
- Creating new variables from existing variables to help a model identify patterns in the data.
- Linear regression still fits a “line”, but to features that can be non-linear version of the original features.
 - For example, predicting a patient’s healthcare cost from their height and weight.
 - Calculate BMI ($\text{height} / \text{weight}^2$) as a new feature.

Healthcare Costs	Height	Weight	BMI
...	...	...	...
...	...	...	...
...	...	...	...

Adding Interaction Terms

Different slopes for different days of the week.

Interaction between Hour and Day



What are Interaction Terms?

- In vanilla linear regression, all variables are independent.
 - Changing x_1 by 1 results a β_1 change to y regardless of other variables.

$$y = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2$$

- Add the product of two variables as a new variable.

$$y = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \beta_{12} \cdot \underbrace{(x_1 \cdot x_2)}_{\text{New variable } x_{12}}$$

A 1 unit increase in x_1 changes y by:
 $\beta_1 + \beta_{12} \cdot x_2$

A 1 unit increase in x_2 changes y by:
 $\beta_2 + \beta_{12} \cdot x_1$

Adding New Variables

- Add 3 new variables/columns by multiplying the hour column (**black**) by each of the day columns (**orange**) to get the new interaction columns (**maroon**)

Hour	Fri	Sat	Sun	Hour:Fri	Hour:Sat	Hour:Sun
5.00	0	0	0	0	0	0
5.25	0	0	0	0	0	0
...	...	...	...	...	...	...
12.50	1	0	0	12.50	0	0
12.75	1	0	0	12.75	0	0
...	...	...	...	...	...	...
23.50	0	0	1	0	0	23.55
23.75	0	0	1	0	0	23.75

Linear Regression with New Vars.

```
model_interaction = smf.ols("pickups ~ hour + day + hour_Fri  
+ hour_Sat + hour_Sun", data = nyc_train).fit()
```

```
=====  
Regression Results  
=====
```

```
R-squared: 0.54
```

```
No. Observations: 2688  
=====
```

```
Coefficients:  
-----
```

	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
Intercept	1072.70	52.30	20.51	0.00	970.16	1175.25	***
day[T.Friday]	228.76	116.94	1.96	0.05	-0.54	458.06	.
day[T.Saturday]	607.03	116.94	5.19	0.00	377.74	836.33	***
day[T.Sunday]	1045.81	116.94	8.94	0.00	816.51	1275.10	***
hour	175.42	3.80	46.12	0.00	167.97	182.88	***
hour_Fri	8.31	8.51	0.98	0.33	-8.37	24.99	
hour_Sat	-41.22	8.51	-4.85	0.00	-57.90	-24.54	***
hour_Sun	-123.46	8.51	-14.52	0.00	-140.14	-106.78	***

```
=====  
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1  
=====
```

Different Slopes for Different Days

$$Y = 1072 + 175 \cdot \text{hour}$$

$$+ 228 \cdot \text{Friday}$$

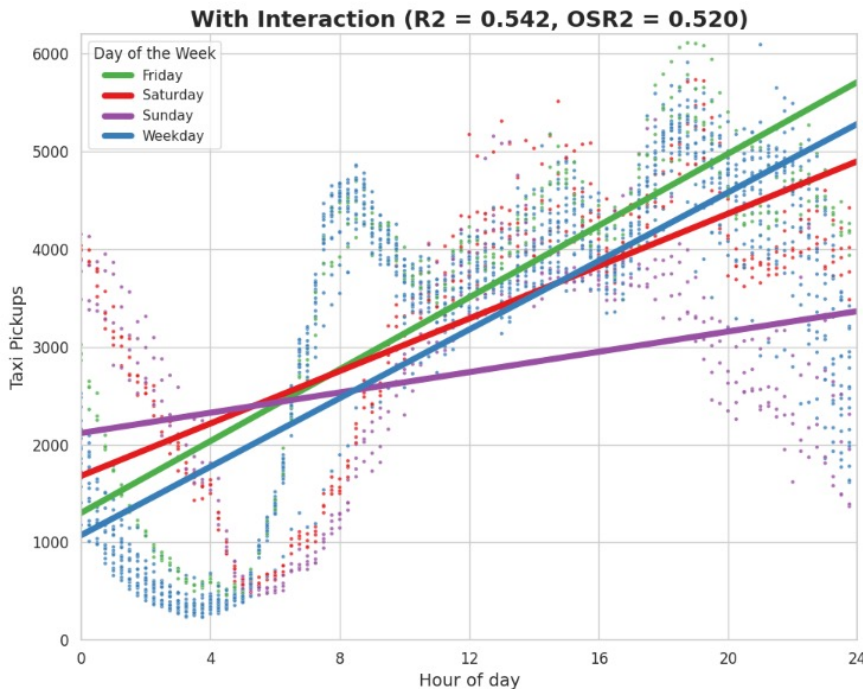
$$+ 8.3 \cdot \text{hour_Friday}$$

$$+ 607 \cdot \text{Sat}$$

$$- 41 \cdot \text{hour_Sat}$$

$$+ 1045 \cdot \text{Sun}$$

$$- 123 \cdot \text{hour_Sun}$$



For Weekday:

- $1072 + 175 \cdot \text{hour}$

For Friday:

- $Y = 1072 + 175 \cdot \text{hour} + 228 + 8.3 \cdot \text{hour_Fri}$

- $1300 + 183.3 \cdot \text{hour}$

For Saturday:

- $Y = 1072 + 175 \cdot \text{hour} + 607 - 41 \cdot \text{hour_Sat}$

- $1679 + 134 \cdot \text{hour}$

For Sunday:

- $Y = 1072 + 175 \cdot \text{hour} + 1045 - 123 \cdot \text{hour_Sun}$

- $2117 + 52 \cdot \text{hour}$

Model Performance

Model	Number of Parameters	In-sample R^2	Out-of-sample R^2
LR with hour	2	0.48	0.47
LR with hour + day	5	0.50	0.48
LR with hour * day interaction	8	0.54	0.52

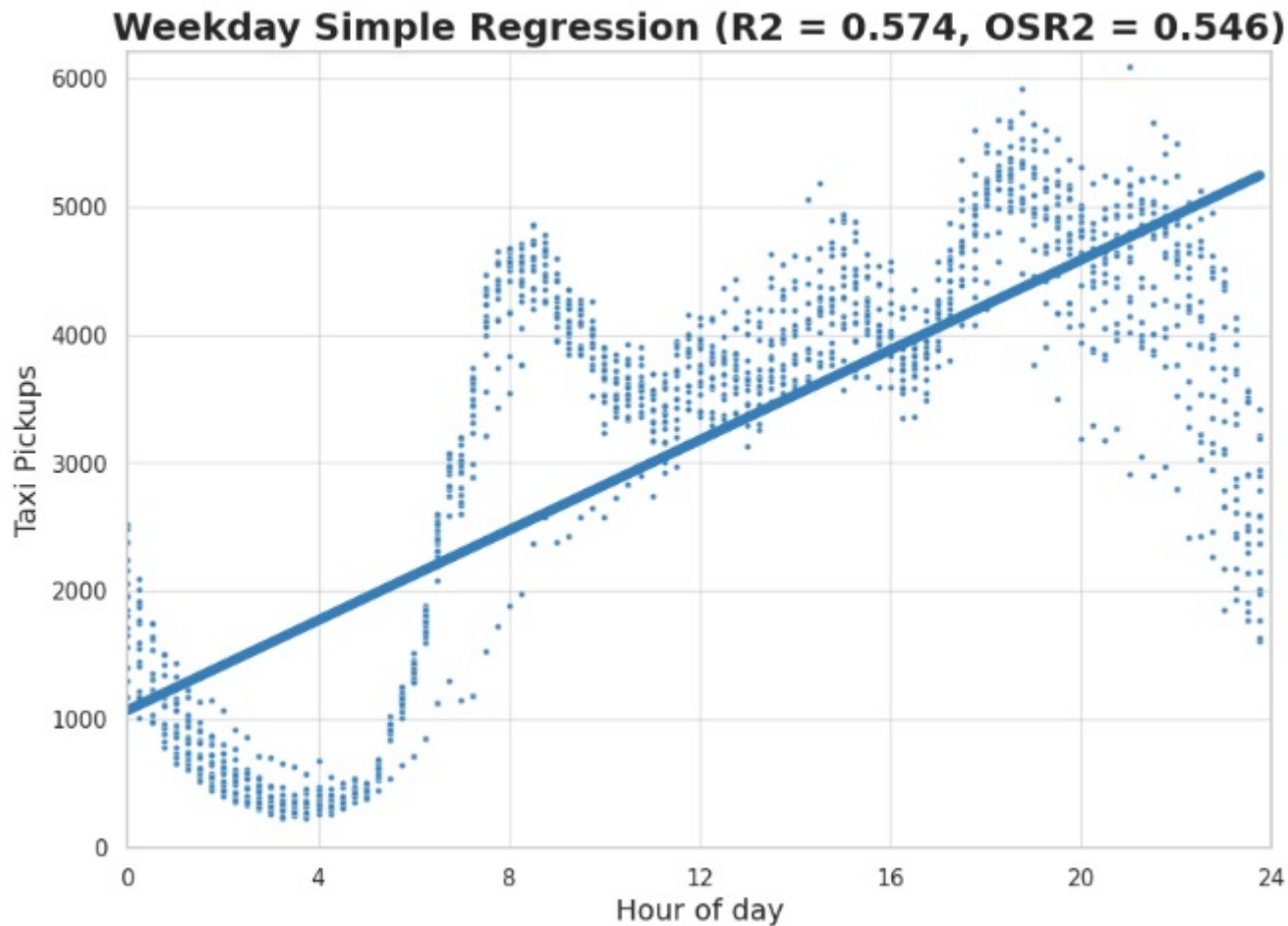
Let's focus on each day separately

- By adding these interaction columns (hour * Friday, hour * Saturday, hour * Sun), we are essentially performing regression for each of the 4 days of the week (Weekday, Fri, Sat, Sun) separately.
- Let's now focus on weekday and see how to improve our fit there.
- Then we will put it all together for weekday, Friday, Saturday, Sunday.

Improvement Opportunities

- **Observation 1.** The line for each day of the week should have different slopes, in addition to having different intercepts.
 - **Solution 1:** Adding interaction terms between hour and day of the week.
- **Observation 2.** A single line is just not a good fit. Allow the curve to “bend” by:
 - **Solution 2a:** Breaking up the x-axis into segments (AM rush hour, PM rush hour...etc.)
 - **Solution 2b:** Adding higher-order polynomial terms (x^2 , x^3 ...etc.)
 - **Solution 2:** Put them together!

A Line is a Poor Fit

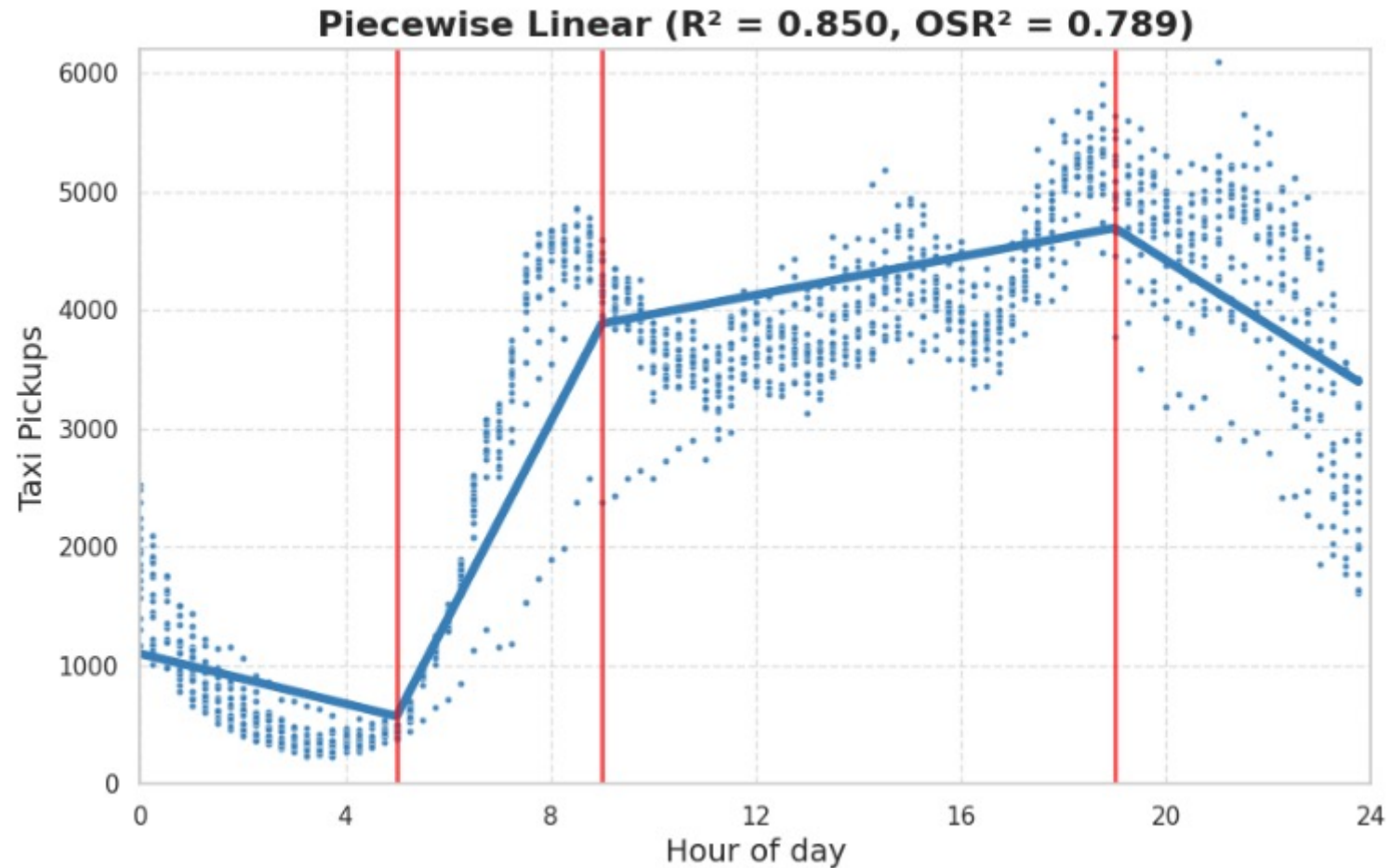


Beyond A Single Line

How to incorporate “bends” in the line.

1. Piece-Wise
2. Polynomial
3. Putting them Together

Piece-wise Line (Line with Bends)



How do we do this?

- Use our domain knowledge:
 - 0:00 – 5:00 Night (decreasing)
 - 5:00 – 9:00 Morning (sharply increasing)
 - 9:00 – 19:00 Daytime (slight increase)
 - 19:00 – 23:59 Evening (decreasing)
- Add 3 dummy variables into the dataset, one for each break point. Transform the “Hour” variable into the following 3:
 - Hour_after_5 = Number of hours since 5am
 - Hour_after_9 = Number of hours since 9am
 - Hour_after_19 = Number of hours since 7pm
- Each dummy variable is 0 before its time (e.g., if hour = 7, then Hour_after_9 = 0), then counts the number of hours since that time.

Adding Variables

Pickups	Hour		Hour_after_5	Hour_after_9	Hour_after_19
2600	0.00	Night	0	0	0
1400	2.00		0	0	0
870	4.00		0	0	0
1700	6.00	Morning	1.00	0	0
4000	7.00		2.00	0	0
3800	8.00		3.00	0	0
4200	10.00	Day	5.00	1.00	0
...	...		...	...	...
4000	16.00		11.00	7.00	0
4800	18.00	Evening	13.00	9.00	0
5100	20.00		15.00	11.00	1.00
4400	22.00		17.00	13.00	3.00

Output with Additional Variables

```
model_pl = smf.ols("pickups ~ hour + hour_after_5 + hour_after_9 +  
    hour_after_19", data = nyc_train_weekday).fit()
```

```
=====
```

```
Regression Results
```

```
=====
```

```
R-squared: 0.85
```

```
No. Observations: 1536
```

```
=====
```

```
Coefficients:
```

```
-----
```

	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
Intercept	1099.65	62.41	17.62	0.0	977.24	1222.07	***
hour	-105.77	18.49	-5.72	0.0	-142.03	-69.51	***
hour_after_5	934.08	32.40	28.83	0.0	870.53	997.64	***
hour_after_9	-747.45	21.74	-34.37	0.0	-790.10	-704.80	***
hour_after_19	-354.13	21.78	-16.26	0.0	-396.85	-311.40	***

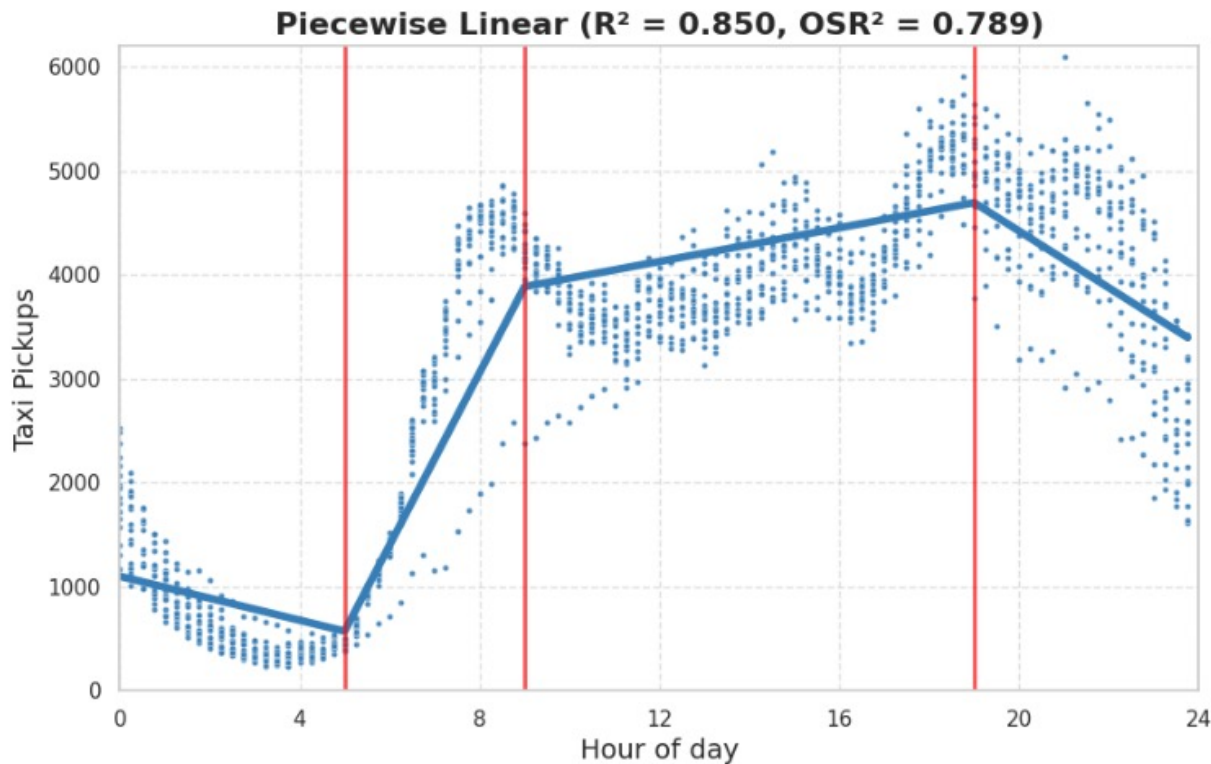
```
=====
```

```
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
```

```
=====
```

Bending the Line

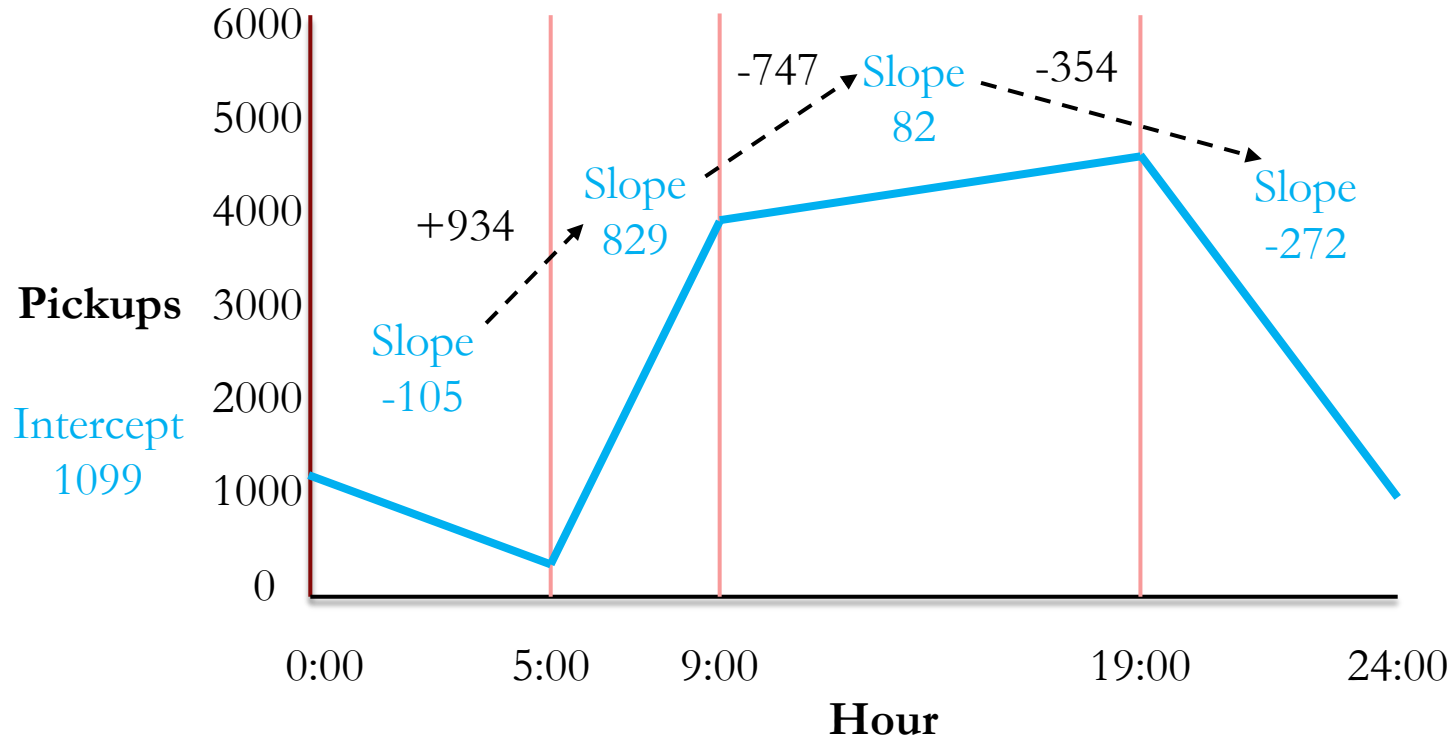
$$y = 1099 - 105 \cdot x_{\text{hour}} + 934 \cdot x_{\text{hour_after_5}} - 747 \cdot x_{\text{hour_after_9}} - 354 \cdot x_{\text{hour_after_19}}$$



	Coef.
Intercept	1099.6542
hour	-105.7656
hour_after_5	934.0847
hour_after_9	-747.4473
hour_after_19	-354.1266

Bending the Line

$$y = 1099 - 105 \cdot x_{\text{hour}} + 934 \cdot x_{\text{hour_after_5}} - 747 \cdot x_{\text{hour_after_9}} - 354 \cdot x_{\text{hour_after_19}}$$



Example in Excel

Sum-product of regression coefficients and each row of data.



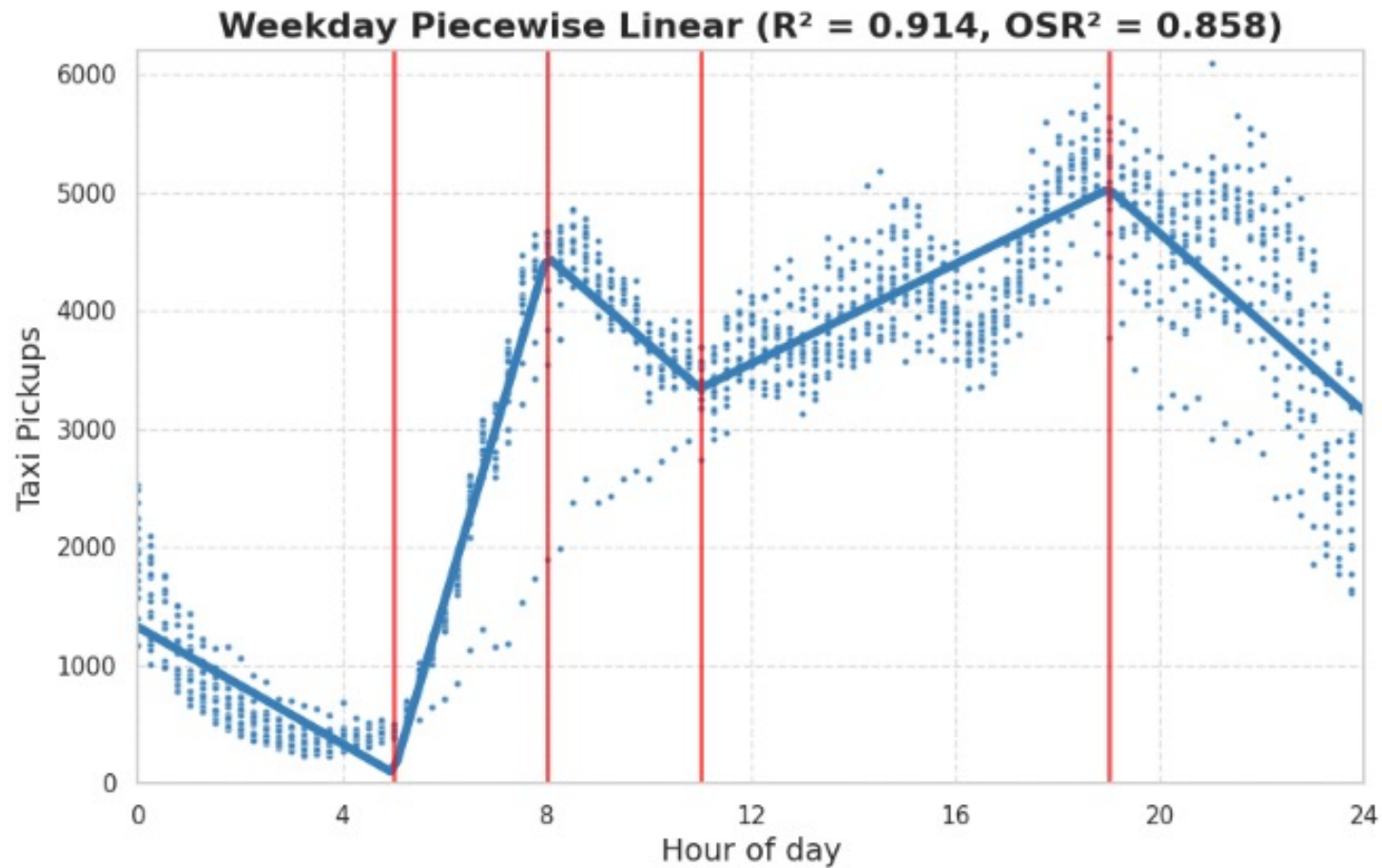
Regression Coefficients							
1099	-105	934	-747	-354			
(Intercept)	Hour	Hour since 5	Hour since 9	Hour since 19	Predicted Y	Difference	
1	0	0	0	0	1099	0	
1	1	0	0	0	994	-105	
1	2	0	0	0	889	-105	
1	3	0	0	0	784	-105	
1	4	0	0	0	679	-105	
1	5	0	0	0	574	-105	
1	6	1	0	0	1403	829	
1	7	2	0	0	2232	829	
1	8	3	0	0	3061	829	
1	9	4	0	0	3890	829	
1	10	5	1	0	3972	82	
1	11	6	2	0	4054	82	
1	12	7	3	0	4136	82	
1	13	8	4	0	4218	82	
1	14	9	5	0	4300	82	
1	15	10	6	0	4382	82	
1	16	11	7	0	4464	82	
1	17	12	8	0	4546	82	
1	18	13	9	0	4628	82	
1	19	14	10	0	4710	82	
1	20	15	11	1	4438	-272	
1	21	16	12	2	4166	-272	

More Generally...

$$y = b_0 - b_1 \cdot x_{\text{hour}} + b_5 \cdot x_{\text{hour after 5}} - b_9 \cdot x_{\text{hour after 9}} - b_{19} \cdot x_{\text{hour after 19}}$$

- The prediction line starts at hour 0 at b_0
- From hour 0 to 5, the prediction line has slope b_1
- From hour 5 to 9, the prediction line has slope $b_1 + b_5$
- From hour 9 to 19, the prediction line has slope $b_1 + b_5 + b_9$
- From hour 19 to 24, the prediction line has slope $b_1 + b_5 + b_9 + b_{19}$
- **Important for later:** Note that if $b_5 = 0$, then the line does not “bend” at 5am...

Other Possible Breaking Points



How should I Choose the Best Breaking Points?

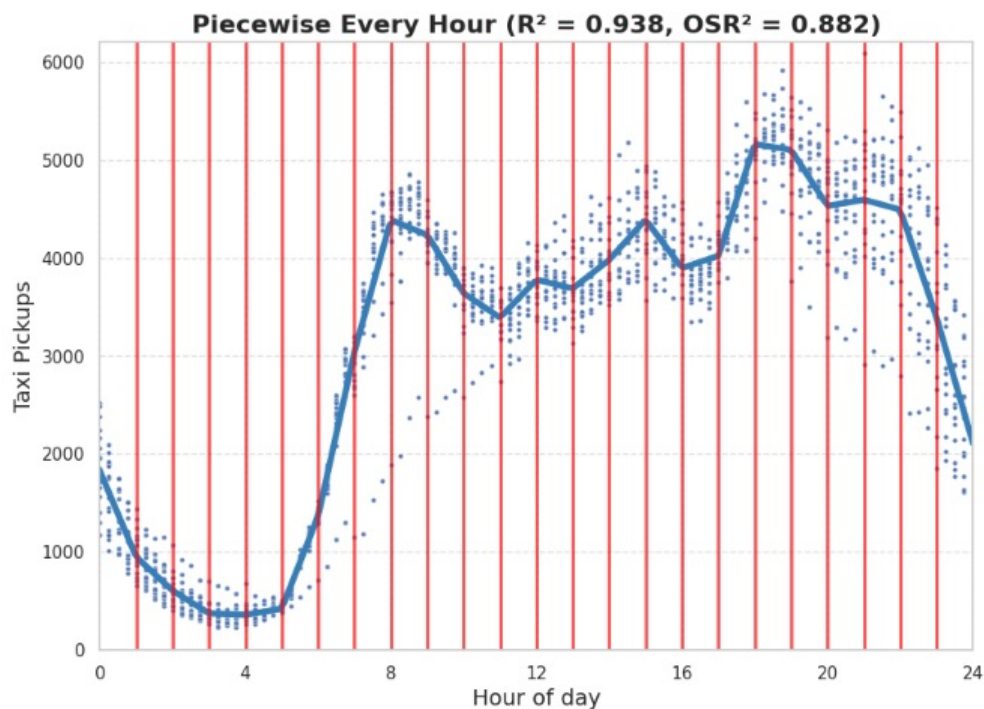
Regression Results

R-squared: 0.94
No. Observations: 1536

Coefficients:

	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
Intercept	1828.91	76.75	23.83	0.00	1678.35	1979.47	***
hour	-878.49	113.90	-7.71	0.00	-1101.90	-655.08	***
hour_after_1	536.01	193.58	2.77	0.01	156.30	915.72	**
hour_after_2	110.64	183.26	0.60	0.55	-248.83	470.12	
hour_after_3	217.38	182.65	1.19	0.23	-140.89	575.65	
hour_after_4	75.62	182.61	0.41	0.68	-282.58	433.82	
hour_after_5	882.78	182.61	4.83	0.00	524.58	1240.97	***
hour_after_6	734.45	182.61	4.02	0.00	376.26	1092.65	***
hour_after_7	-335.57	182.61	-1.84	0.07	-693.76	22.63	.
hour_after_8	-1497.03	182.61	-8.20	0.00	-1855.23	-1138.84	***
hour_after_9	-439.79	182.61	-2.41	0.02	-797.99	-81.60	*
hour_after_10	344.96	182.61	1.89	0.06	-13.23	703.16	.
hour_after_11	636.18	182.61	3.48	0.00	277.98	994.37	***
hour_after_12	-476.22	182.61	-2.61	0.01	-834.41	-118.02	**
hour_after_13	380.17	182.61	2.08	0.04	21.98	738.37	*
hour_after_14	120.60	182.61	0.66	0.51	-237.60	478.79	
hour_after_15	-907.25	182.61	-4.97	0.00	-1265.44	-549.05	***
hour_after_16	619.22	182.61	3.39	0.00	261.03	977.42	***
hour_after_17	1016.96	182.61	5.57	0.00	658.77	1375.16	***
hour_after_18	-1199.01	182.61	-6.57	0.00	-1557.21	-840.82	***
hour_after_19	-513.61	182.61	-2.81	0.00	-871.81	-155.41	**
hour_after_20	634.94	182.62	3.48	0.00	276.72	993.16	***
hour_after_21	-164.00	182.82	-0.90	0.37	-522.62	194.61	
hour_after_22	-999.81	186.27	-5.37	0.00	-1365.19	-634.43	***
hour_after_23	-179.79	238.15	-0.75	0.45	-646.93	287.36	

Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1



Model Performance for Weekday

Model	Number of Parameters	In-sample R^2	Out-of-sample R^2
Linear with no bends	2	0.574	0.546
Linear with 3 Bends (5, 9, 19)	5	0.850	0.789
Linear with 4 Bends (5, 8, 11, 19)	6	0.914	0.858
Linear with 23 Bends (All Hours)	25	0.938	0.882

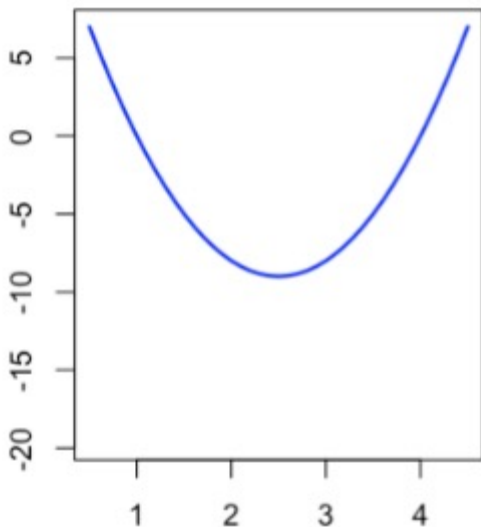
Polynomial Regression

How to incorporate “bends” in the line.

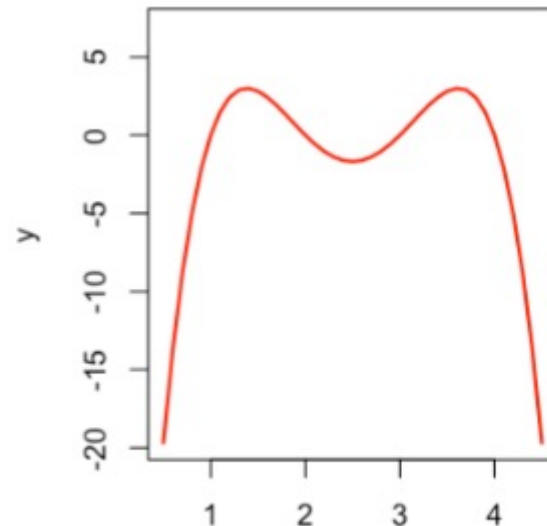
1. Piece-wise
2. **Polynomial**

Polynomial Regression

- **Polynomial:** $y = \beta_0 + \beta_1 \cdot x + \beta_2 \cdot x^2 + \dots + \beta_d \cdot x^d$
 - A line is a polynomial of degree 1
 - A polynomial of degree d can “bend” $d - 1$ times.



$$y = 16 - 20 \cdot x - 4 \cdot x^2$$



$$y = -24 + 50 \cdot x - 35 \cdot x^2 + 10 \cdot x^3 + x^4$$

Creating Polynomial Features

- Create polynomial features and then run vanilla linear regression.

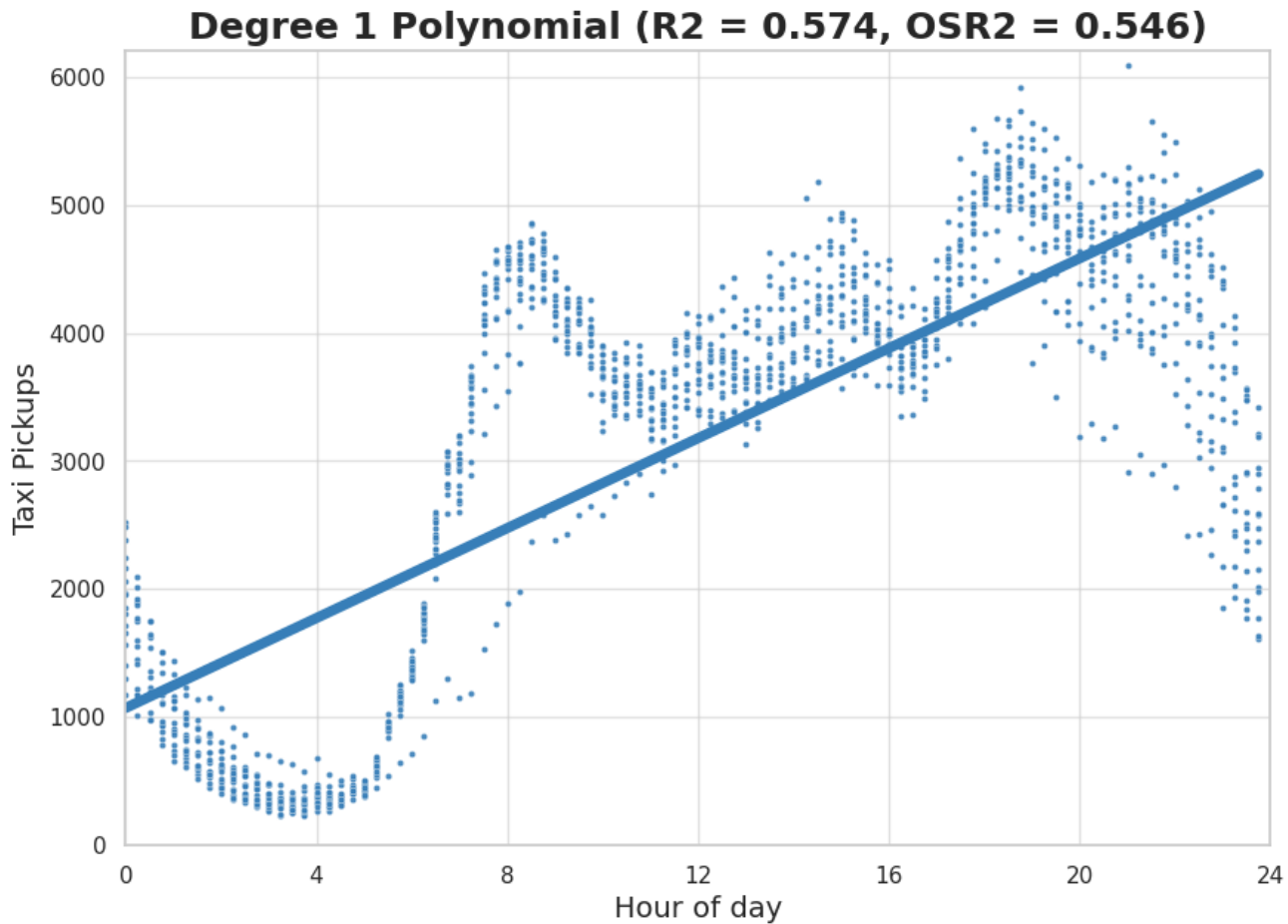
Pickups	Hour	Hour ²	Hour ³
2600	0.00	0.00	0.00
1400	2.00	4.00	8.00
870	4.00	16.00	64.00
1700	6.00	36.00	216.00

$$y = \beta_0 + \beta_1 \cdot x_{\text{hour}}$$

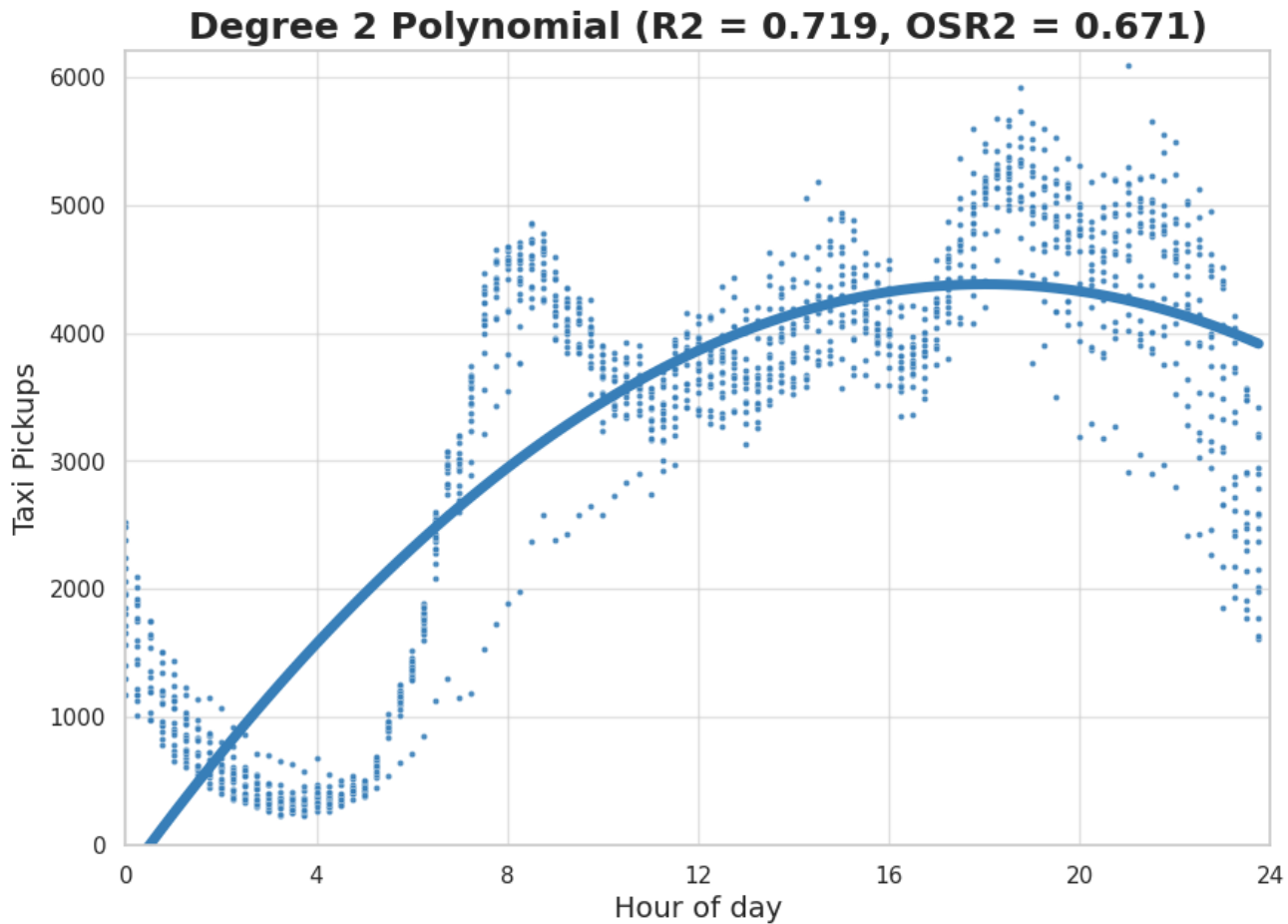
$$y = \beta_0 + \beta_1 \cdot x_{\text{hour}} + \beta_2 \cdot x_{\text{hour}}^2$$

$$y = \beta_0 + \beta_1 \cdot x_{\text{hour}} + \beta_2 \cdot x_{\text{hour}}^2 + \beta_3 \cdot x_{\text{hour}}^3$$

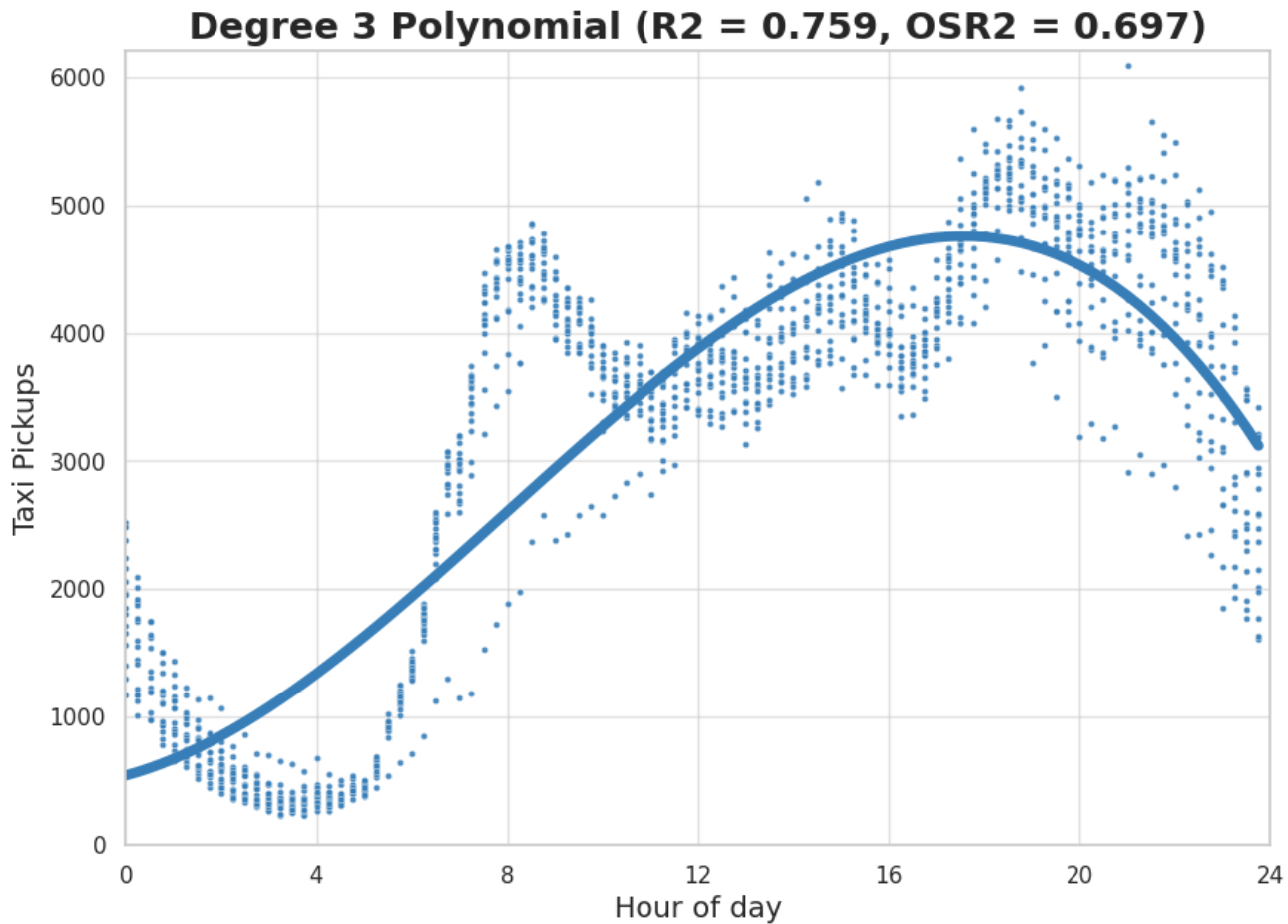
Degree 1 Polynomial



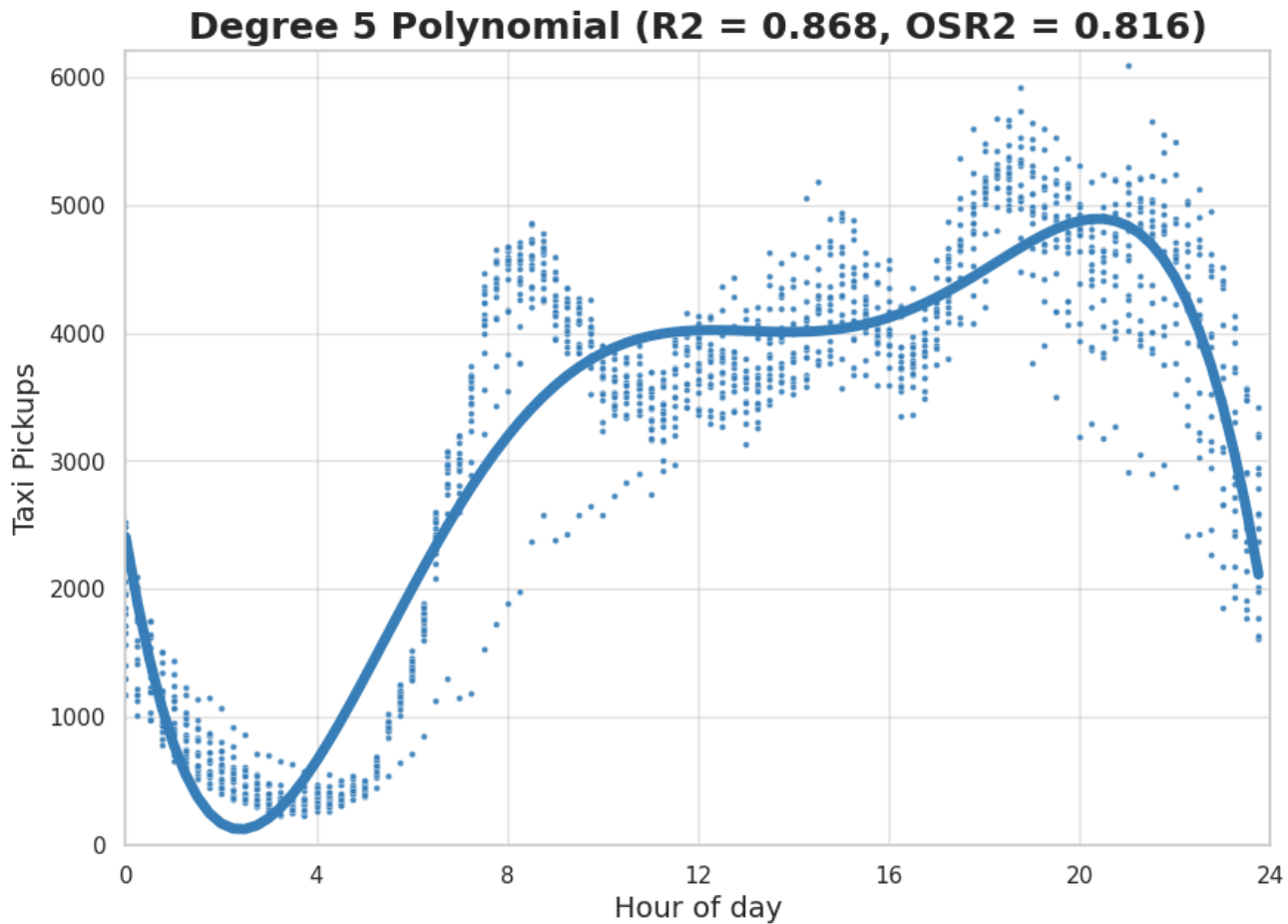
Degree 2 Polynomial



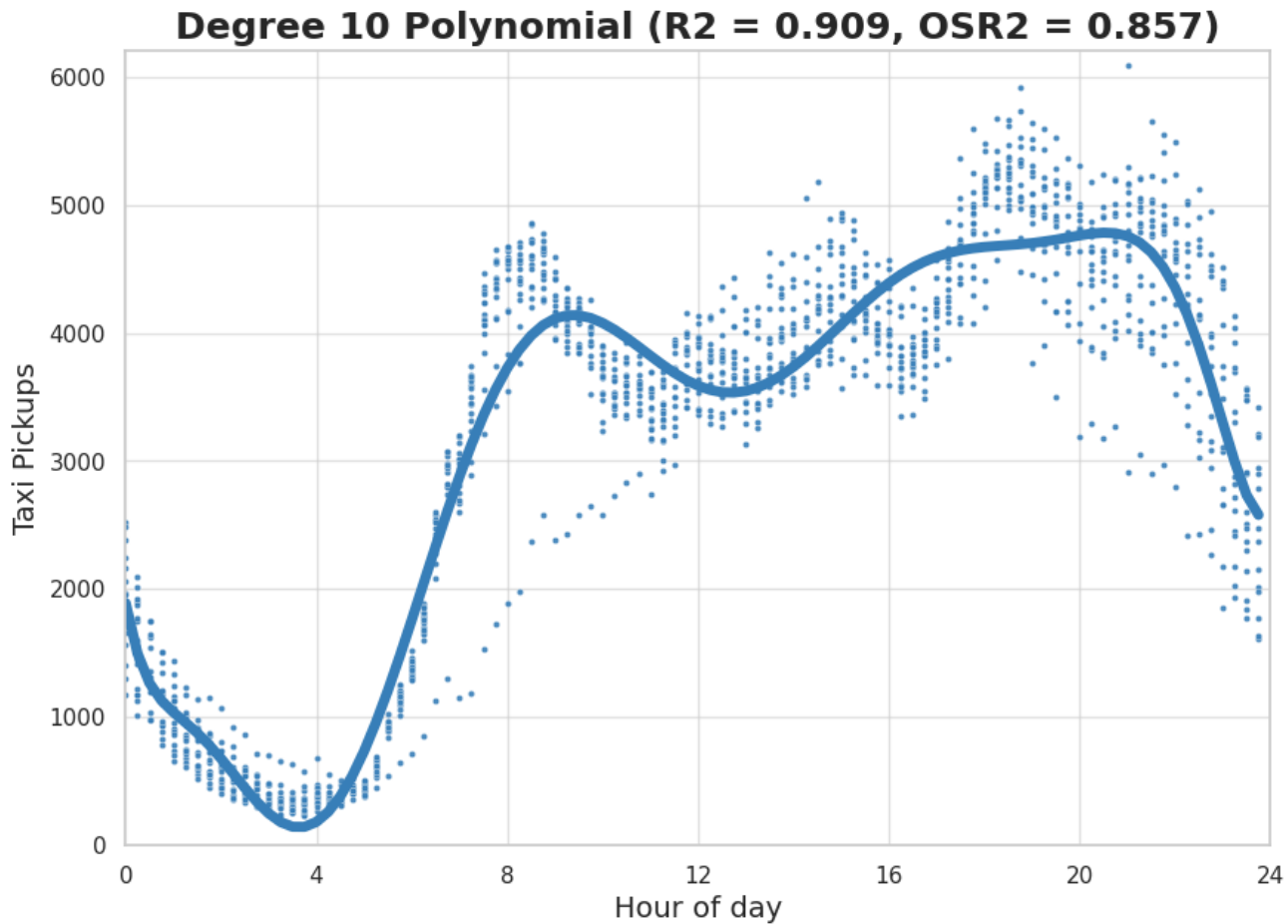
Degree 3 Polynomial



Degree 5 Polynomial



Degree 10 Polynomial



Model Performance for Weekday

Model	Number of Parameters	In-sample R^2	Out-of-sample R^2
Linear with no bends	2	0.574	0.546
Linear with 3 Bends (5, 9, 19)	5	0.850	0.789
Linear with 4 Bends (5, 8, 11, 19)	6	0.914	0.858
Linear with 23 Bends (All Hours)	25	0.938	0.882
Polynomial (Degree = 2)	3	0.719	0.671
Polynomial (Degree = 3)	4	0.759	0.697
Polynomial (Degree = 5)	6	0.868	0.816
Polynomial (Degree = 10)	11	0.909	0.857

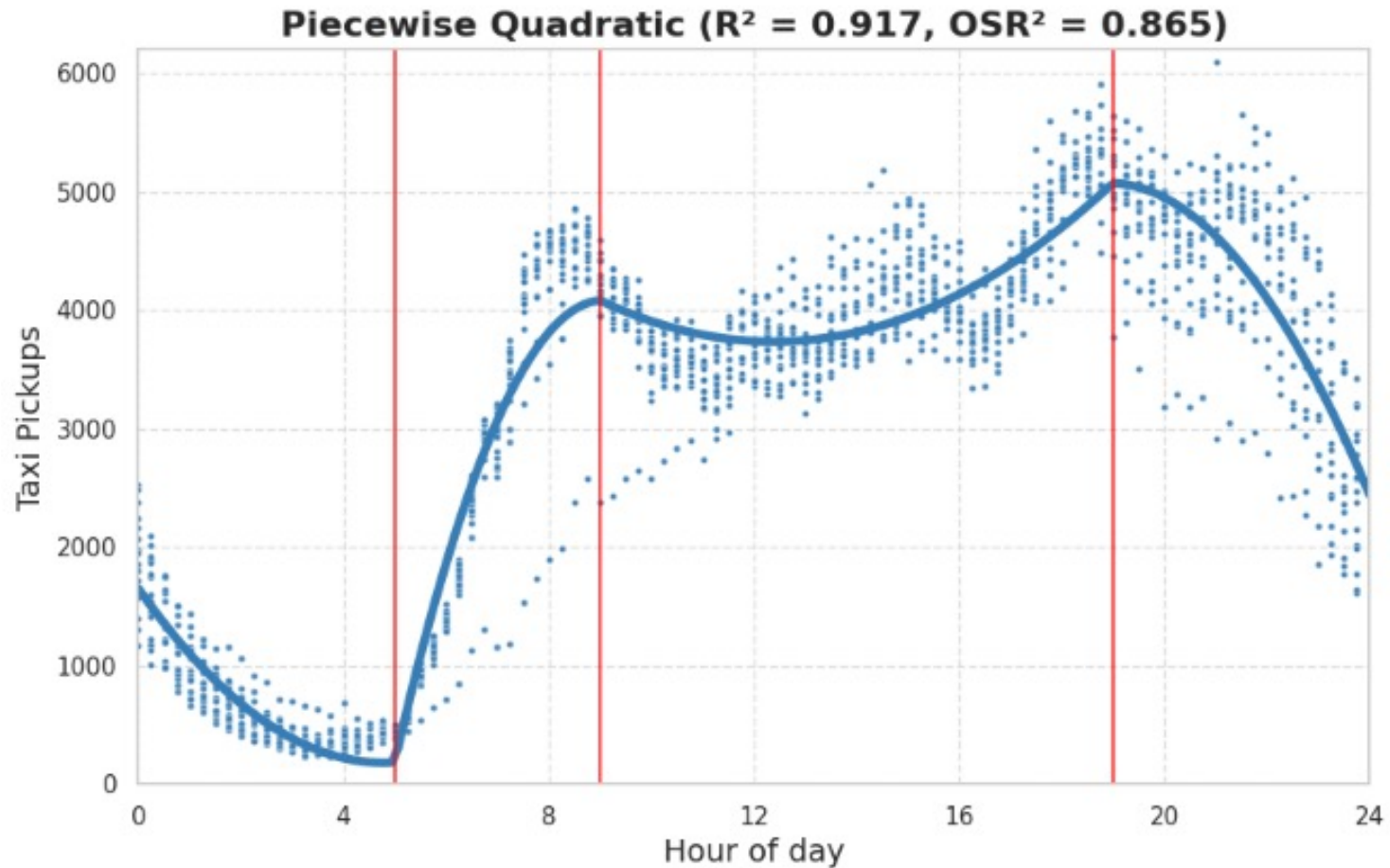
Combining Piecewise + Polynomial

- Although a high-degree polynomial can capture the curvy behavior of our data, it is generally better practice to use polynomials of lower degree.
 - **Reason 1:** Interpretability/Simplicity
 - **Reason 2:** Stability, less prone to overfitting

$$\begin{aligned}\hat{y} = & 1904 - 2033 \cdot x + 2092 \cdot x^2 - 1232 \cdot x^3 + 373.9 \cdot x^4 \\ & - 62.19 \cdot x^5 + 6.089 \cdot x^6 + 0.3617 \cdot x^7 + .0128 \cdot x^8 \\ & - .0002509 \cdot x^9 + .000002077 \cdot x^{10}\end{aligned}$$

- Let's combine the two ideas!
 - 1. Piecewise (dividing the x-axis into regions)
 - 2. Using (slightly) higher-order polynomials in each region

Piecewise Quadratic Fit



Combining Piecewise + Polynomial

Piecewise

hour



hour

hour_after_5

hour_after_9,

hour_after_19,

Polynomial

hour



hour

hour²,

hour³,

hour⁴,

Piecewise + Polynomial

hour



hour

hour²

hour³

hour_since_5

(hour_since_5)²

(hour_since_5)³...

hour_since_9

(hour_since_9)²

(hour_since_9)³

hour_since_19

(hour_since_19)²

(hour_since_19)³

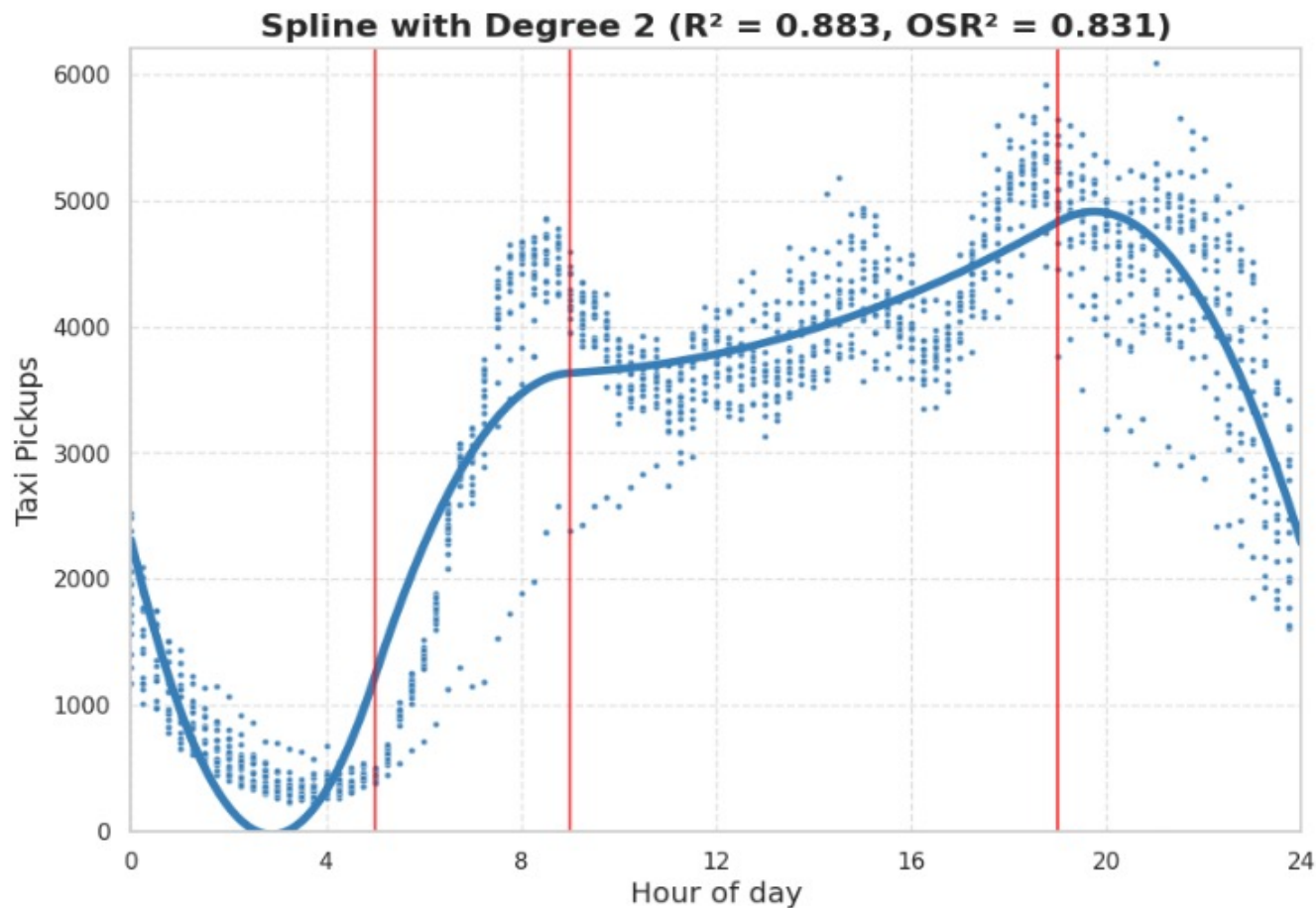
Model Performance for Weekday

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Linear with no bends	2	0.574	0.546
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Linear with 23 Bends (All Hours)	25	0.938	0.882
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Polynomial (Degree = 3)	4	0.759	0.697
Polynomial (Degree = 5)	6	0.868	0.816
Polynomial (Degree = 10)	11	0.909	0.857
Piecewise Quadratic (5, 9, 19)	12	0.917	0.865

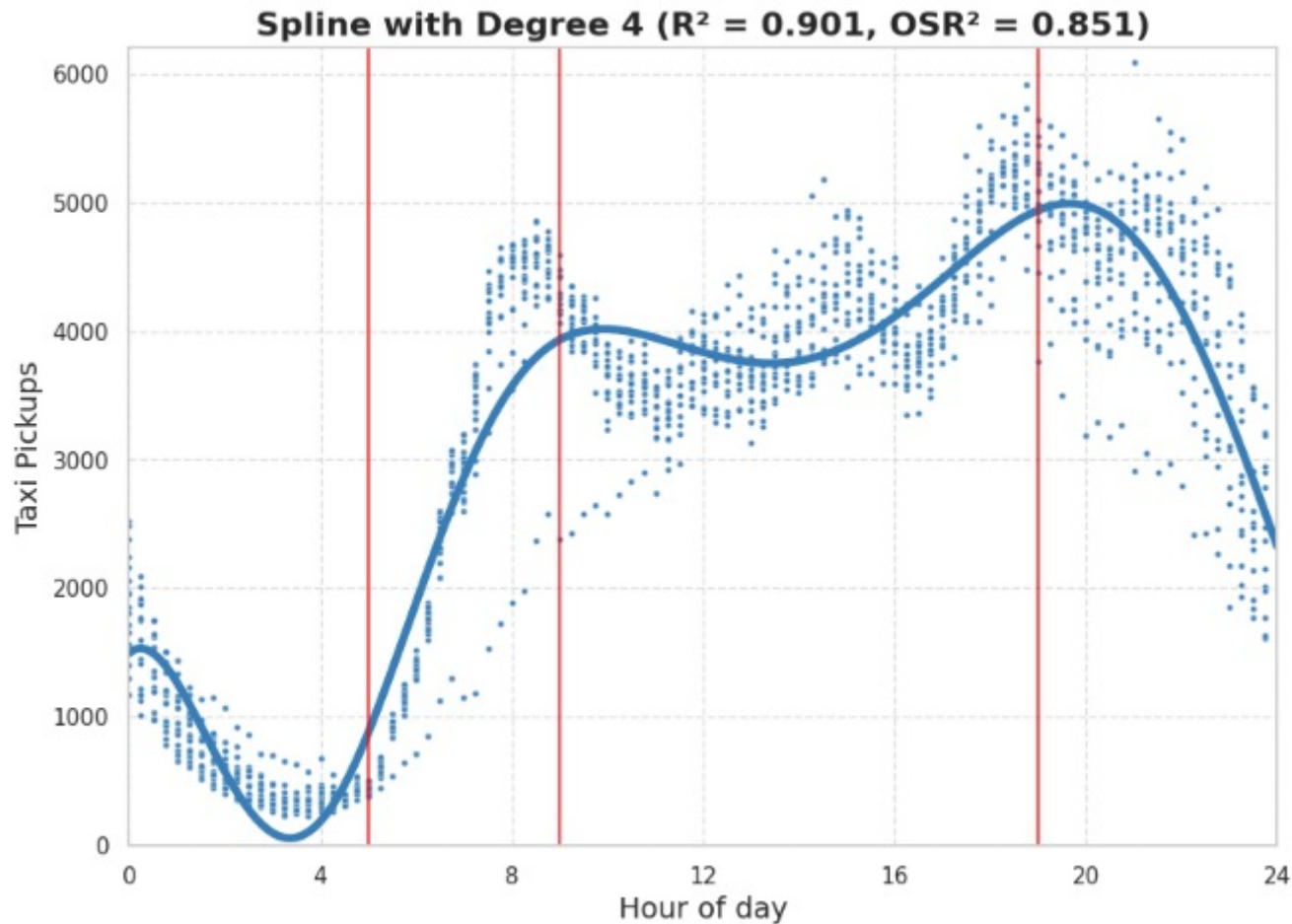
Advanced Topic (Optional): Splines

- A piecewise quadratic fit like the one we saw above is likely not smooth, i.e. contains sharp corners at the break points.
- **Splines Idea:** Adding constraints to the coefficients so that the derivative stays constant between regions.
- This is an advanced topic and brings us outside the world of “feature engineering for non-linear regression”.

Spline with Degree 2 Polynomials



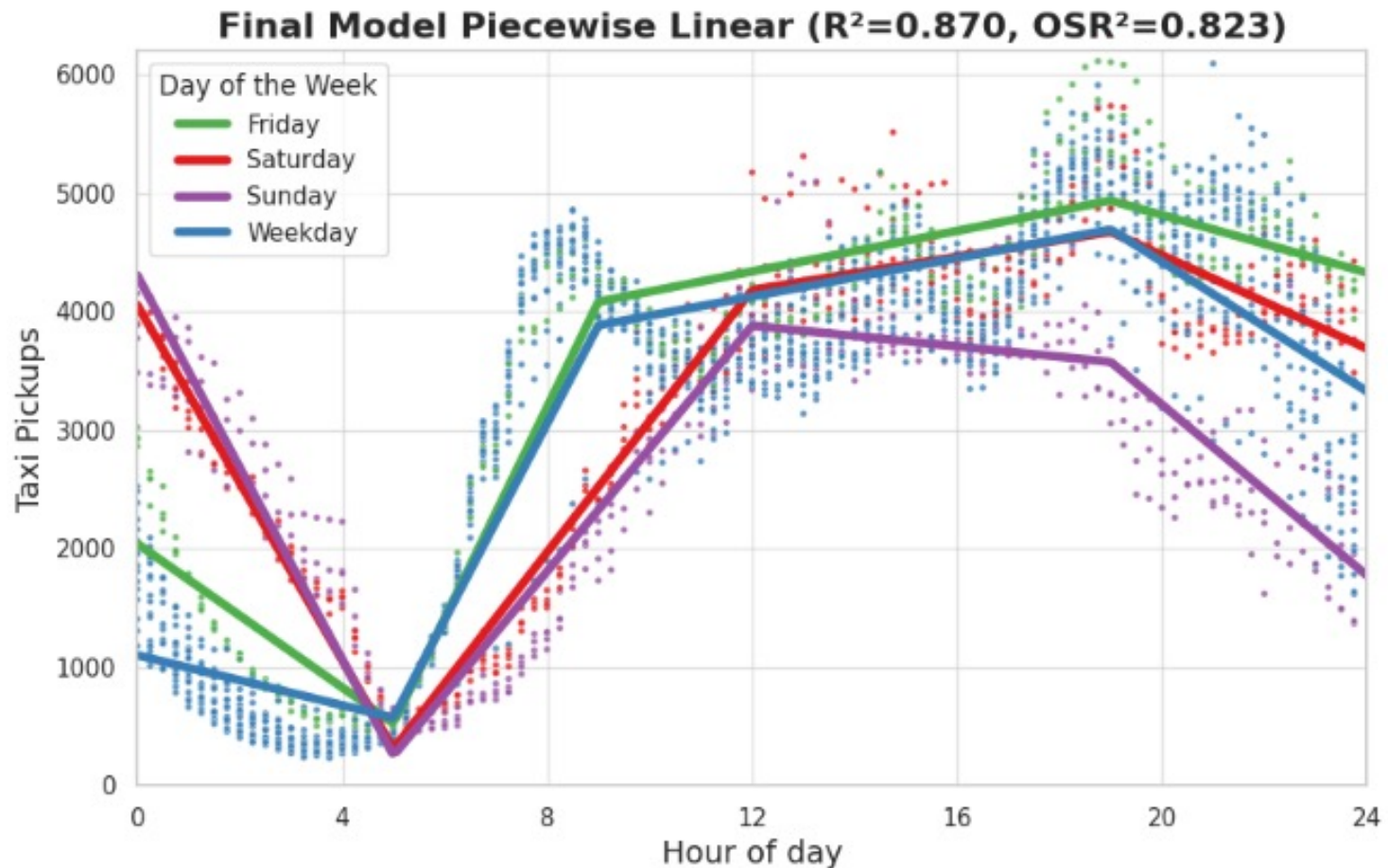
Spline with Degree 4 Polynomials



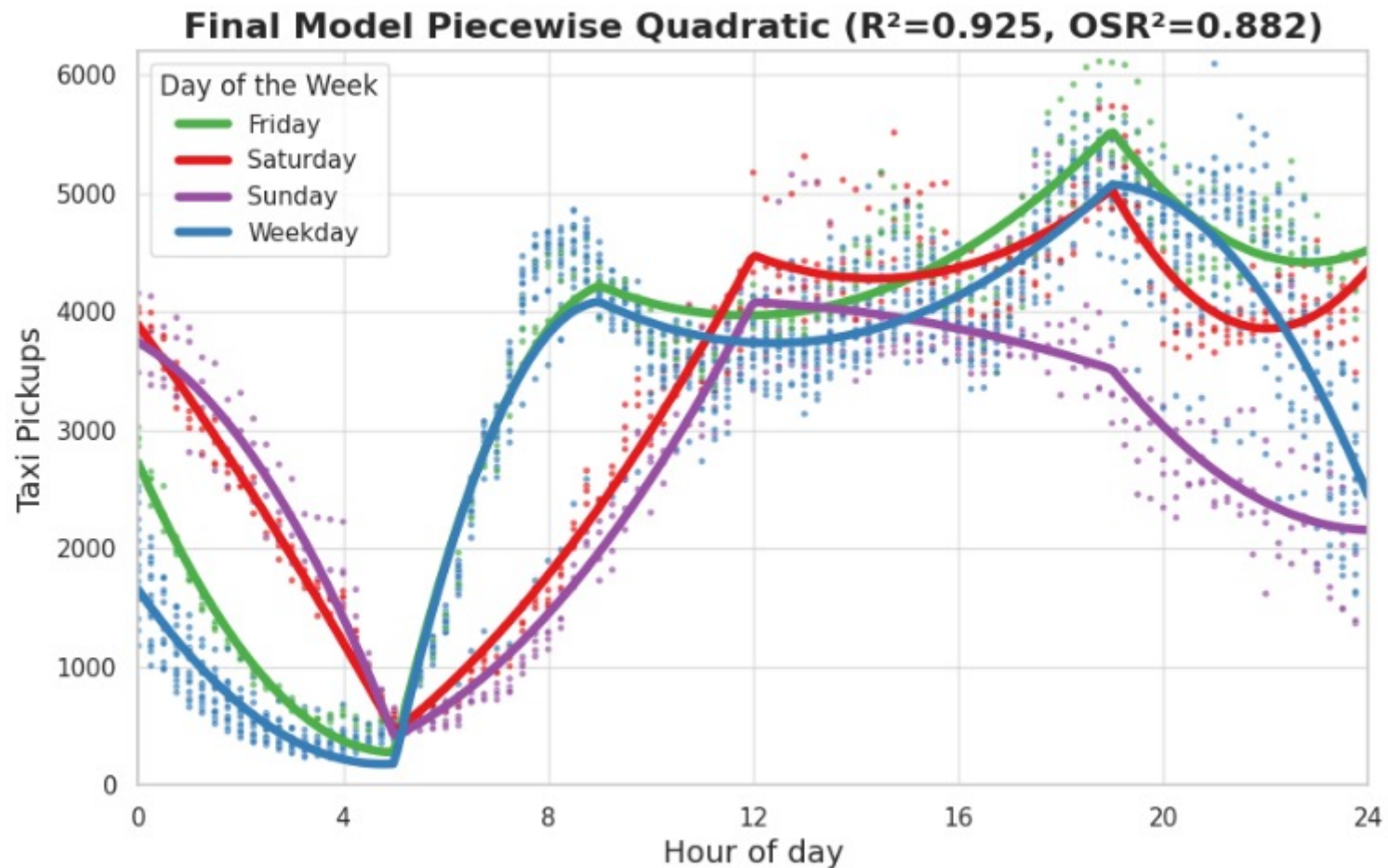
Putting it all Together

Overall model for all days of the week

Piecewise Linear



Piecewise Quadratic



Model Performance on All Days

Model	Number of Parameters	In-sample R^2	Out-of-sample R^2
LR with hour	2	0.48	0.47
LR with hour + day	5	0.50	0.48
LR with hour * day interaction	8	0.54	0.52
Piecewise Lines (3 breaks)	24	0.87	0.82
Piecewise Quadratics (3 breaks)	48	0.93	0.88

Feature Engineering Multiple Variables

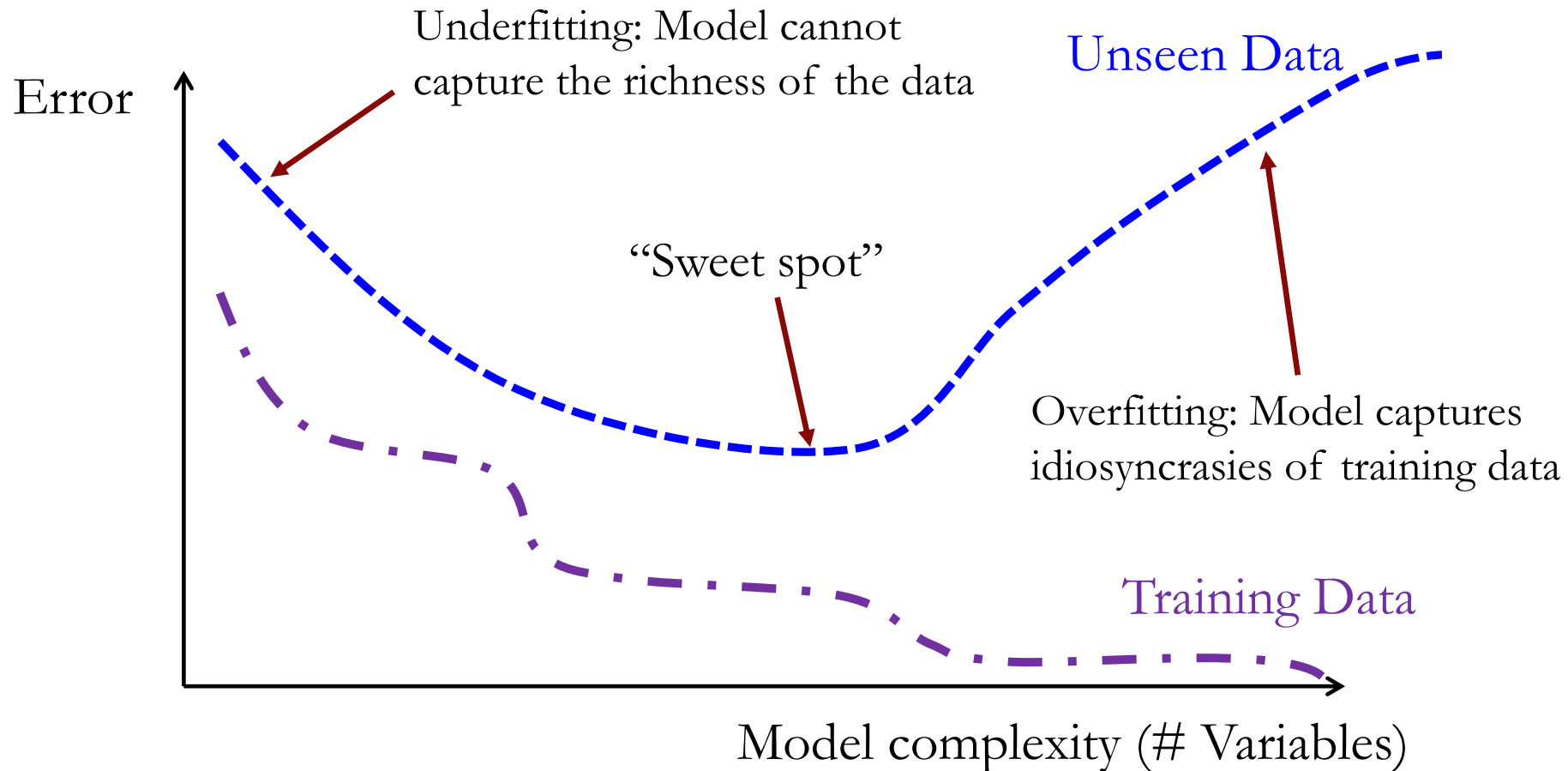
- In the NYC taxi example, there was only 1 numeric variable (hour).
- All feature engineering concepts we discussed can be applied independently to multiple numeric variables. For example:
 - Piecewise quadratic for temperature
 - Piecewise linear for humidity
 - Piecewise linear for hour
 - Interaction term for pair: temperature * humidity
 - Interaction term for the triple: temperature * humidity * hour



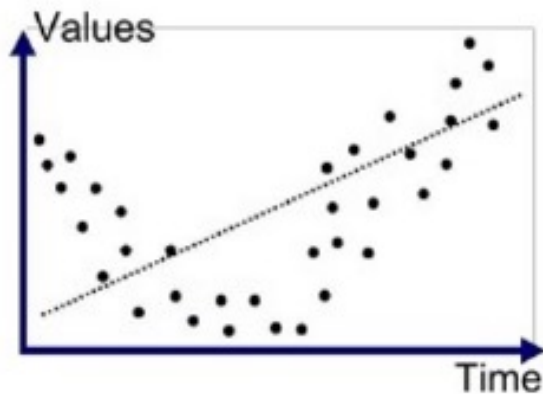
But we must be careful...

We can't go “too crazy” with feature engineering.

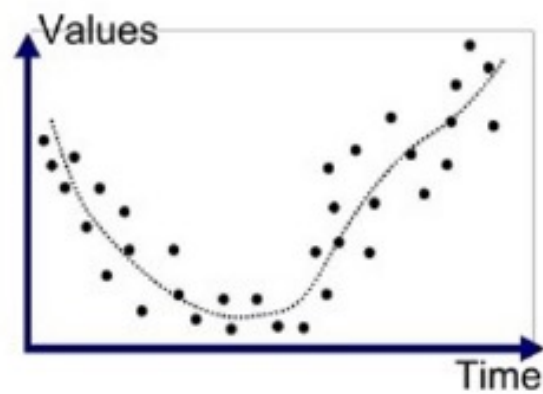
Performance vs. Model Complexity



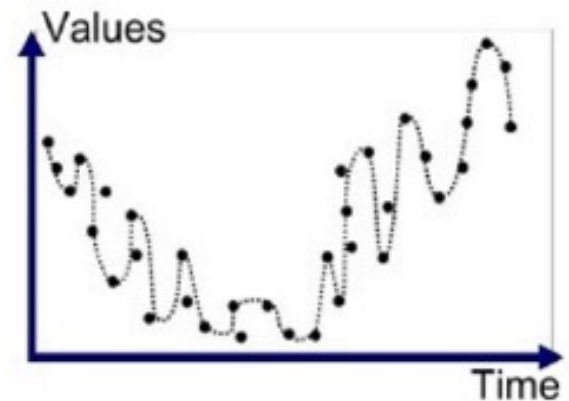
Performance vs. Model Complexity



Underfitted

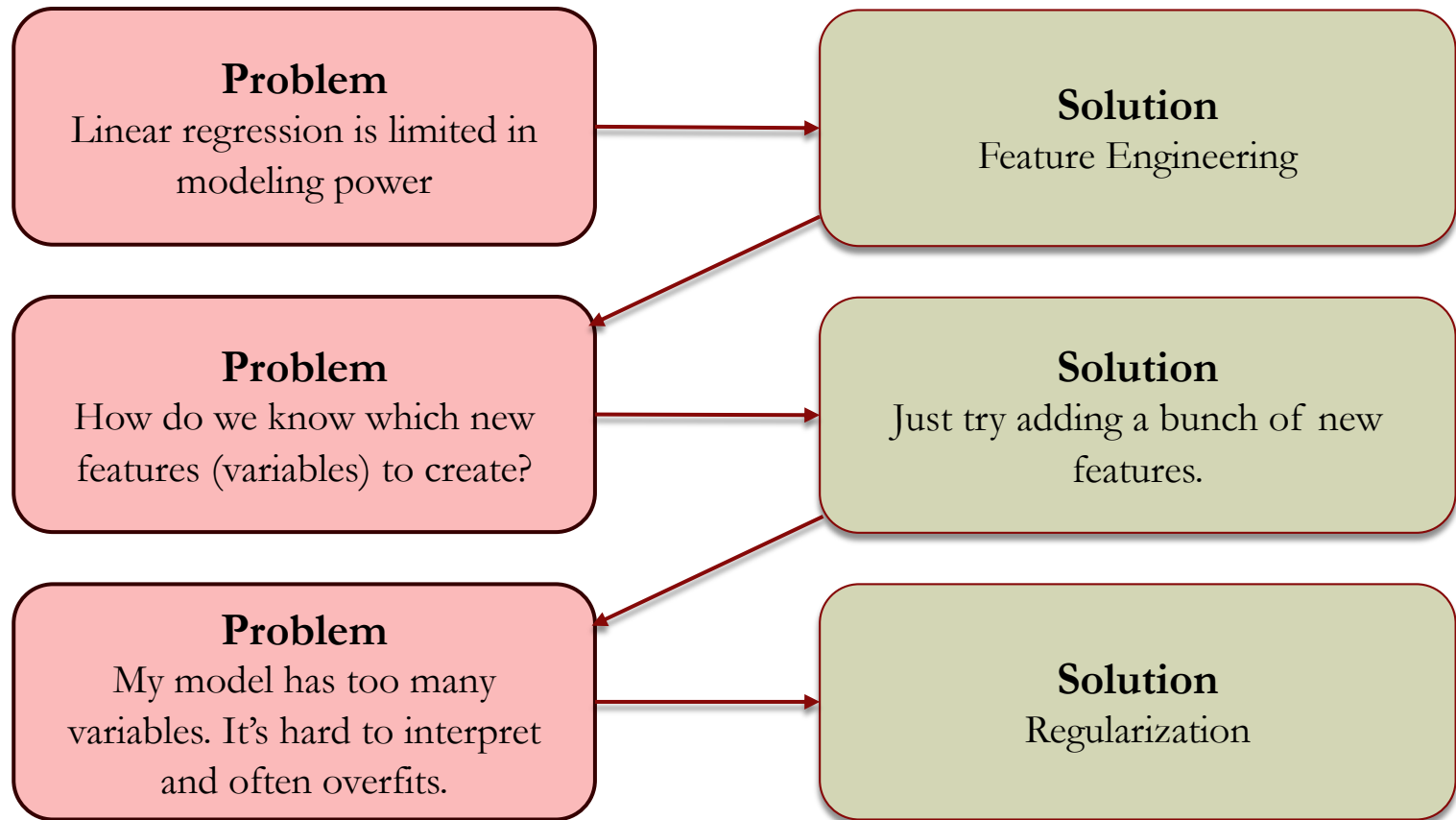


Good Fit/Robust

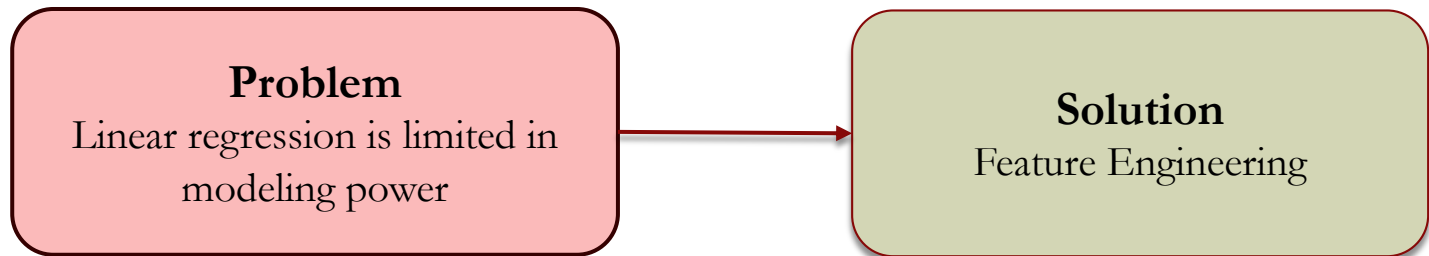


Overfitted

The Big Picture Again



Conclusion on Feature Engineering



- Feature engineering (adding new variables from existing variables) is a powerful technique that allows linear regression to capture non-linear relationships.
 - We saw 3 types of non-linearities: interaction terms, piecewise, and polynomial.
- Feature engineering is challenging because one needs to have domain knowledge to know what features to try.
 - If we just add features ad-hoc, we obtain models that are complex, hard to interpret, and likely overfit.



Part II: Regularization

Diabetes Prediction

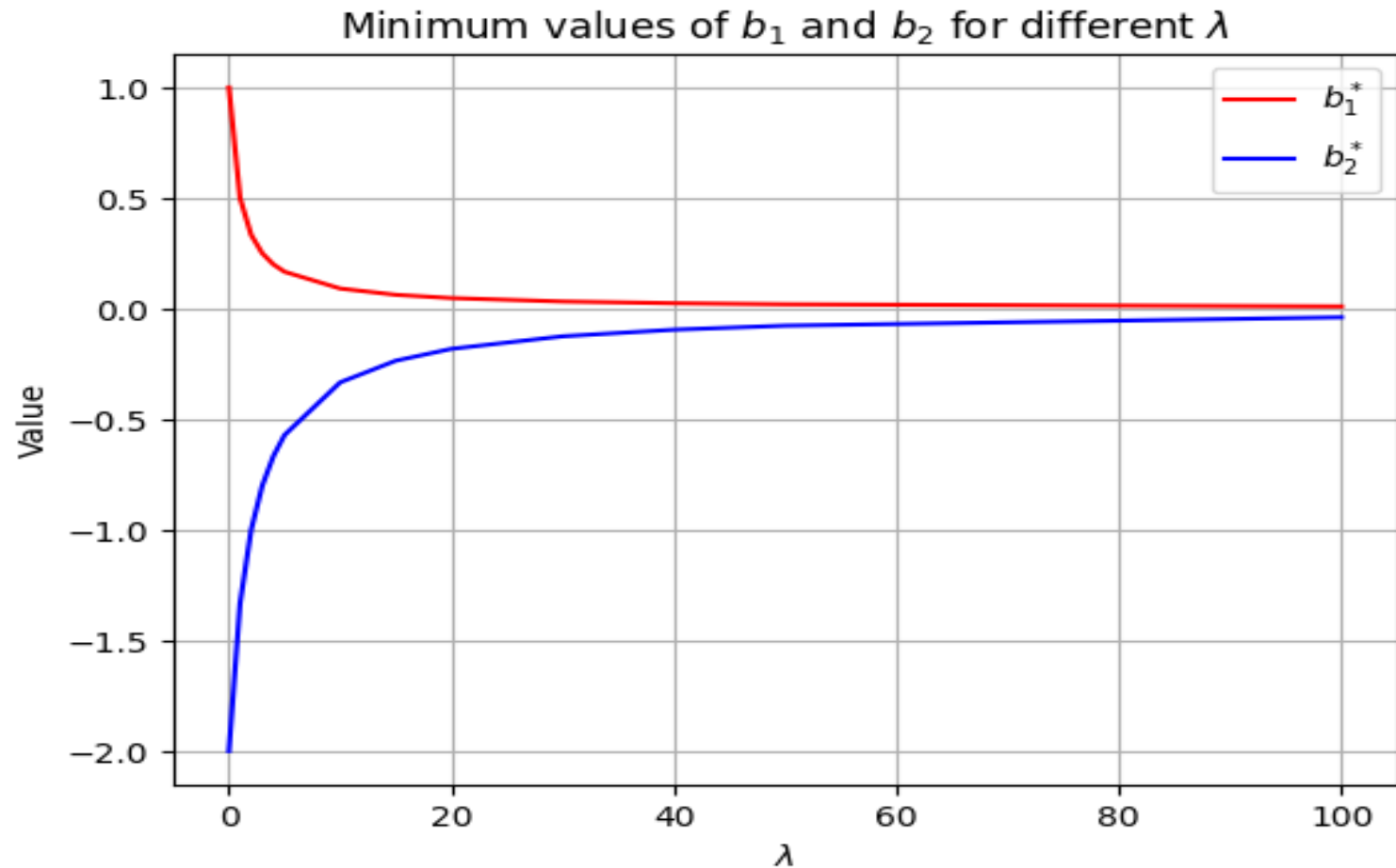
An Optimization Problem, Shrinkage, and Coefficient Profile

An optimization problem

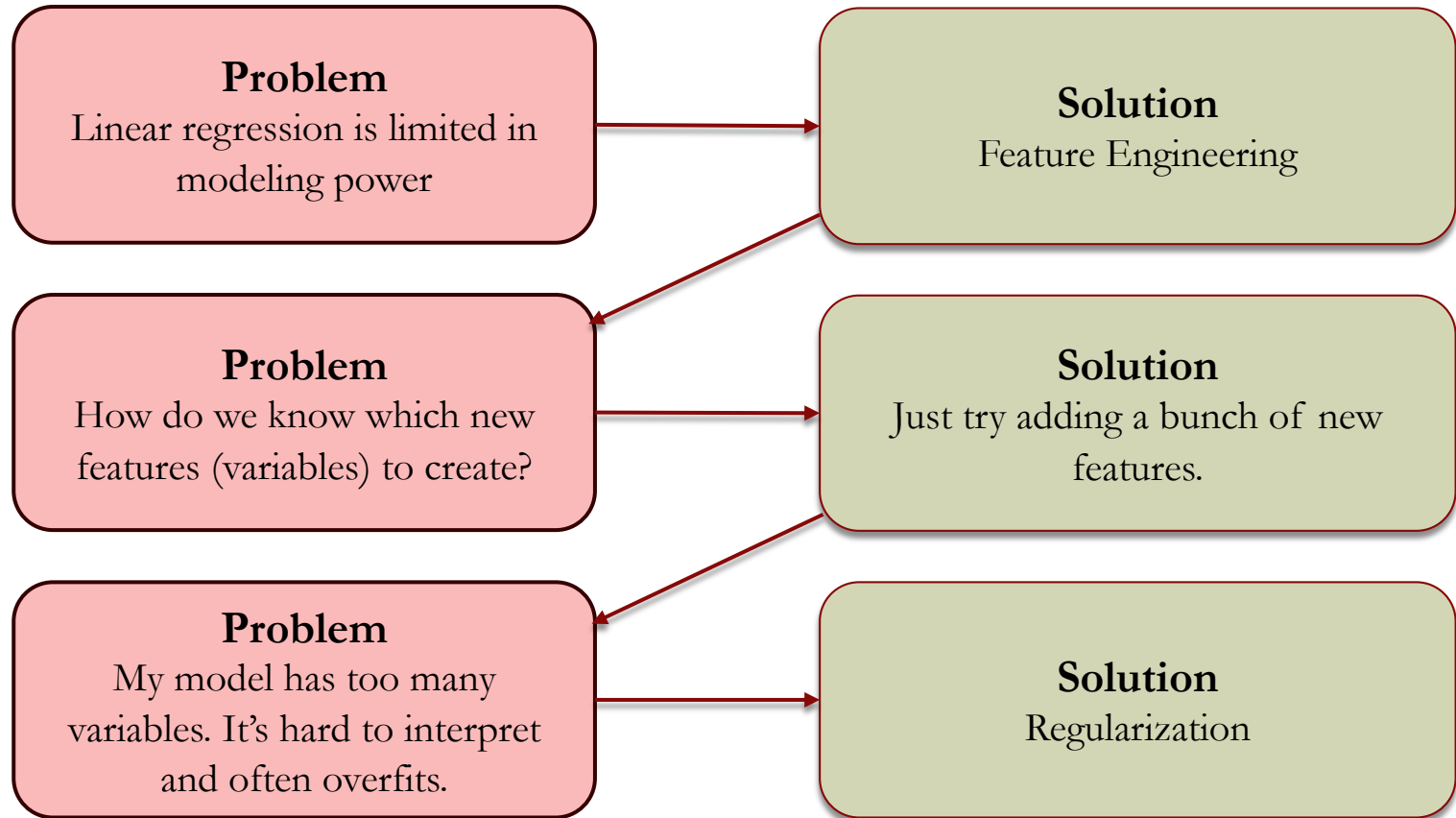
$$\min_{b_1, b_2} [(b_1 - 1)^2 + 2(b_2 + 2)^2] + \lambda \cdot (b_1^2 + b_2^2)$$

λ	Optimal Solution		$(b_1)^2 + (b_2)^2$
	b_1	b_2	
0	1.00	-2.00	5.00
1	0.50	-1.33	2.03
2	0.33	-1.00	1.11
3	0.25	-0.80	0.70
4	0.20	-0.67	0.48
5	0.17	-0.57	0.35
10	0.09	-0.33	0.12
15	0.06	-0.24	0.06
20	0.05	-0.18	0.04
30	0.03	-0.13	0.02
40	0.02	-0.10	0.01
50	0.02	-0.08	0.01
100	0.01	-0.04	0.00

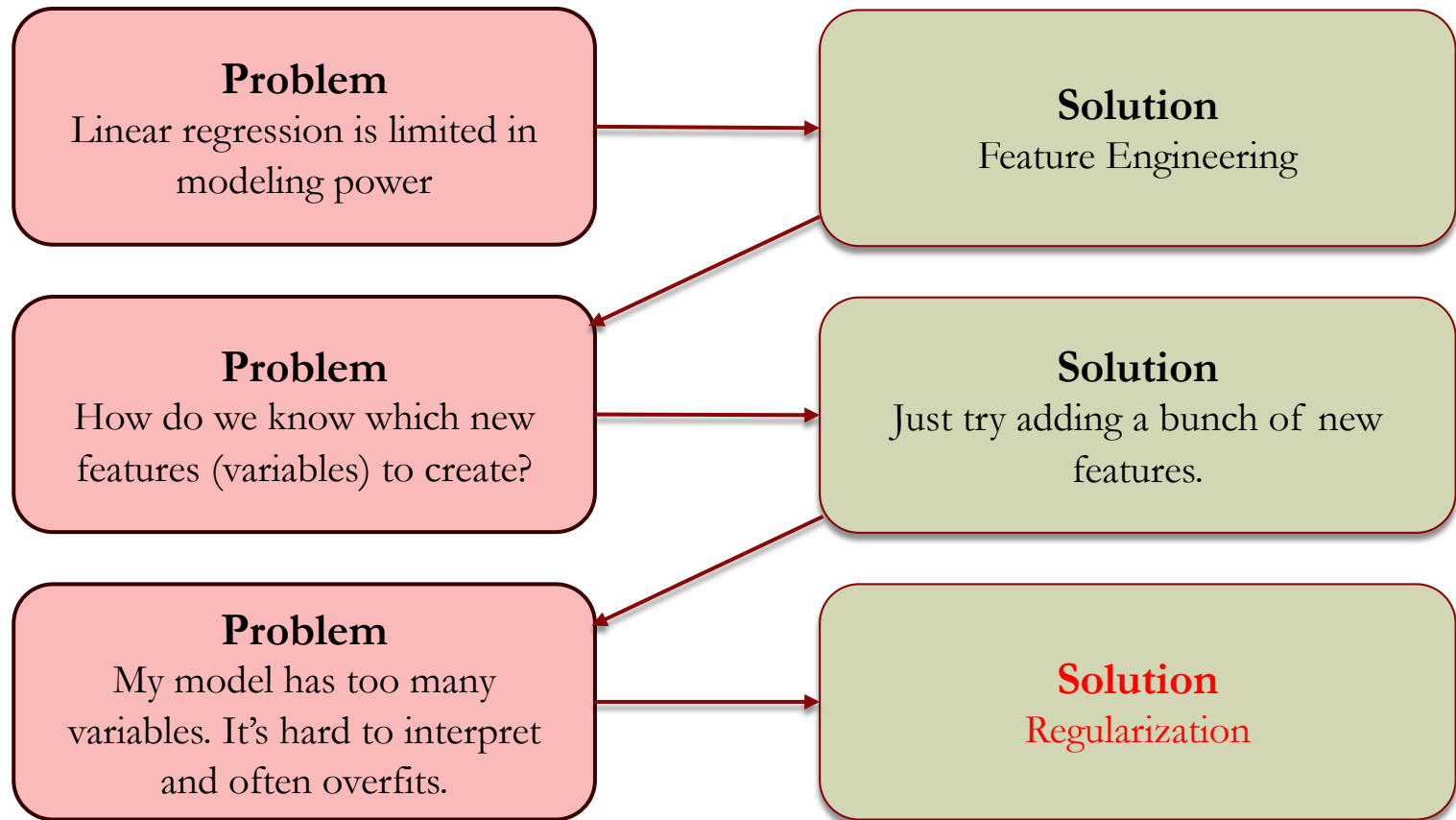
“Coefficient Profile”



Recall Big Picture of the Two Lectures



The Big Picture



Regularization

- Diabetes Prediction Dataset and Feature Engineering
- Recap on Value of Simple Models: Overfitting
- **Regularization Paradigm**
 - Balancing between Two Objectives
 - Linear Regression with Best Subset Selection (BSS) Penalty
 - Cross-Validation for Picking Lambda
- **Regularization Implementation**
 - Computational Tractability
 - LASSO and Ridge Penalties

Questions to Keep in Mind

1. Value of Simple Models

- Why are simple models less prone to overfitting than complex models?

2. Regularization Paradigm

- How does regularization conceptually allow us to balance between two objectives: model fit and complexity?
- How do we choose the optimal balance between the two (i.e., choice of λ)?

3. Regularization in Practice

- How is regularization used in practice with LASSO and Ridge penalties?

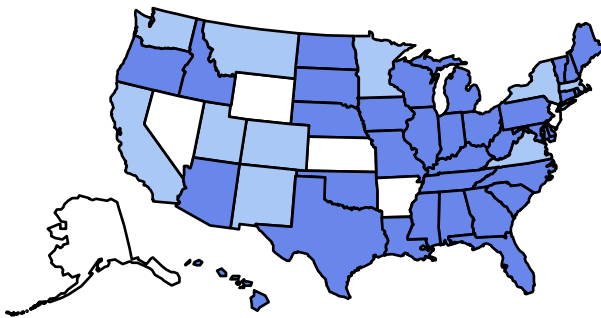
Diabetes Dataset and Preprocessing

Obesity in the United States

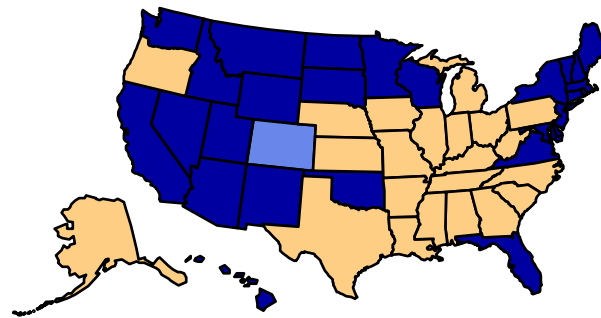
- Obesity has become increasingly prevalent in the United States, with major implications for health and life expectancy

Prevalence of Self-Reported Obesity Among U.S. Adults by State

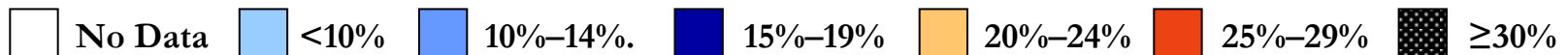
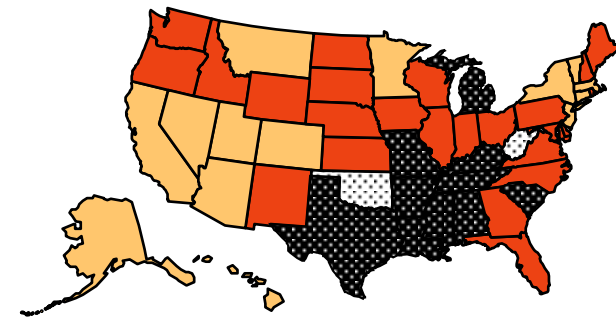
1990



2000



2010

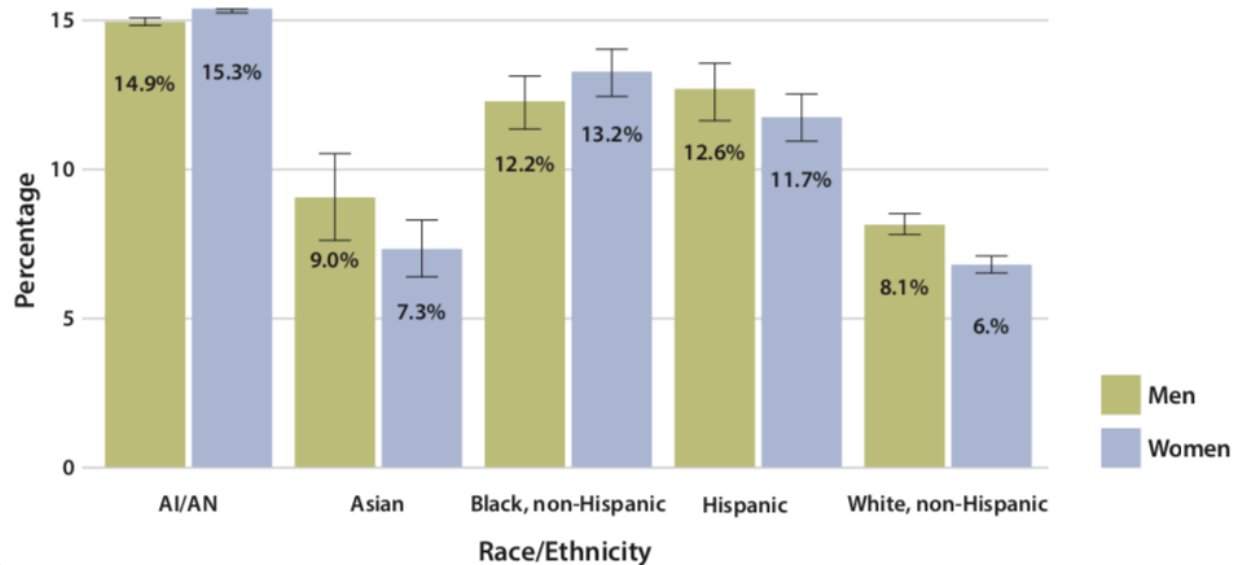


Source: Center for Disease Control, Behavioral Risk Factor Surveillance System

Diabetes in the United States

- Diabetes is a metabolic disorder characterized by high blood sugar
 - In 2022, 37.3 million Americans had diabetes and 96 million had prediabetes
 - 1.5 million Americans are diagnosed with diabetes every year
 - Diabetes has been the 7th leading cause of death in the United States (non-COVID)

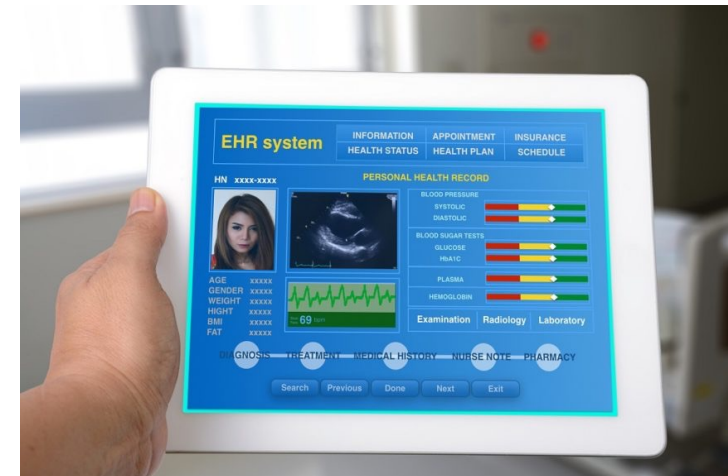
Rates of diagnosed diabetes in adults in the United States



Source: American Diabetes Association

Diabetes Prediction

- Significant health benefits when diabetes is treated early
- Thus the need for accurate and reliable prediction models to diagnose diabetes and anticipate its progression over time
- *Electronic Health Records*
 - Demographics and personal characteristics: age, weight, etc.
 - Medical history: medication, allergies, immunization, lab tests, etc.
 - other billing information
- Use of demographic and clinical data to predict diabetes



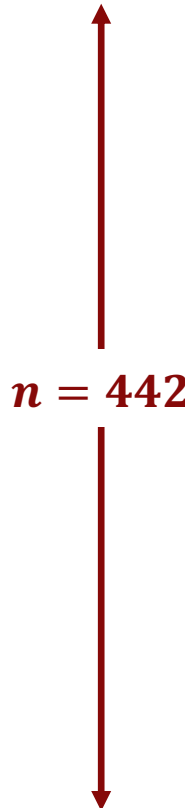
Diabetes Dataset

- **Dataset*** : a sample of 442 patients with diabetes
- **Dependent variable Y**: quantitative measure of disease progression one year after baseline
- **Independent Variables:**
 - age
 - sex
 - bmi (body mass index)
 - map (mean arterial pressure)
 - tc (total cholesterol)
 - ldl (low density lipoprotein level – “bad cholesterol”)
 - hdl (high density lipoprotein level – “good cholesterol”)
 - tch (ratio of triglycerides divided by hdl)
 - ltg (low-tension glaucoma)
 - glu (glucose level) ...

* Source: https://hastie.su.domains/StatLearnSparsity_files/DATA/diabetes.html

Diabetes Dataset

k : # of independent variables ($k = 10$); n : # of observations ($n = 442$)



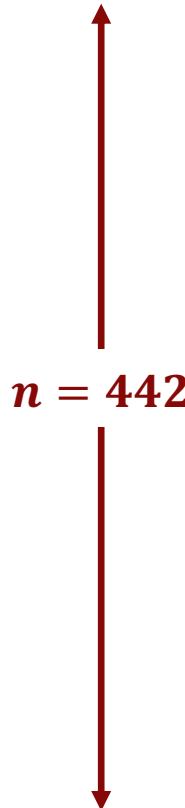
	y	age	sex	bmi	map	tc	ldl	hdl	tch	ltg	glu
0	151	59	2	32.1	101	157	93.2	38	4.00	4.86	87
1	75	48	1	21.6	87	183	103.2	70	3.00	3.89	69
2	141	72	2	30.5	93	156	93.6	41	4.00	4.67	85
3	206	24	1	25.3	84	198	131.4	40	5.00	4.89	89
4	135	50	1	23.0	101	192	125.4	52	4.00	4.29	80
5	97	23	1	22.6	89	139	64.8	61	2.00	4.19	68
6	138	36	2	22.0	90	160	99.6	50	3.00	3.95	82
7	63	66	2	26.2	114	255	185.0	56	4.55	4.25	92
8	110	60	2	32.1	83	179	119.4	42	4.00	4.48	94
9	310	29	1	30.0	85	180	93.4	43	4.00	5.38	88
10	101	22	1	18.6	97	114	57.6	46	2.00	3.95	83
11	69	56	2	28.0	85	184	144.8	32	6.00	3.58	77
12	179	53	1	23.7	92	186	109.2	62	3.00	4.30	81
13	185	50	2	26.2	97	186	105.4	49	4.00	5.06	88
14	118	61	1	24.0	91	202	115.4	72	3.00	4.29	73
...	...	...	...	...	...	...	...	...	...	...	...
437	178	60	2	28.2	112	185	113.8	42	4.00	4.98	93
438	104	47	2	24.9	75	225	166.0	42	5.00	4.44	102
439	132	60	2	24.9	99.67	162	106.6	43	3.77	4.13	95
440	220	36	1	30.0	95	201	125.2	42	4.79	5.13	85
441	57	36	1	19.6	71	250	133.2	97	3.00	4.60	92

Feature Engineering

- Unlike the NYC Taxicab example, we do not have domain knowledge here. So, let's just try adding a bunch of features!
- We can go “fishing” for good features to consider in our model
- We can take existing features like age and bmi and create lots of new features such as:
 - age^2
 - bmi^2
 - $\text{age} * \text{bmi}$
 - ...etc.
- May not look scientific, but we might discover some relationships and then try to understand them...

Diabetes Dataset with New Features

k : # of independent variables ($k = 64$); n : # of observations ($n = 442$)



	y	age		glu	age^2		glu^2	age:glu		ltg:glu
0	151	59	...	87	3481	...	7569	5133	...	422.80
1	75	48	...	69	2304	...	4761	3312	...	268.53
2	141	72	...	85	5184	...	7225	6120	...	397.19
3	206	24	...	89	576	...	7921	2136	...	435.24
4	135	50	...	80	2500	...	6400	4000	...	343.24
5	97	23	...	68	529	...	4624	1564	...	284.90
6	138	36	...	82	1296	...	6724	2952	...	324.00
7	63	66	...	92	4356	...	8464	6072	...	390.86
8	110	60	...	94	3600	...	8836	5640	...	420.87
9	310	29	...	88	841	...	7744	2552	...	473.84
10	101	22	...	83	484	...	6889	1826	...	327.95
11	69	56	...	77	3136	...	5929	4312	...	275.93
12	179	53	...	81	2809	...	6561	4293	...	348.63
13	185	50	...	88	2500	...	7744	4400	...	445.51
14	118	61	...	73	3721	...	5329	4453	...	313.21
...	...	...	...	...	...	...	...	...	...	...
437	178	60	...	93	3600	...	8649	5580	...	463.47
438	104	47	...	102	2209	...	10404	4794	...	453.16
439	132	60	...	95	3600	...	9025	5700	...	392.07
440	220	36	...	85	1296	...	7225	3060	...	436.04
441	57	36	...	92	1296	...	8464	3312	...	422.75

Adding Quadratic and Interaction Terms

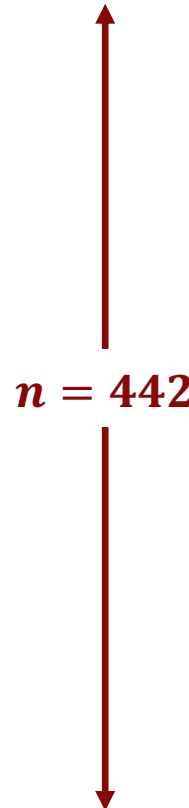
- Quadratic and interaction terms allow extra flexibility in modeling with our given features
- Quadratic terms (age^2 , bmi^2 , ...)
 - If the effects of age increase with age, then age^2 is a better feature variable
 - Note that for binary variables (specifically, sex in our dataset), quadratic terms do not add value. We therefore will add $10 - 1 = 9$ quadratic terms
- Interaction terms ($\text{age}*\text{bmi}$, $\text{sex}*\text{bmi}$, ...)
 - If the effects of bmi also depend on age or sex , then $\text{age}*\text{bmi}$ is a useful feature variable
 - We have $10*9/2 = 45$ interaction terms
- Now the total number of feature variables is $64 = 10 + 9 + 45$
- Note that models that incorporate quadratic and/or interaction terms are still linear regression models! (They are linear in the original and the newly created variables)
- Multicollinearity is likely to be an issue when we add these new variables ...

Centering and Scaling the Independent Variables

- The (regularization) methods we will learn today work best when the regression coefficients are comparable in terms of scale. This is accomplished by “standardizing” the variables...
- We re-center each independent variable column by subtracting its sample mean, and then re-scaling it so that the variance in each column is 1.0 (or choose any value that is the same for each variable)
- This yields the “standardized” dataset shown on the next slide.
- You should accept that such re-centering and scaling yields a completely equivalent linear regression model and that we can easily undo the scaling and re-centering at the end.

Standardized Expanded Diabetes Dataset

k : # of independent variables ($k = 64$); n : # of observations ($n = 442$)



	y	age		glu	age^2		glu^2		age:glu		ltg:glu
0	151	0.0381	...	-0.0176	-0.0149	...	-0.0275	...	-0.0303	...	-0.0278
1	75	-0.0019	...	-0.0922	-0.0413	...	0.0880	...	-0.0114	...	0.1040
2	141	0.0853	...	-0.0259	0.0916	...	-0.0224	...	-0.0647	...	-0.0223
3	206	-0.0891	...	-0.0094	0.1036	...	-0.0307	...	0.0034	...	-0.0250
4	135	0.0054	...	-0.0466	-0.0408	...	-0.0012	...	-0.0209	...	0.0087
5	97	-0.0927	...	-0.0963	0.1157	...	0.0990	...	0.1844	...	0.0578
6	138	-0.0455	...	-0.0384	-0.0036	...	-0.0112	...	0.0237	...	0.0270
7	63	0.0635	...	0.0031	0.0324	...	-0.0318	...	-0.0109	...	-0.0230
8	110	0.0417	...	0.0113	-0.0096	...	-0.0301	...	-0.0047	...	-0.0242
9	310	-0.0709	...	-0.0135	0.0505	...	-0.0293	...	0.0061	...	-0.0390
10	101	-0.0963	...	-0.0342	0.1283	...	-0.0154	...	0.0584	...	0.0218
11	69	0.0272	...	-0.0591	-0.0279	...	0.0173	...	-0.0511	...	0.0920
12	179	0.0163	...	-0.0425	-0.0365	...	-0.0064	...	-0.0307	...	0.0051
13	185	0.0054	...	-0.0135	-0.0408	...	-0.0293	...	-0.0169	...	-0.0311
14	118	0.0453	...	-0.0756	-0.0038	...	0.0488	...	-0.0919	...	0.0271
...	...	...	...	...	...	...	...	...	...	...	...
437	178	0.0417	...	0.0072	-0.0096	...	-0.0312	...	-0.0085	...	-0.0164
438	104	-0.0055	...	0.0445	-0.0408	...	-0.0040	...	-0.0207	...	-0.0368
439	132	0.0417	...	0.0155	-0.0096	...	-0.0285	...	-0.0008	...	-0.0352
440	220	-0.0455	...	-0.0259	-0.0036	...	-0.0224	...	0.0111	...	-0.0437
441	57	-0.0455	...	0.0031	-0.0036	...	-0.0318	...	-0.0184	...	-0.0211

Training/Testing (Random Sampling)

Full Dataset ($n = 442$)

	y	age		age^2		age:glu		ltg:glu
0	151	0.0381	...	-0.0149	...	-0.0303	...	-0.0278
1	75	-0.0019	...	-0.0413	...	-0.0114	...	0.1040
2	141	0.0853	...	0.0916	...	-0.0647	...	-0.0223
3	206	-0.0891	...	0.1036	...	0.0034	...	-0.0250
4	135	0.0054	...	-0.0408	...	-0.0209	...	0.0087
5	97	-0.0927	...	0.1157	...	0.1844	...	0.0578
6	138	-0.0455	...	-0.0036	...	0.0237	...	0.0270
7	63	0.0635	...	0.0324	...	-0.0109	...	-0.0230
8	110	0.0417	...	-0.0096	...	-0.0047	...	-0.0242
9	310	-0.0709	...	0.0505	...	0.0061	...	-0.0390
10	101	-0.0963	...	0.1283	...	0.0584	...	0.0218
11	69	0.0272	...	-0.0279	...	-0.0511	...	0.0920
12	179	0.0163	...	-0.0365	...	-0.0307	...	0.0051
13	185	0.0054	...	-0.0408	...	-0.0169	...	-0.0311
14	118	0.0453	...	-0.0038	...	-0.0919	...	0.0271
...	...	...	...	...	...	...	...	...
437	178	0.0417	...	-0.0096	...	-0.0085	...	-0.0164
438	104	-0.0055	...	-0.0408	...	-0.0207	...	-0.0368
439	132	0.0417	...	-0.0096	...	-0.0008	...	-0.0352
440	220	-0.0455	...	-0.0036	...	0.0111	...	-0.0437
441	57	-0.0455	...	-0.0036	...	-0.0184	...	-0.0211

70% Training Set ($n = 309$)

	y	age		age^2		age:glu		ltg:glu
1	75	-0.0019	...	-0.0413	...	-0.0114	...	0.1040
3	206	-0.0891	...	0.1036	...	0.0034	...	-0.0250
4	135	0.0054	...	-0.0408	...	-0.0209	...	0.0087
6	138	-0.0455	...	-0.0036	...	0.0237	...	0.0270
9	310	-0.0709	...	0.0505	...	0.0061	...	-0.0390
10	101	-0.0963	...	0.1283	...	0.0584	...	0.0218
12	179	0.0163	...	-0.0365	...	-0.0307	...	0.0051
13	185	0.0054	...	-0.0408	...	-0.0169	...	-0.0311
...	...	...	...	...	...	...	...	...
438	104	-0.0055	...	-0.0408	...	-0.0207	...	-0.0368
439	132	0.0417	...	-0.0096	...	-0.0008	...	-0.0352

30% Testing Set ($n = 133$)

	y	age		age^2		age:glu		ltg:glu
0	151	0.0381	...	-0.0149	...	-0.0303	...	-0.0278
2	141	0.0853	...	0.0916	...	-0.0647	...	-0.0223
5	97	-0.0927	...	0.1157	...	0.1844	...	0.0578
7	63	0.0635	...	0.0324	...	-0.0109	...	-0.0230
8	110	0.0417	...	-0.0096	...	-0.0047	...	-0.0242
11	69	0.0272	...	-0.0279	...	-0.0511	...	0.0920
14	118	0.0453	...	-0.0038	...	-0.0919	...	0.0271
...	...	...	...	...	...	...	...	...
437	178	0.0417	...	-0.0096	...	-0.0085	...	-0.0164
440	220	-0.0455	...	-0.0036	...	0.0111	...	-0.0437
441	57	-0.0455	...	-0.0036	...	-0.0184	...	-0.0211

Manual Variable Selection Attempts

Bottom Line: Challenging

Regression model with all 64 variables

Regression Results

R-squared: 0.63

No. Observations: 309

Coefficients:

	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
const	154.25	3.19	48.32	0.00	147.96	160.53	***
age	138.81	83.03	1.67	0.10	-24.74	302.36	.
sex	-196.66	85.21	-2.31	0.02	-364.51	-28.81	*
bmi	560.60	109.43	5.12	0.00	345.06	776.13	***
map	290.25	93.48	3.10	0.00	106.12	474.37	**
tc	-989.91	79936.43	-0.01	0.99	-158443.41	156463.58	
ldl	867.34	70258.28	0.01	0.99	-137522.78	139257.45	
hdl	39.14	29883.58	0.00	1.00	-58823.56	58901.84	
tch	-227.84	341.15	-0.67	0.50	-899.81	444.13	
ltg	953.54	26282.71	0.04	0.97	-50816.41	52723.49	
glu	6.75	90.36	0.07	0.94	-171.24	184.74	
age^2	57.78	89.11	0.65	0.52	-117.73	233.30	
bmi^2	75.77	109.61	0.69	0.49	-140.14	291.68	
map^2	2.75	87.48	0.03	0.97	-169.55	175.06	
tc^2	6829.26	8926.69	0.77	0.44	-10753.94	24412.46	
ldl^2	3384.14	6815.31	0.50	0.62	-10040.20	16808.49	
hdl^2	876.36	1933.42	0.45	0.65	-2931.97	4684.69	
tch^2	994.60	745.45	1.33	0.18	-473.74	2462.93	
ltg^2	1330.03	2231.47	0.60	0.55	-3065.36	5725.42	
glu^2	73.43	122.21	0.60	0.55	-167.29	314.16	
age:sex	197.89	94.25	2.10	0.04	12.25	383.53	*
age:bmi	76.33	101.98	0.75	0.45	-124.54	277.20	
age:map	-22.65	98.77	-0.23	0.82	-217.21	171.90	
age:tc	-332.06	852.22	-0.39	0.70	-2010.70	1346.58	
age:ldl	142.85	702.48	0.20	0.84	-1240.86	1526.56	
age:hdl	289.05	365.61	0.79	0.43	-431.10	1009.20	
age:tch	182.45	282.37	0.65	0.52	-373.74	738.64	
age:ltg	134.72	302.24	0.45	0.66	-460.61	730.05	
age:glu	96.67	105.15	0.92	0.36	-110.45	303.80	
sex:bmi	138.81	105.59	1.31	0.19	-69.16	346.79	

sex:map	32.68	96.08	0.34	0.73	-156.57	221.92
sex:tc	348.81	698.05	0.50	0.62	-1026.17	1723.79
sex:ldl	-193.31	557.69	-0.35	0.73	-1291.81	905.18
sex:hdl	-212.97	319.61	-0.67	0.51	-842.52	416.57
sex:tch	-443.91	258.23	-1.72	0.09	-952.55	64.73
sex:ltg	-10.86	280.11	-0.04	0.97	-562.61	540.89
sex:glu	0.30	94.19	0.00	1.00	-185.22	185.83
bmi:map	79.39	116.60	0.68	0.50	-150.29	309.07
bmi:tc	-127.41	773.81	-0.16	0.87	-1651.61	1396.79
bmi:ldl	7.64	663.40	0.01	0.99	-1299.08	1314.35
bmi:hdl	20.03	388.28	0.05	0.96	-744.78	784.84
bmi:tch	-131.95	285.97	-0.46	0.64	-695.25	431.34
bmi:ltg	133.08	312.92	0.43	0.67	-483.28	749.44
bmi:glu	30.94	111.23	0.28	0.78	-188.16	250.04
map:tc	642.07	894.05	0.72	0.47	-1118.96	2403.10
map:ldl	-256.76	741.11	-0.35	0.73	-1716.54	1203.03
map:hdl	-275.41	407.82	-0.68	0.50	-1078.71	527.88
map:tch	-197.56	268.03	-0.74	0.46	-725.51	330.39
map:ltg	-140.10	349.06	-0.40	0.69	-827.65	547.46
map:glu	-100.73	115.58	-0.87	0.38	-328.40	126.94
tc:ldl	-9404.01	14973.95	-0.63	0.53	-38898.70	20090.69
tc:hdl	-3024.36	4709.81	-0.64	0.52	-12301.44	6252.72
tc:tch	-1193.33	2203.49	-0.54	0.59	-5533.61	3146.96
tc:ltg	-4067.04	17321.79	-0.23	0.81	-38186.36	30052.28
tc:glu	-472.50	704.68	-0.67	0.50	-1860.53	915.53
ldl:hdl	1927.49	3940.76	0.49	0.63	-5834.76	9689.74
ldl:tch	499.21	1796.47	0.28	0.78	-3039.35	4037.78
ldl:ltg	2901.50	14432.49	0.20	0.84	-25526.67	31329.68
ldl:glu	294.32	579.10	0.51	0.61	-846.36	1434.99
hdl:tch	598.99	1260.17	0.48	0.63	-1883.21	3081.19
hdl:ltg	1203.76	6051.16	0.20	0.84	-10715.42	13122.94
hdl:glu	400.05	386.88	1.03	0.30	-362.00	1162.09
tch:ltg	-128.66	882.48	-0.15	0.88	-1866.91	1609.58
tch:glu	393.55	326.49	1.21	0.23	-249.56	1036.65
ltg:glu	185.62	329.55	0.56	0.57	-463.51	834.75

Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1

What if we just use the four variables identified as significant?

```
smf.ols(formula="y ~ sex + bmi + map + age:sex", data=train).fit()
```

```
=====
Regression Results
=====
R-squared: 0.45
No. Observations: 309
=====
Coefficients:
-----

```

	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
Intercept	150.99	3.38	44.65	0.00	144.34	157.65	***
sex	-40.02	72.41	-0.55	0.58	-182.50	102.47	
bmi	856.93	74.82	11.45	0.00	709.71	1004.16	***
map	391.68	79.25	4.94	0.00	235.73	547.63	***
age:sex	3949.77	1496.48	2.64	0.01	1005.00	6894.55	**

```
=====
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====
```

Simple model using only 6 measurements of cholesterol, ldl, etc.

```
smf.ols(formula="y ~ tc + ldl + hdl + tch + ltg + glu", data=train).fit()
```

```
=====
Regression Results
=====
```

```
R-squared: 0.37
```

```
No. Observations: 309
=====
```

```
Coefficients:
```

```
-----
              Coef.  Std.Err.    t  P>|t|  [0.025  0.975]  Signif
Intercept  152.97      3.59  42.63   0.00   145.91   160.03   ***
tc         -491.12    566.42  -0.87   0.39  -1605.75   623.50
ldl         371.52    456.99   0.81   0.42   -527.78  1270.82
hdl        -184.79    292.68  -0.63   0.53   -760.74   391.16
tch         -39.42    231.81  -0.17   0.87   -495.58   416.74
ltg         923.14    234.29   3.94   0.00   462.08  1384.19   ***
glu         161.50     89.19   1.81   0.07   -14.01   337.00    .
=====
```

```
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====
```

“Backward elimination” of variables

```
=====
Regression Results
=====
```

```
R-squared: 0.59
```

```
No. Observations: 309
=====
```

```
Coefficients:
-----
```

	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
const	153.99	2.97	51.91	0.00	148.15	159.82	***
bmi	700.84	75.21	9.32	0.00	552.82	848.86	***
map	256.57	70.00	3.67	0.00	118.80	394.34	***
tc	-183.19	76.42	-2.40	0.02	-333.59	-32.79	*
ltg	715.10	100.78	7.10	0.00	516.77	913.44	***
tch^2	285.66	98.73	2.89	0.00	91.34	479.97	**
ltg^2	623.49	204.98	3.04	0.00	220.06	1026.91	**
age:sex	206.58	65.01	3.18	0.00	78.64	334.52	**
age:glu	174.21	68.66	2.54	0.01	39.09	309.33	*
sex:bmi	176.54	70.62	2.50	0.01	37.54	315.53	*
sex:tch	-200.66	78.39	-2.56	0.01	-354.94	-46.37	*
map:tc	184.11	69.22	2.66	0.01	47.87	320.34	**
tc:ltg	-1042.26	309.00	-3.37	0.00	-1650.39	-434.13	***
ldl:ltg	702.48	231.26	3.04	0.00	247.34	1157.61	**
hdl:ltg	497.87	136.32	3.65	0.00	229.58	766.16	***

```
=====
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====
```

- Find the variable whose elimination decreases R² the least.
- Remove it.
- Keep doing that until all variables are significant.

Performance Assessment

Model	Number of Variables	R^2	OSR^2
Linear Regression Model (all vars)	64	0.6308	0.3399
Significant Vars Only	4	0.4467	0.3115
The six blood measurements	6	0.3666	0.3880
Backward elimination reduced model	14	0.5851	0.3582

- Model with 64 variables is grossly overfitting the training data.
- Models with fewer variables overfit less.
 - Note: training/testing gap is large for all models. Our test set only has 100 rows in total, so this could indicate that the testing set is “harder” to predict for some reason.
- **How can we obtain the best OSR^2 ?** We must balance in-sample R^2 and model complexity (number of variables).

Sneak Peak

Model	Number of Variables	R^2	OSR^2
Linear Regression Model (all vars)	64	0.6308	0.3399
Significant Vars Only	4	0.4467	0.3115
The six blood measurements	6	0.3666	0.3880
Backward elimination reduced model	14	0.5851	0.3582
Best Model	16	0.5622	0.4439

- Best model has a moderate number of variables.
- But these variables and their coefficients are *optimized* to reduce overfitting.
 - Overfitting is caused by not just the number of variables, but also their coefficient values.

Sneak Peak

Model	Number of Variables	R ²	OSR ²
Linear Regression Model (all vars)	64	0.6308	0.3399
Significant Vars Only	4	0.4467	0.3115
The six blood measurements	6	0.3666	0.3880
Backward elimination reduced model	14	0.5851	0.3582
LASSO-2.94	16	0.5622	0.4439

LASSO-2.94 is the solution to the following augmented least squares problem:

$$\min_{b_0 \dots b_k} \sum_{i=1}^n [y_i - (b_0 + b_1 x_{i1} + b_2 x_{i2} + \dots + b_k x_{ik})]^2 + 2.94 \cdot \sum_{j=1}^k |b_j|$$

Variable Selection

- These diabetes linear regression models illustrate the challenge in performing variable selection – choosing which variables/features to be used to build a model
- Variable selection is particularly important for datasets containing very many variables ($k \approx n$, or even $k > n$)

Model Complexity and Overfitting

Finding the “Sweet Spot”

Questions to Keep in Mind

1. Value of Simple Models

- Why are simple models less prone to overfitting than complex models?

2. Regularization Paradigm

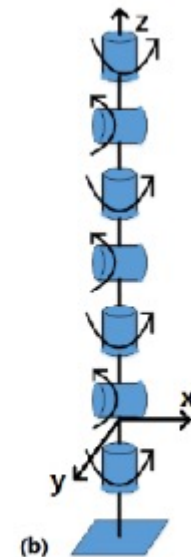
- How does regularization conceptually allow us to balance between two objectives: model fit and complexity?
- How do we choose the optimal balance between the two (i.e., choice of λ)?

3. Regularization in Practice

- How is regularization used in practice with LASSO and Ridge penalties?

What is model complexity?

- Model complexity is broadly defined as how many *parameters* the model controls when fitting to data.
- More parameters means the model can learn more complex patterns in the data.



Linear Regression: # of Variables

R-squared: 0.85
Adjusted R-squared: 0.85
No. Observations: 1536

=====
Coefficients:

	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
Intercept	1099.65	62.41	17.62	0.0	977.24	1222.07	***
hour	-105.77	18.49	-5.72	0.0	-142.03	-69.51	***
hour_after_5	934.08	32.40	28.83	0.0	870.53	997.64	***
hour_after_9	-747.45	21.74	-34.37	0.0	-790.10	-704.80	***
hour_after_19	-354.13	21.78	-16.26	0.0	-396.85	-311.40	***

=====
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====

4 Variables

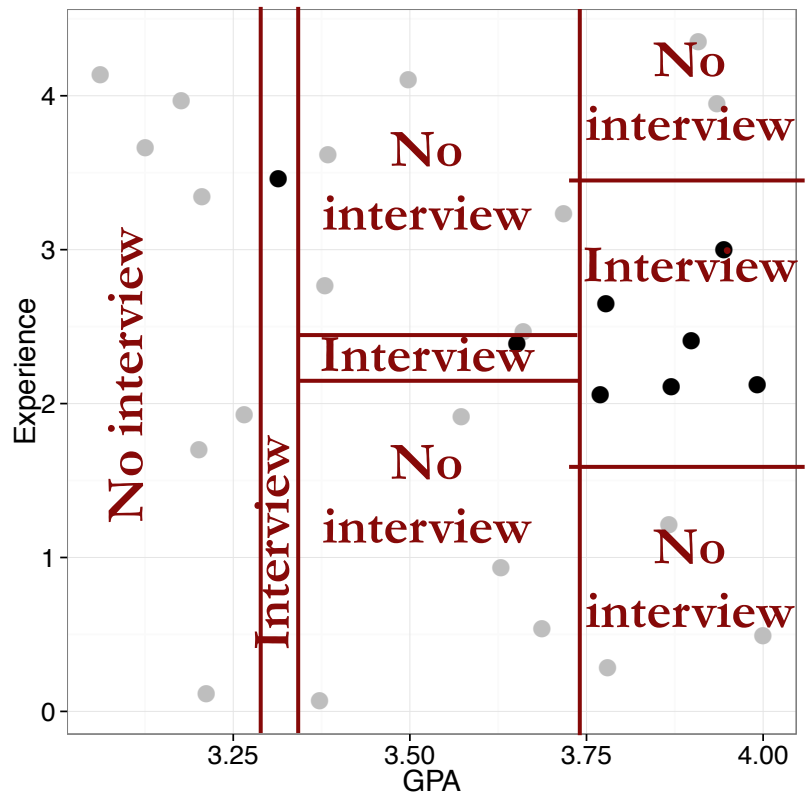
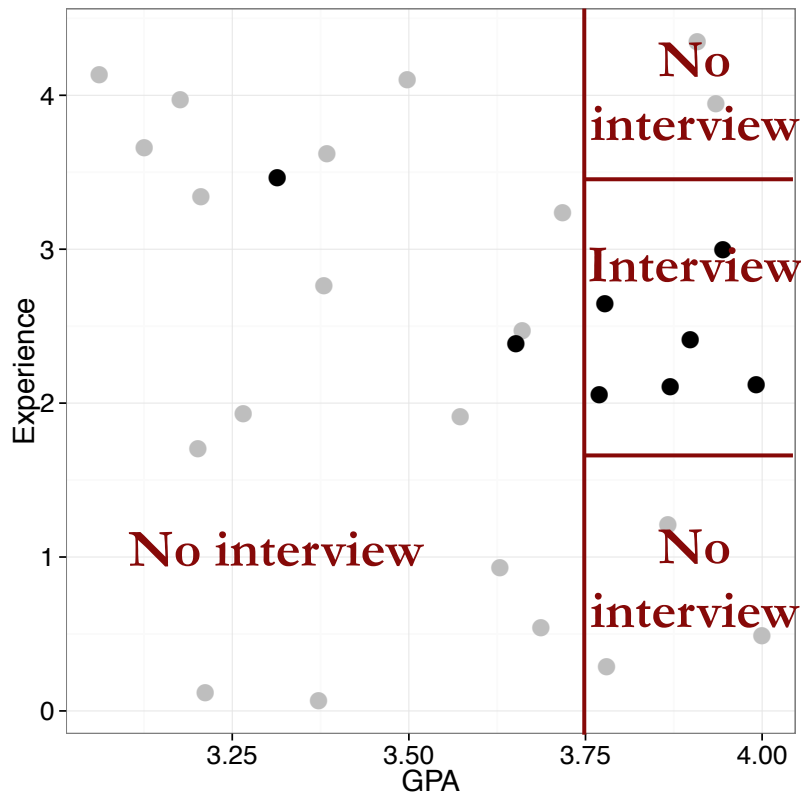
=====
Coefficients:

	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
Intercept	1828.91	76.75	23.83	0.00	1678.35	1979.47	***
hour	-878.49	113.90	-7.71	0.00	-1101.90	-655.08	***
hour_after_1	536.01	193.58	2.77	0.01	156.30	915.72	**
hour_after_2	110.64	183.26	0.60	0.55	-248.83	470.12	
hour_after_3	217.38	182.65	1.19	0.23	-140.89	575.65	
hour_after_4	75.62	182.61	0.41	0.68	-282.58	433.82	
hour_after_5	882.78	182.61	4.83	0.00	524.58	1240.97	***
hour_after_6	734.45	182.61	4.02	0.00	376.26	1092.65	***
hour_after_7	-335.57	182.61	-1.84	0.07	-693.76	22.63	.
hour_after_8	-1497.03	182.61	-8.20	0.00	-1855.23	-1138.84	***
hour_after_9	-439.79	182.61	-2.41	0.02	-797.99	-81.60	*
hour_after_10	344.96	182.61	1.89	0.06	-13.23	703.16	.
hour_after_11	636.18	182.61	3.48	0.00	277.98	994.37	***
hour_after_12	-476.22	182.61	-2.61	0.01	-834.41	-118.02	**
hour_after_13	380.17	182.61	2.08	0.04	21.98	738.37	*
hour_after_14	120.60	182.61	0.66	0.51	-237.60	478.79	
hour_after_15	-907.25	182.61	-4.97	0.00	-1265.44	-549.05	***
hour_after_16	619.22	182.61	3.39	0.00	261.03	977.42	***
hour_after_17	1016.96	182.61	5.57	0.00	658.77	1375.16	***
hour_after_18	-1199.01	182.61	-6.57	0.00	-1557.21	-840.82	***
hour_after_19	-513.61	182.61	-2.81	0.00	-871.81	-155.41	**
hour_after_20	634.94	182.62	3.48	0.00	276.72	993.16	***
hour_after_21	-164.00	182.82	-0.90	0.37	-522.62	194.61	
hour_after_22	-999.81	186.27	-5.37	0.00	-1365.19	-634.43	***
hour_after_23	-179.79	238.15	-0.75	0.45	-646.93	287.36	

=====
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====

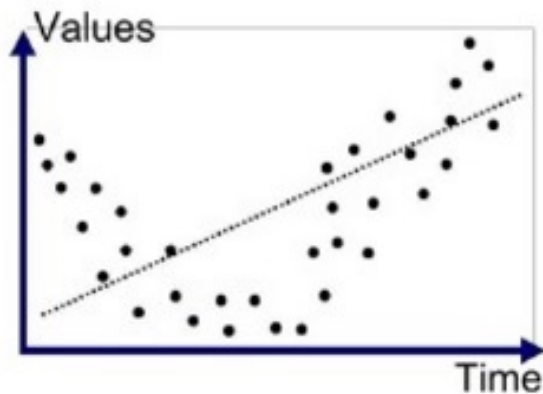
24 Variables

CART: Number of Splits/Leaves

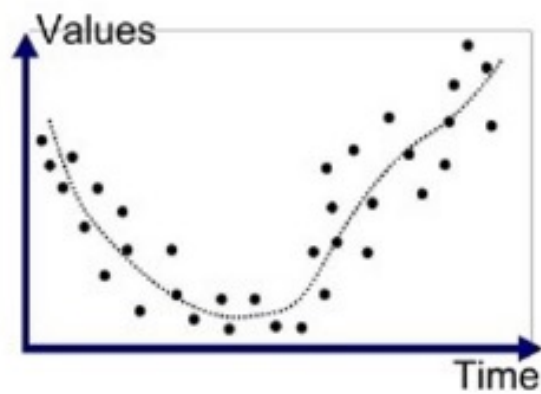


- Which model is more accurate on the training data? test data?

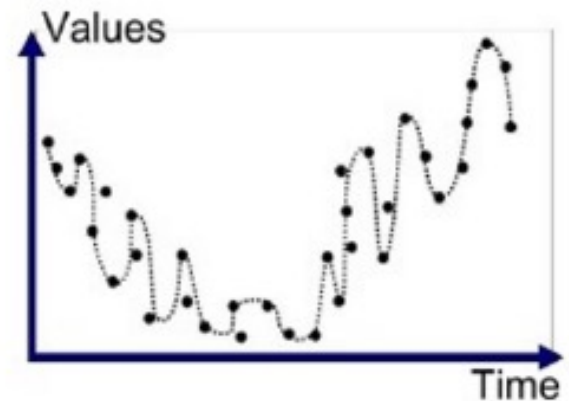
Underfitting and Overfitting



Underfitted

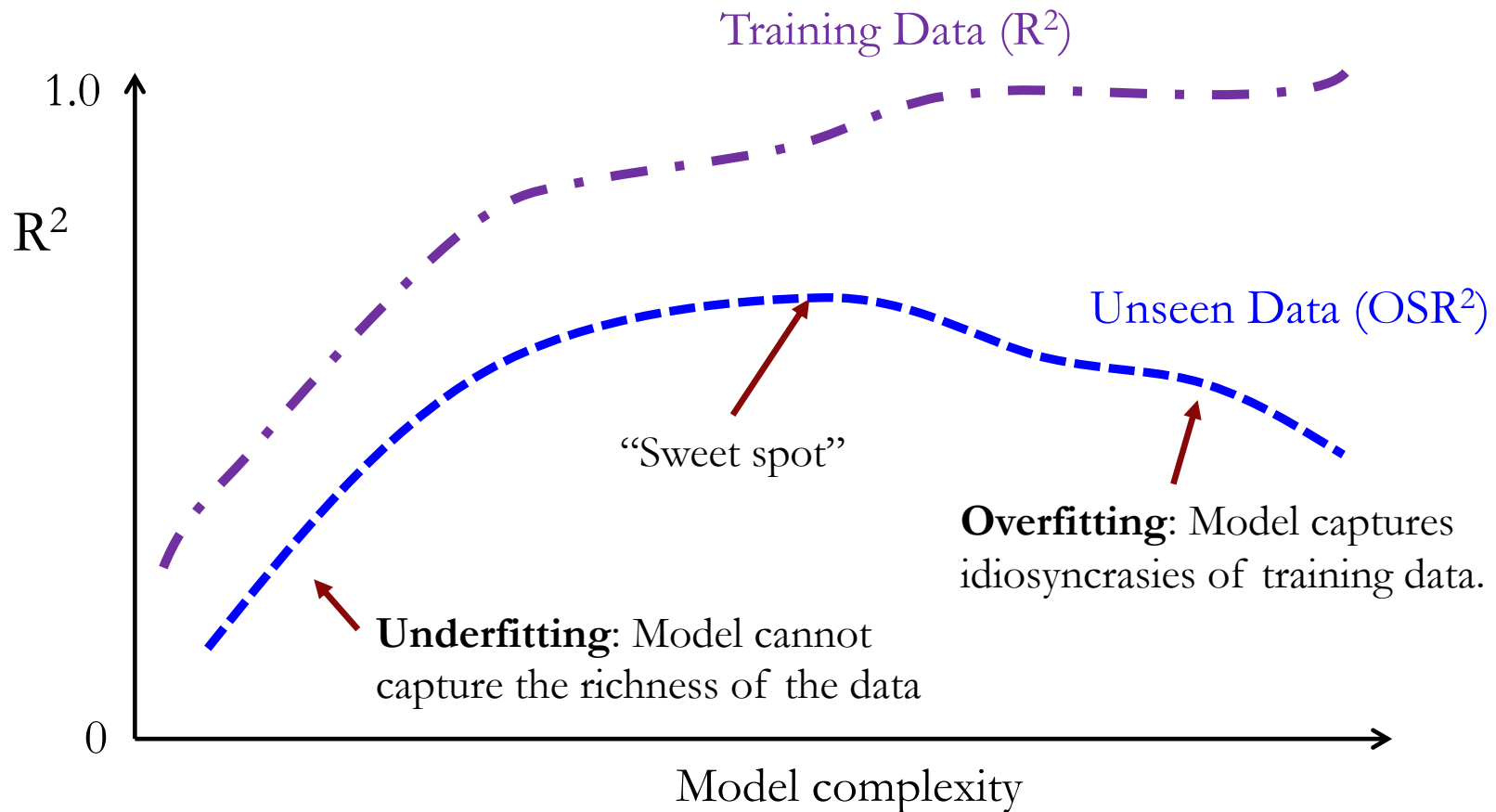


Good Fit/Robust

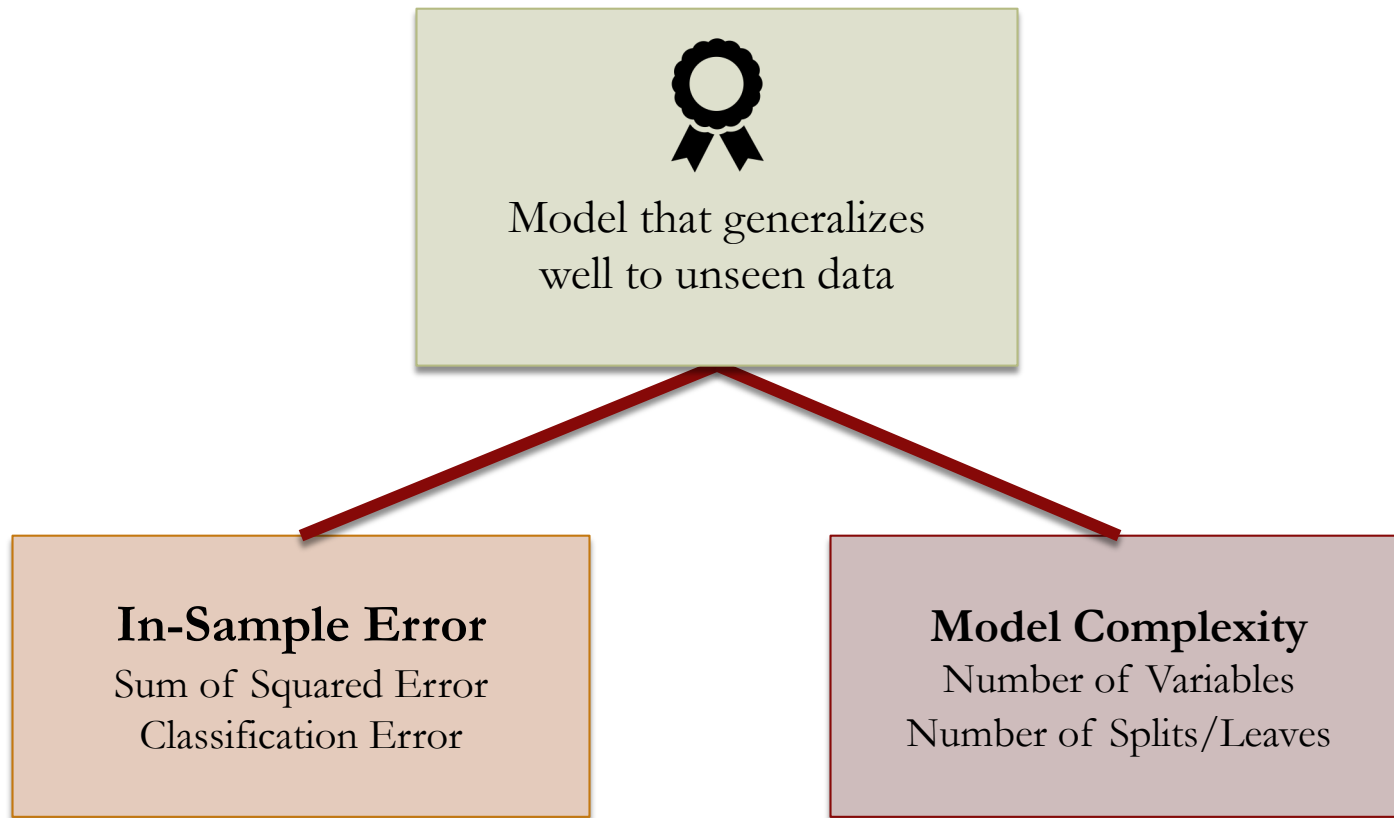


Overfitted

Finding the Sweet Spot



Balancing Two Objectives



Regularization Paradigm

Questions to Keep in Mind

1. Value of Simple Models

- Why are simple models less prone to overfitting than complex models?

2. Regularization Paradigm

- How does regularization conceptually allow us to balance between model fit and complexity?
- How do we choose the optimal balance between the two (i.e., choice of λ)?

3. Regularization in Practice

- How is regularization used in practice with LASSO and Ridge penalties?

Multi-objective Optimization

You are a fresh graduate out of college. You are deciding the best career for you. You want to maximize your happiness, which you know is influenced by (1) Salary, and (2) Work-life balance.

You are young and don't know yourself yet... What kind of balance leads to the best happiness for me?

Maximize: $\text{Salary} + \lambda \cdot \text{Hours of Free Time}$

Here's what you do:

- Year 1: Try $\lambda = 1$ (burn yourself out, but very lucrative)
- Year 2: Try $\lambda = 25$ (decent balance)
- Year 3: Try $\lambda = 2000$ (paycheck-to-paycheck, but pursue hobbies)

After the 3 years, you realize that Year 2 gives you the most happiness. You choose this balance for the rest of your career.

Regularization as Optimization

- Old Paradigm

Minimize In-Sample Error
over Model Parameters

- New Paradigm (with Regularization)

Minimize In-Sample Error
over Model Parameters

+ λ · Model Complexity

Shrinkage Parameter

Controls the weight (trade-off) between
measure of fit vs. measure of complexity.

Vanilla Linear Regression

$$\min_{b_0, b_1, \dots, b_k} \sum_{i \in \text{Train Data}} (\hat{y}_i - y_i)^2$$

Measure of Fit/Error

- Find coefficients $b_0, b_1, \dots, b_k$ to optimize fit and complexity.
- **Measure of Fit/Error:** Minimize Sum of Squared Error (SSE), equivalent to R^2

.

Linear Regression with BSS Regularization

The diagram illustrates the BSS regularization equation with several annotations:

- Predicted**: A blue arrow points to $\hat{y}_i$ in the equation.
- Actual**: A red arrow points to y_i in the equation.
- Shrinkage Parameter**: A green arrow points to the λ coefficient in the equation.
- Controls the weight (trade-off) between measure of fit vs. measure of complexity.**: A text annotation explaining the role of the shrinkage parameter.
- Measure of Fit/Error**: A bracket under the first part of the equation, $\sum_{i \in \text{Train Data}} (\hat{y}_i - y_i)^2$.
- Measure of Complexity**: A bracket under the second part of the equation, $\lambda \cdot \text{Number of Non-Zero}(b_0, \dots, b_k)$.

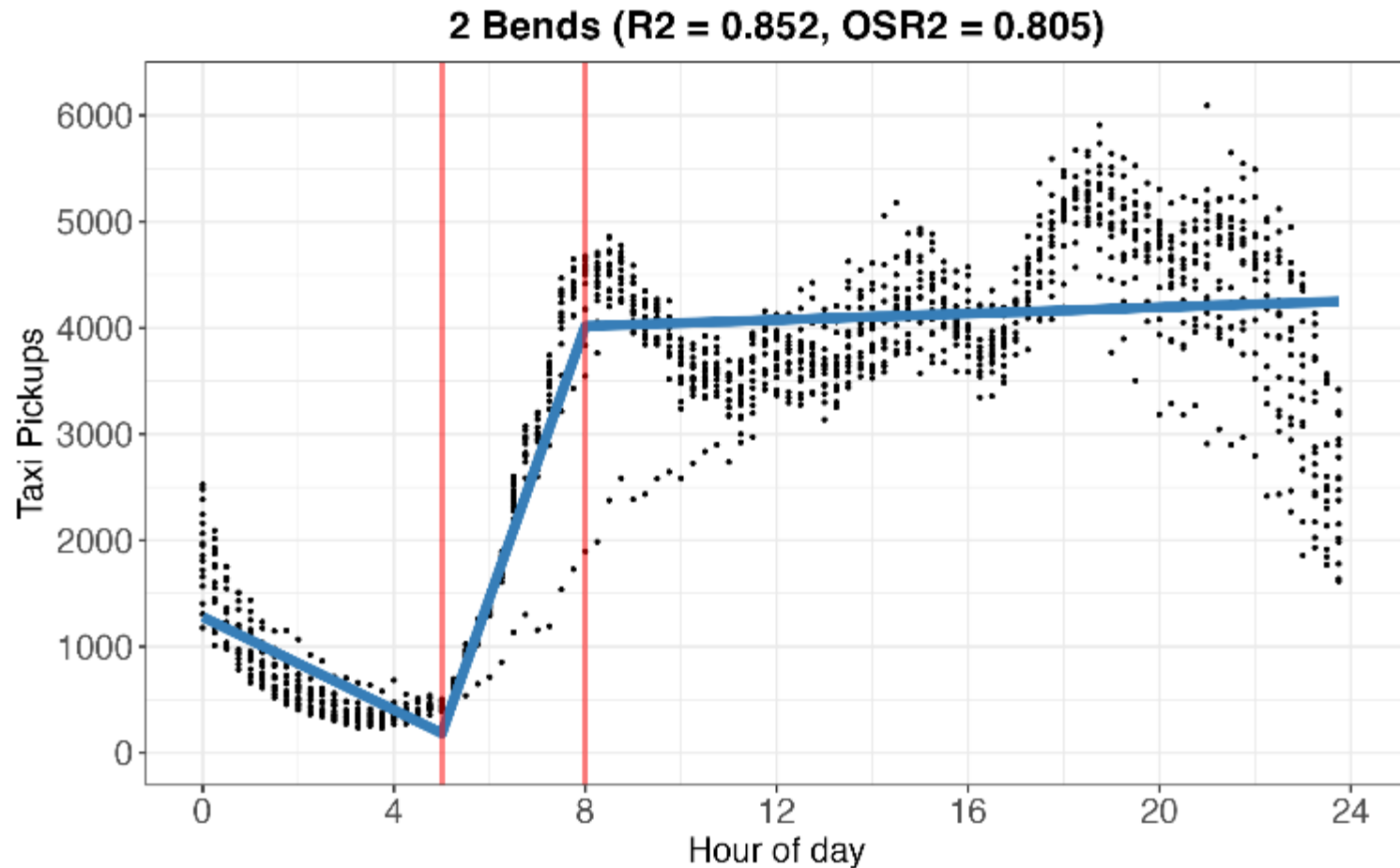
$$\min_{b_0, b_1, \dots, b_k} \underbrace{\sum_{i \in \text{Train Data}} (\hat{y}_i - y_i)^2}_{\text{Measure of Fit/Error}} + \underbrace{\lambda \cdot \text{Number of Non-Zero}(b_0, \dots, b_k)}_{\text{Measure of Complexity}}$$

- Find coefficients $b_0, b_1, \dots, b_k$ to optimize fit and complexity.
 - **Measure of Fit/Error**: Minimize Sum of Squared Error (SSE), equivalent to R^2
 - **Measure of Complexity**: Number of Non-Zero Coefficients
- Best Subset Selection (BSS) refers to the measure of complexity.

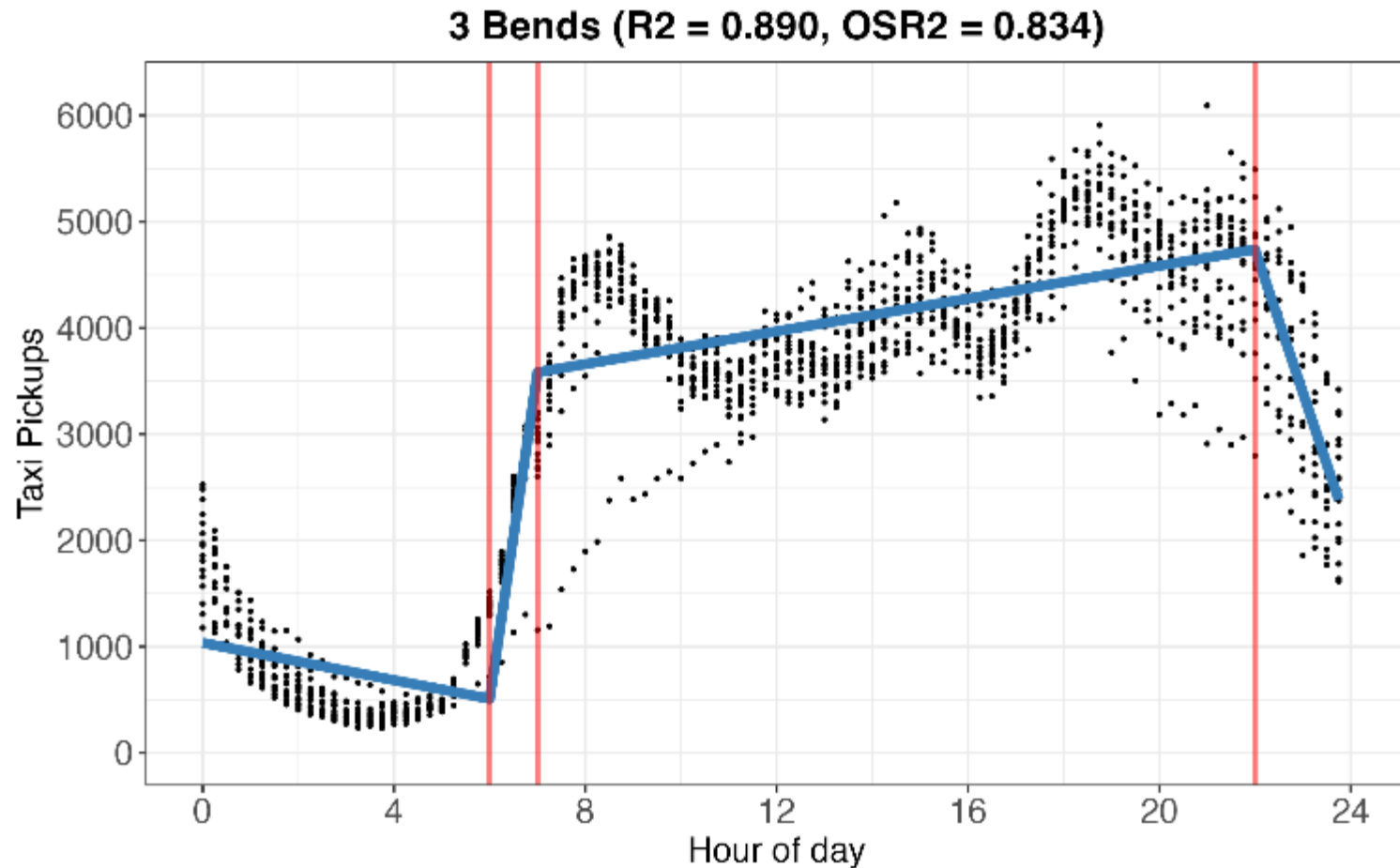
Recall our Line Bending Example

- We had 24 variables, each of which controlled how much our piece-wise line bended at each hour of the day.
 - A variable being 0 means the line does not bend.
- Without Regularization:
 - Choose the change in slope at 24 bends to minimize sum of squared error (SSE).
- With Regularization:
 - Choose the change in slopes at 24 bends to minimize $SSE + \text{Lambda} * (\# \text{ of Bends You Choose})$.
 - Each Bend is equivalent to a Lambda increase in SSE.

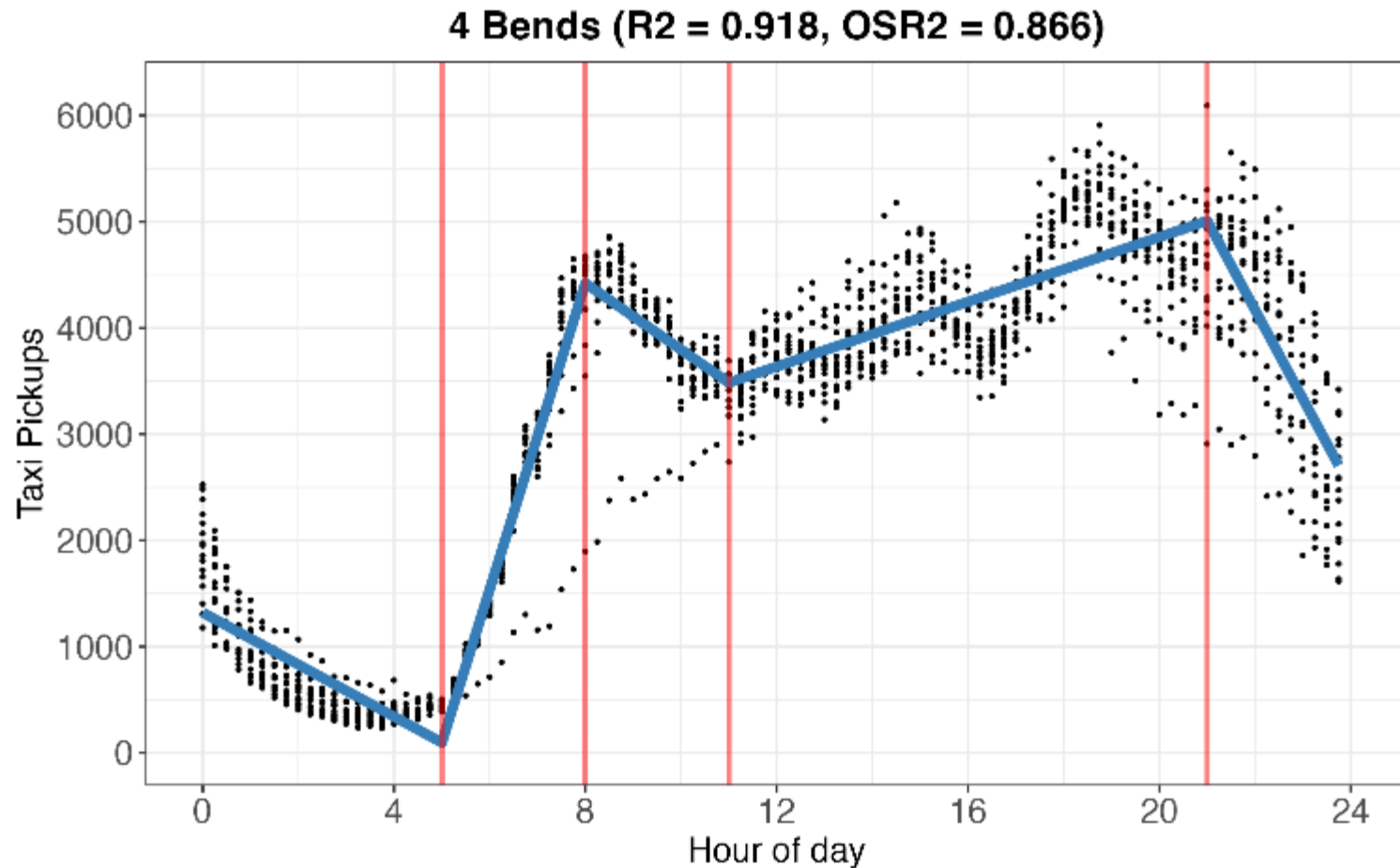
$\lambda = 10^5$ leads to 2 Bends



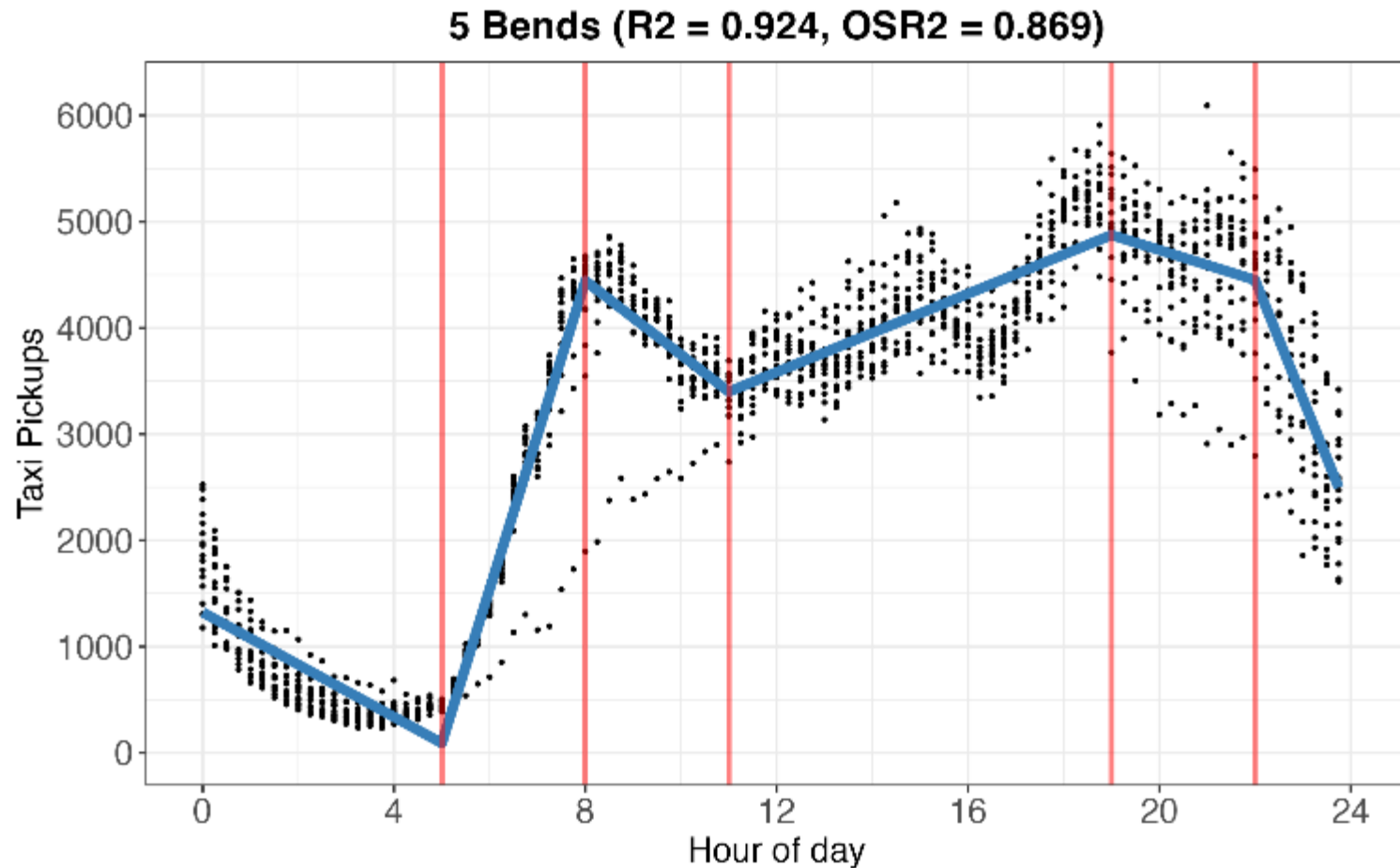
$\lambda = 6 \cdot 10^4$ leads to 3 Bends



$\lambda = 3 \cdot 10^4$ leads to 4 Bends



$\lambda = 2 \cdot 10^4$ leads to 5 Bends



Best Subset Selection for Diabetes

$$\min_{b_0, b_1, \dots, b_k} \sum_{i \in \text{Train Data}} (\hat{y}_i - y_i)^2 + \lambda \cdot \text{Number of Non-Zero}(b_0, \dots, b_k)$$


Penalty “Tradeoff” Parameter (λ)	SSE	Number of Non-Zero Coefs	In-Sample R^2	Out-of- Sample R^2
3.27×10^6	1.08×10^6	0	0	~ 0
4.42×10^5	8.71×10^5	1	0.3910	0.2207
1.63×10^5	6.22×10^5	2	0.4830	0.3936
22,026	5.08×10^5	5	0.5371	0.4161
2,981	4.79×10^5	9	0.5650	0.4470

How to Pick Lambda

- What choice of trade-off parameter leads to the best OSR^2 ?
 - There is no one-size-fits-all answer.
 - Depends on the dataset
- **Question:** How can we pick lambda to perform the best on the testing set?
 - **Answer:** Cross Validation!
 - **Intuition:** Giving yourself a “practice test” before the real test.
 - Note: Cross-validation can be used for any parameter selection (e.g., selecting the best `ccp_alpha` in CART).

Connection

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You are young and don't know yourself yet... What kind of balance leads to the best happiness for me?

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Here's what you do:

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Cross Validation

After the 3 years, you realize that Year 2 gives you the most happiness. You choose this balance for the rest of your career.

Salary = In-Sample Fit
Free Time = Model Complexity
Happiness = OSR^2

Regularization in Practice

Practical Implementation in Linear Regression

Best Subset Selection is Hard to Optimize

Number of Variables (k)	Best Subset Selection Computation Time
1	0.06 second
2	0.18 second
3	0.28 second
4	0.69 second
5	2.33 second
6	9.75 second
7	74.14 seconds
8	330.21 seconds
9	1636.40 seconds
10	Hours?
20	Years?

- It is challenging to find an optimal solution to the Best Subset Selection problem.
- There's not much we can do besides enumerate all possible subsets of k variables out of say 64 (in the diabetes dataset).
- In practice, we use some alternative penalties that try to mimic Best Subset Selection.

Alternative Measures of Complexity

1. BSS – Best Subset Selection

$$\min_{b_0, b_1, \dots, b_k} \sum_{i \in \text{Train Data}} (\hat{y}_i - y_i)^2 + \lambda \cdot \text{Number of Non-Zero}(b_0, \dots, b_k)$$

2. LASSO

$$\min_{b_0, b_1, \dots, b_k} \sum_{i \in \text{Train Data}} (\hat{y}_i - y_i)^2 + \lambda \cdot \sum_{j=1}^k |b_j|$$

3. Ridge

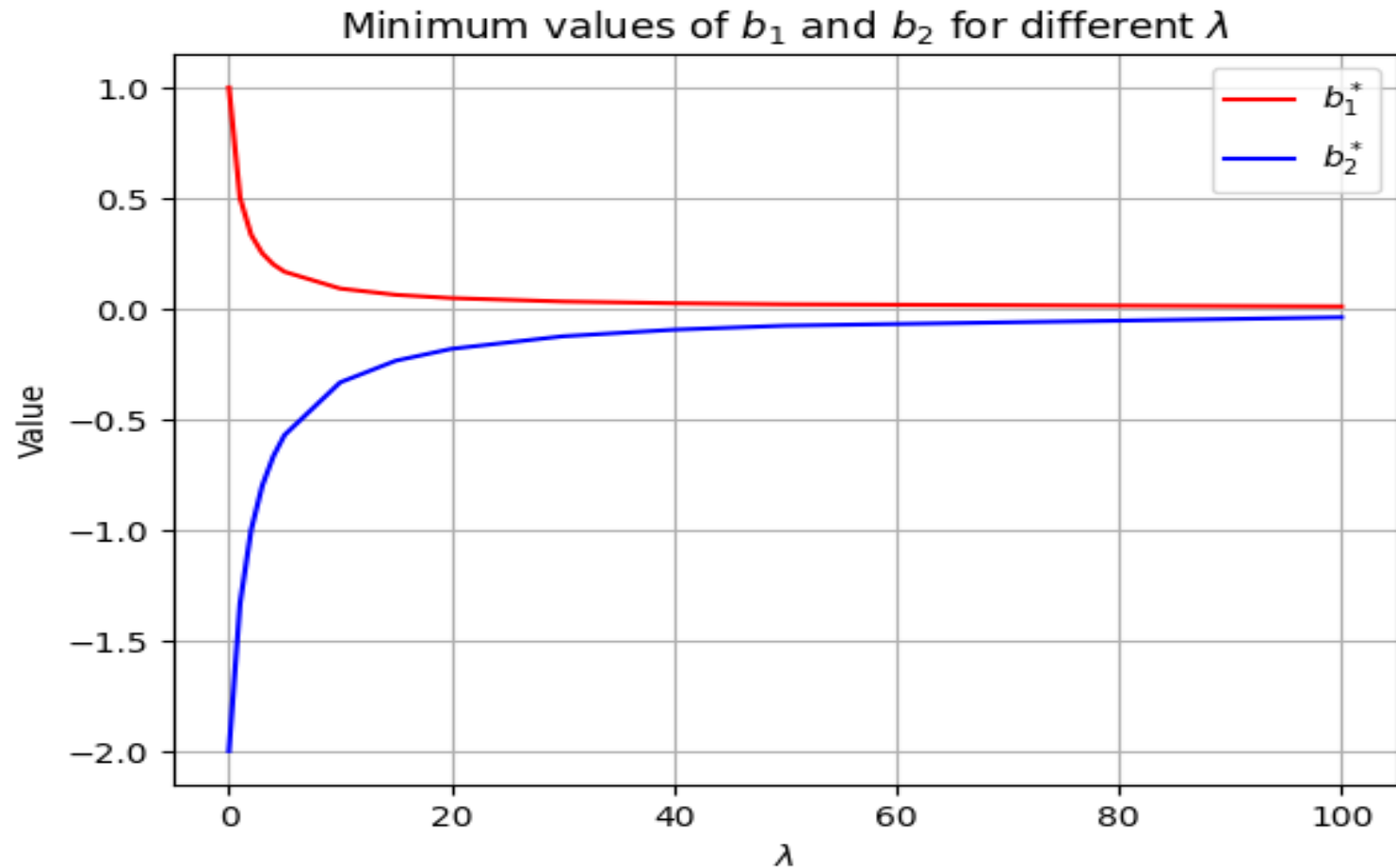
$$\min_{b_0, b_1, \dots, b_k} \sum_{i \in \text{Train Data}} (\hat{y}_i - y_i)^2 + \lambda \cdot \sum_{j=0}^k (b_j)^2$$

Encourages many b_i
to be close to 0.

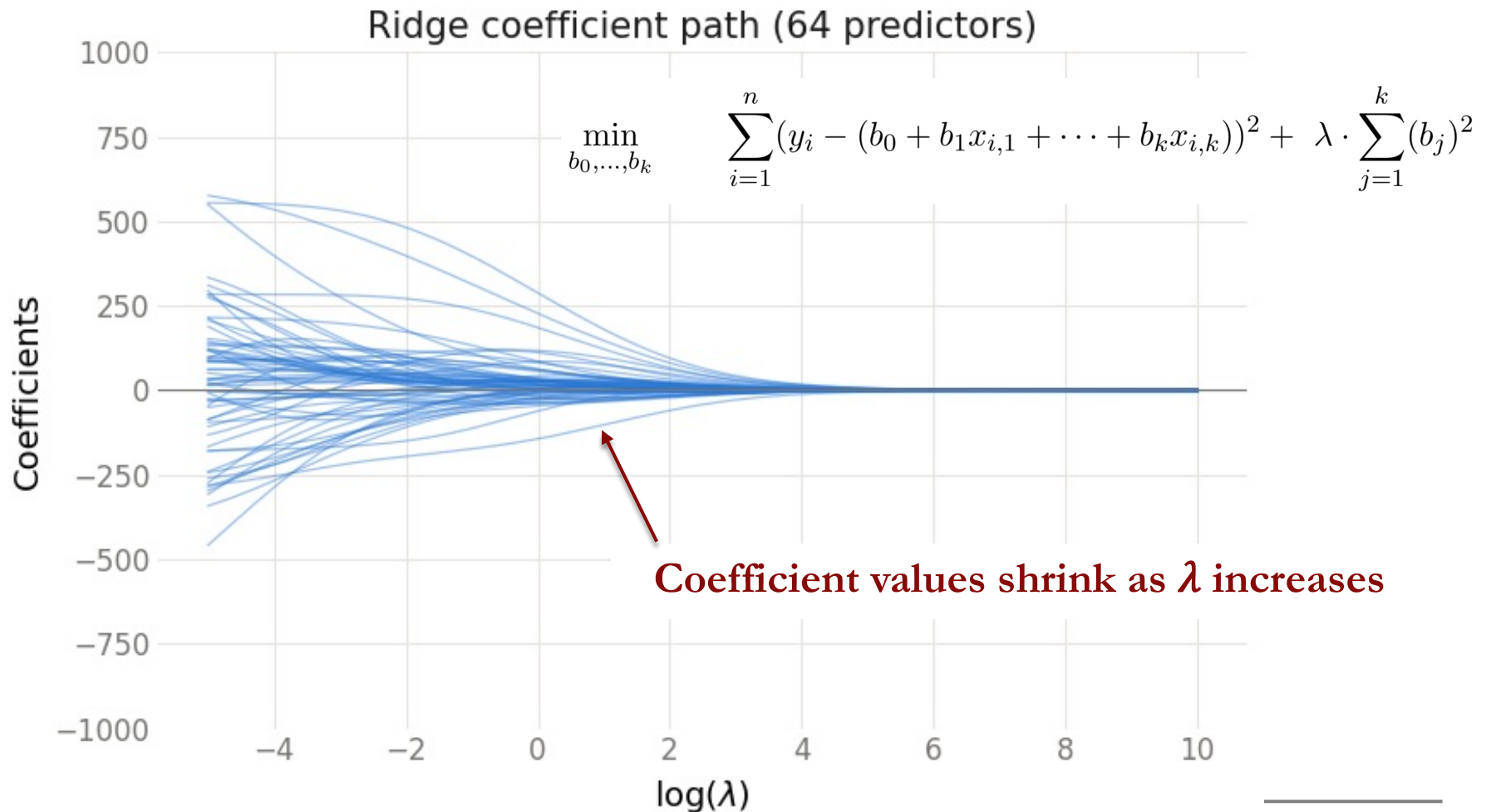
Measure of Fit/Error

Measure of Complexity

“Coefficient Profile”



Ridge Coefficient Profile



“LASSO”

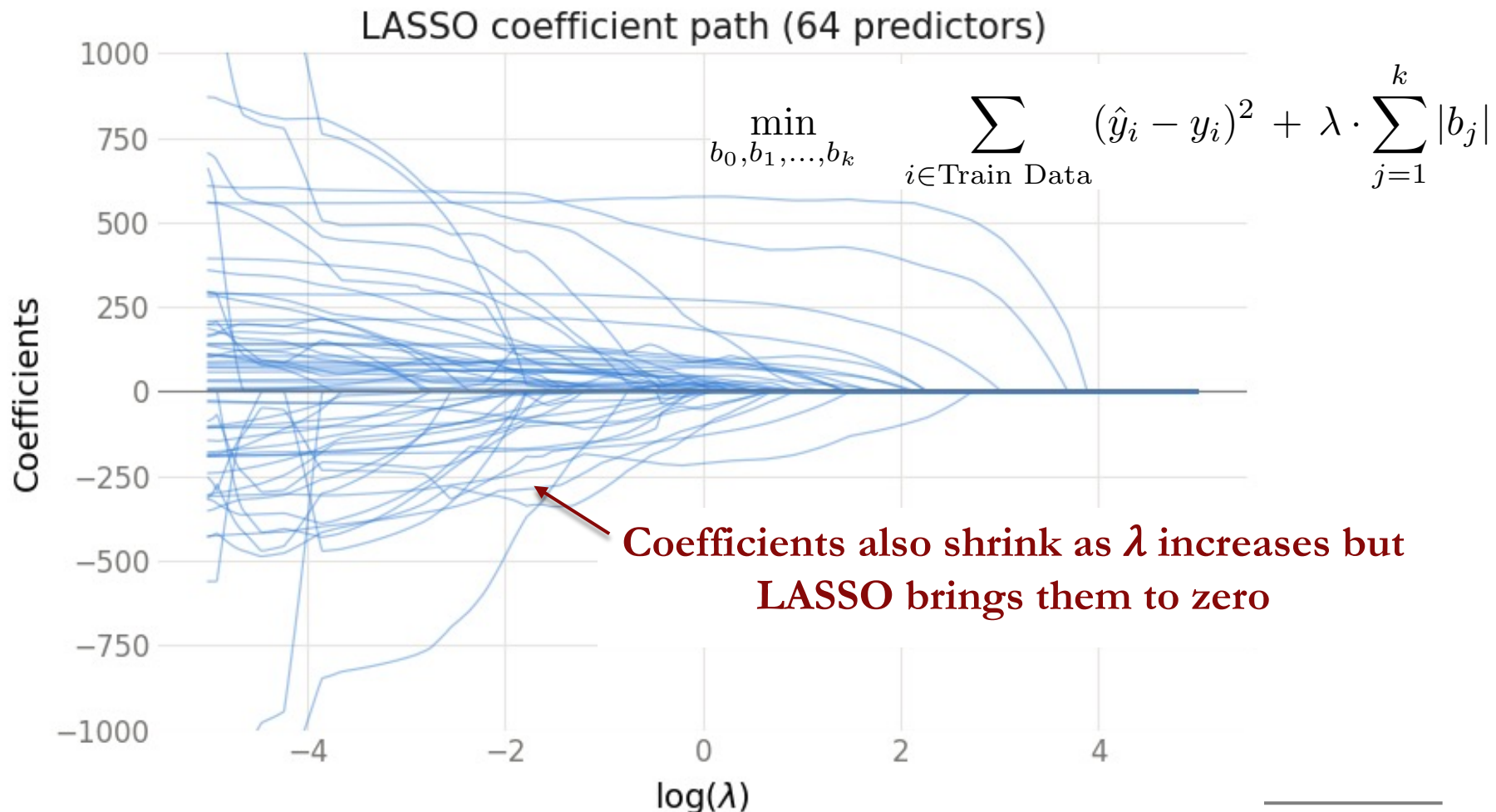
- LASSO (“Least Absolute Shrinkage and Selection Operator”) encourages coefficients close to 0 by penalizing their absolute value.

$$\min_{b_0, b_1, \dots, b_k} \sum_{i \in \text{Train Data}} (\hat{y}_i - y_i)^2 + \lambda \cdot \sum_{j=1}^k |b_j|$$

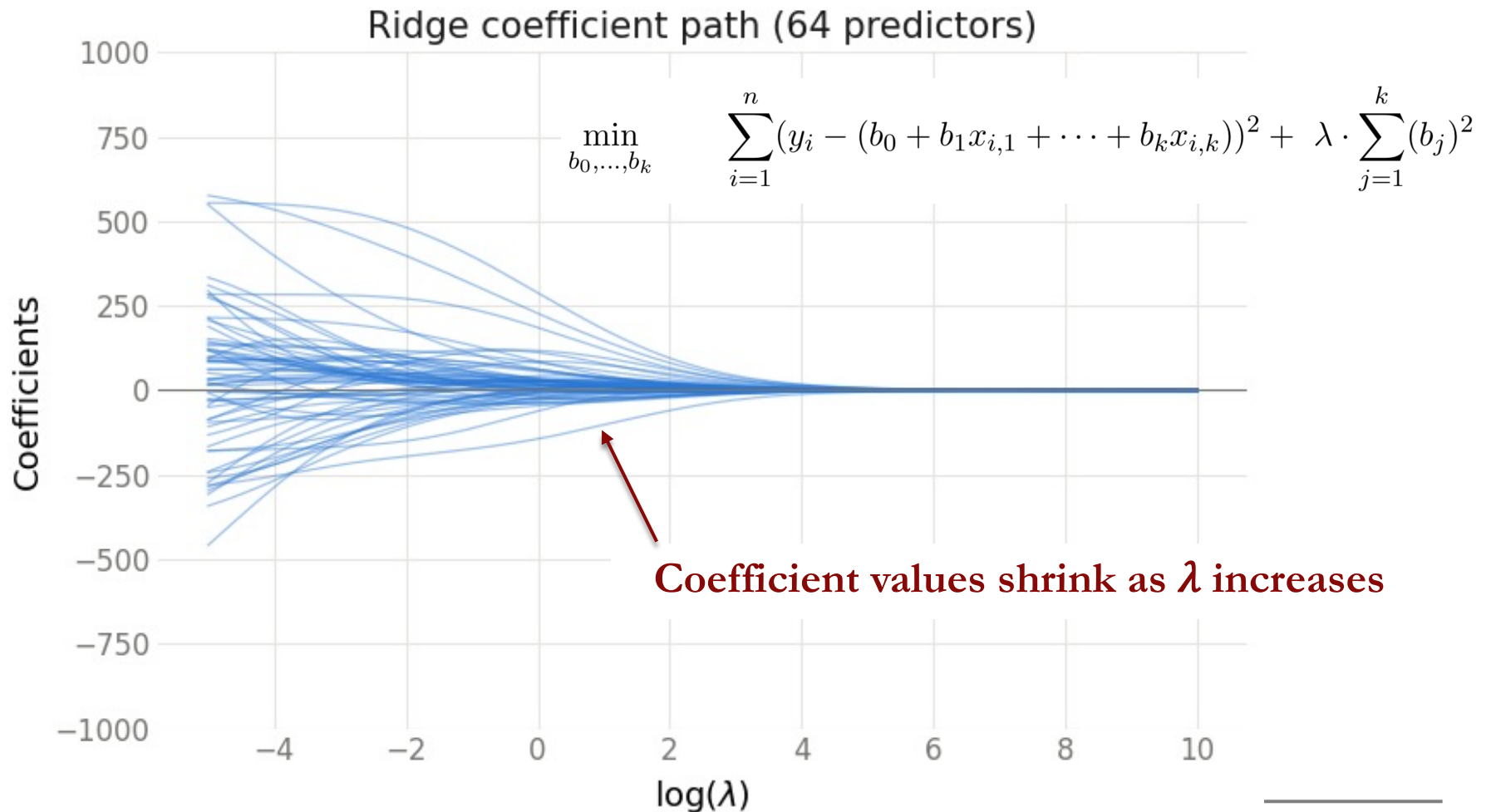
- Properties
 - **Computation:** Much better than BSS, but still a bit challenging.
 - **Results:** As lambda increases, coefficients b_j shrink and many are equal to 0 (hence called the “selection operator”)

LASSO Coefficient Profile

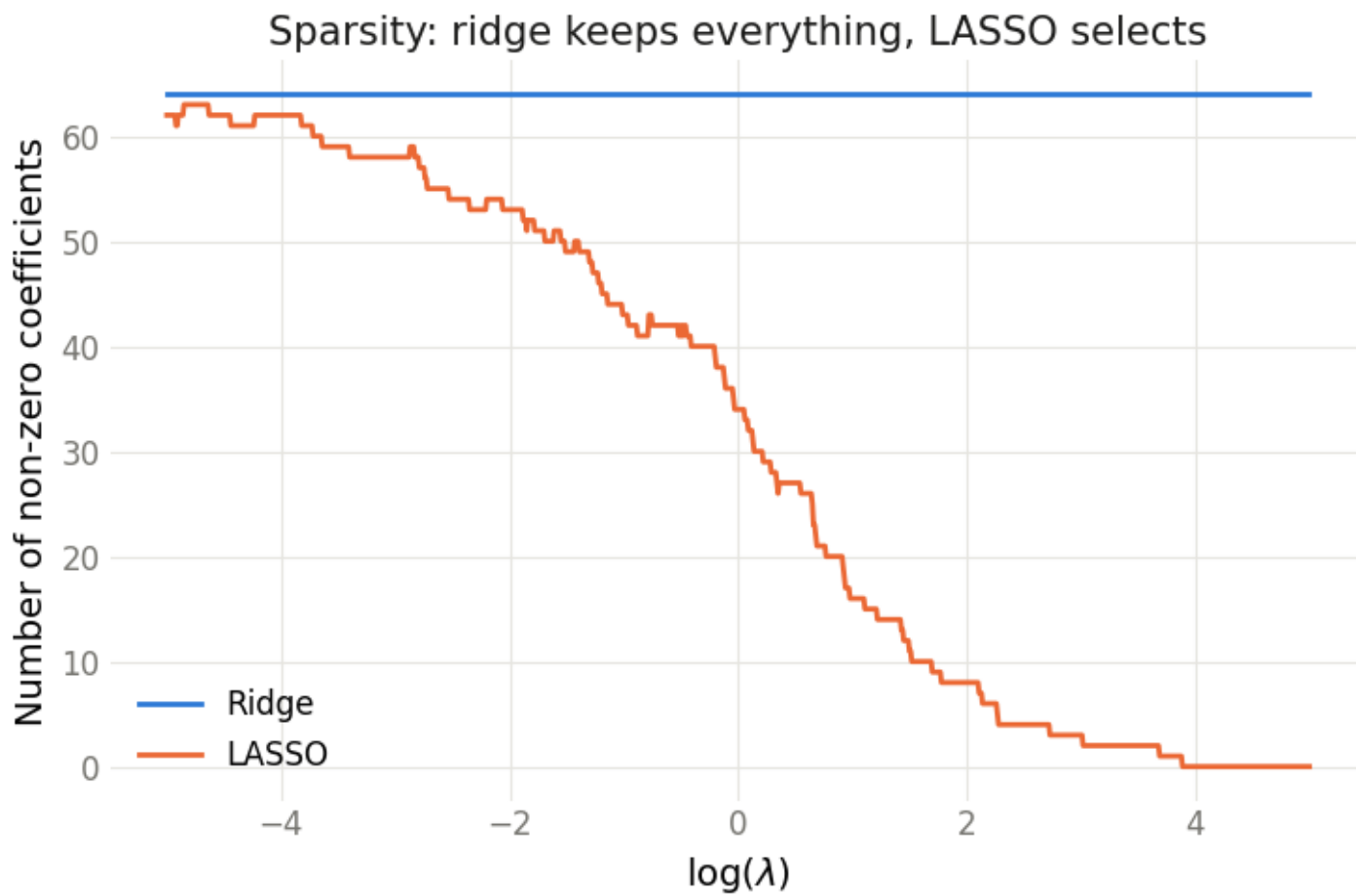
for the regularized diabetes model



Ridge Coefficient Profile

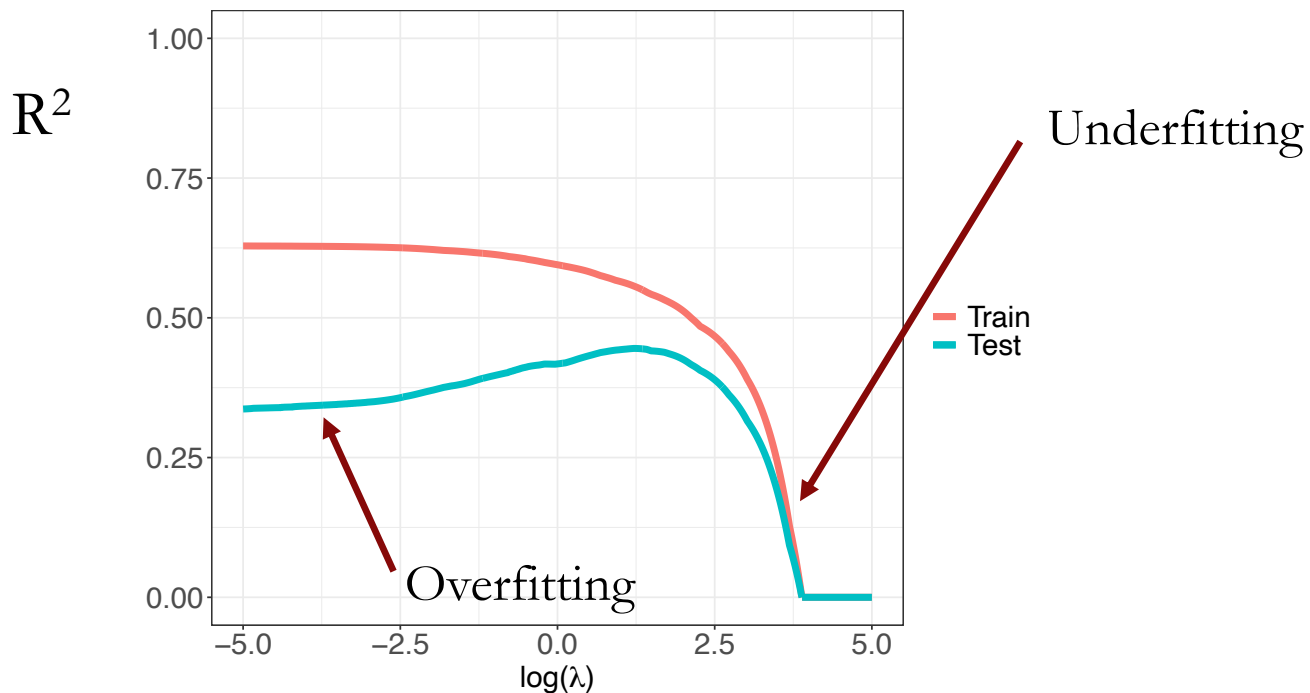


LASSO: Higher λ yields Simpler Models



LASSO: Performance

- In-sample performance: smaller λ always results in a better fit to Training data
- Out-of-sample performance: the best fit is typically achieved with intermediate values of λ , achieving a “sweet spot” between overfitting and underfitting



Python call λ “alpha”

alpha = 1.0: 34 non-zero coef

Intercept	: 153.3451
age	: 91.8553
sex	: -129.9518
bmi	: 575.6187
map	: 269.5742
tc	: -43.7085
hdl	: -213.1209
ltg	: 450.5594
age^2	: 44.3134
bmi^2	: 77.3966
tch^2	: 190.4212
glu^2	: 45.8530
age:sex	: 181.6209
age:bmi	: 29.1560
age:ldl	: -7.1227
age:hdl	: 16.2974
age:ltg	: 2.7179
age:glu	: 89.9253
sex:bmi	: 98.7207
sex:map	: 16.2283
sex:hdl	: 70.8751
sex:tch	: -99.1359
bmi:map	: 85.5033
bmi:tc	: -52.2914
bmi:ldl	: -7.4845
bmi:ltg	: 10.2341

...

alpha = 2.9447: 16 non-zero coef

Intercept	: 152.8508
age	: 37.9020
sex	: -53.9072
bmi	: 564.6246
map	: 241.2214
hdl	: -175.2877
ltg	: 422.1609
bmi^2	: 70.4747
tch^2	: 9.7551
glu^2	: 22.6446
age:sex	: 138.2954
age:map	: 0.7935
age:glu	: 102.9532
sex:bmi	: 35.4915
sex:hdl	: 42.3630
bmi:map	: 57.0496
map:tc	: 51.7285

alpha = 10: 4 non-zero coef

Intercept	: 153.1317
bmi	: 543.3172
map	: 136.0266
hdl	: -72.0403
ltg	: 364.3673

Impact

- Analytics can process vast amounts of data to help diagnose patients with critical diseases as early as possible
 - “Signs and symptoms of heart failure are often documented years before a diagnosis. [This] can offer [...] signals for early detection of this deadly disease” S. Steinhubl (2013), Cardiologist MD
- Design and use of predictive algorithms requires collaborations of analytics experts and domain experts (nurses, doctors, etc.)
- Predictive diagnosis can be leveraged to implement...
 - early treatment
 - personalized treatment: specific treatments to specific patients

Conclusions

- **Value of Simple Models.**
 - Complex models can fit to noise, i.e. patterns in the training data that do not generalize to the testing data.
- **Balancing Fit and Complexity.**
 - Regularization is a paradigm for balancing between in-sample fit and complexity, two objectives useful in constructing a model with high OSR^2 .
 - Cross-validation is used to pick the optimal balance between the two objectives
- **Regularization in Practice.**
 - For linear regression, LASSO is very popular in practice due to its computational tractability and good performance.